

All but one of the following atmospheric pollutants is matched with its probable environmental effect. Which agent is matched to an unlikely environmental consequence?

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- A) chlorofluorocarbons - stratospheric ozone depletion
- B) carbon monoxide - spruce forest die-off
- C) nitrogen oxides - photochemical smog
- D) carbon dioxide - global warming
- E) sulfur dioxide - acid precipitation

Molybdenum toxicity in cattle can be treated with which of the following?

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- A) chromium
- B) atropine
- C) copper sulfate
- D) neostigmine
- E) d-penicillamine

Which of the following is NOT true for turpentine?

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- A) it belongs to a group of compounds known as alicyclic hydrocarbons (terpenes)
- B) it causes depression of the central nervous system
- C) it is nephrotoxic
- D) it resembles oxides of nitrogen in its ability to directly induce lipid peroxidation
- E) its vapors can cause primary irritation of the skin and mucous membranes

Which of the following is NOT characteristic of atropine toxicity?

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- A) tachycardia
- B) CNS stimulation
- C) hypothermia
- D) dry mouth and skin
- E) flushing of the skin

Febrile illness and symptoms like those of acute respiratory disorder occur with inhalation of the fumes of which of the following compounds?

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- A) chromates
- B) hydrogen sulfide
- C) zinc
- D) beryllium
- E) formaldehyde

Which one of the following causes mineralization of soft tissues (heart, lung, arterial walls) after chronic ingestion of high doses?

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- A) vitamin A
- B) saxitoxin
- C) vitamin E
- D) vitamin D
- E) lycopene

Which of the following is NOT associated with salicylate poisoning?

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- A) tinnitus
- B) diplopia
- C) metabolic alkalosis
- D) petechiae
- E) none of the above

Fibrotic lung changes and malignancies of the bronchial tree and pleura are associated with which of the following conditions?

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- A) silicosis
- B) anthracosis
- C) asbestosis
- D) siderosis
- E) berylliosis



Which of the following causes functional alterations in monkeys and man that closely resemble the signs and symptoms of Parkinsonism?

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- A) lead
- B) mercury
- C) manganese
- D) thallium
- E) tellurium

After inhaling fumes, a 14-year-old boy developed severe respiratory irritation, pulmonary edema and conjunctivitis. On the basis of this information, which of the following substances could be eliminated as the cause of these symptoms?

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- A) ammonia
- B) sulfur dioxide
- C) dichlorodifluoromethane (freon)
- D) methyl chloride
- E) methyl bromide

Which of the following are true with respect to radiation?

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- 1) cancer can be increased in frequency by exposure to radiation
- 2) the hematopoietic system is especially sensitive to radiation
- 3) laboratory animals exposed to single or divided doses of radiation in the sublethal range have demonstrated a shortened life-span
- 4) radiation-induced cancers can be distinguished from cancers caused by exposure to carcinogenic chemicals

By definition, intentional food additives are used for which of the following?

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- 1) maintain freshness
- 2) increase nutritional value of food
- 3) flavor food
- 4) increase the esthetic quality of food by controlling damaging insect infestations

Which of the following plants owe their poisonous quality to their content of digitalis glycosides?

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- 1) lily of the valley
- 2) crocus
- 3) oleander
- 4) rhubarb

Which of the following statements is/are correct?

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- 1) carbon monoxide impairs hemoglobin release of O<sub>2</sub>
- 2) cyanide inhibits cytochrome oxidase
- 3) CCl<sub>4</sub> toxicity can be antagonized by antioxidants
- 4) hereditary methemoglobinemia results from a deficiency in NADH methemoglobin reductase

Severe respiratory depression in barbiturate poisoning is treated most effectively and safely with which of the following?

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- A) pentylenetetrazole
- B) amphetamine
- C) artificial respiration
- D) caffeine
- E) aminophylline

Which of the following form coordination complexes with certain metals?

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- A) sodium citrate
- B) dimercaprol (BAL)
- C) both
- D) neither



Which of the following is NOT correct in regard to phthalic acid esters?

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- A) they may produce polyneuritis in exposed human workers
- B) they prolong hexobarbital sleeping times in rats
- C) they interfere with survival of chick embryo heart cells in tissue culture
- D) they were responsible for the "Yusho" poisoning disease incidents
- E) they may be released from polymerized plastic resins

Death is caused by direct neuromuscular effects of which group of toxins?

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- A) staphylococcus toxin, saxitoxin, scorpion venom
- B) rattlesnake venom, saxitoxin, botulinum toxin
- C) cobra venom, scorpion venom, black widow spider venom
- D) rattlesnake venom, bee venom, black widow spider venom
- E) cobra venom, botulinum toxin, saxitoxin

Which of the following plants, when ingested, is most likely to cause symptoms of thirst, dilated pupils, rapid pulse, delirium, vertigo and coma?

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- A) autumn crocus
- B) foxglove
- C) rhubarb
- D) mushrooms
- E) jimson weed

Which of the following is NOT true for nitrite poisoning?

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- A) it may result from dietary ingestion of nitrates and nitrites in infants and young animals
- B) it is commonly associated with the therapeutic use of nitrite vasodilators
- C) it may arise from ingestion of plants grown on nitrate rich soil
- D) it results from the oxidation of hemoglobin to methemoglobin by nitrite ion
- E) it may be successfully treated by the intravenous administration of methylene blue

Which of the following is a characteristic of carbon monoxide?

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- A) it accumulates in blood by irreversible combination with hemoglobin
- B) it is detoxified by metabolism to form carbon dioxide
- C) it increases oxyhemoglobin dissociation
- D) it oxidizes the heme iron of hemoglobin from the ferrous to the ferric state
- E) it impairs the dissociation of oxygen from hemoglobin

Oximes, such as pralidoxime methiodide (2-PAM), have antidotal action in organophosphate by which of the following mechanisms?

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- A) competitive inhibition with acetylcholine at receptor sites
- B) physiological antagonism
- C) nucleophilic displacement of the organophosphorus inhibitor from the active site of acetylcholinesterase
- D) hydrolysis of phenylacetate
- E) none of the above

Most of the organophosphates used on a large scale in agriculture require biotransformation to unfold their full insecticidal potency. The responsible reaction occurring in insects and mammalian species is:

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- A) dealkylation
- B) hydrolysis of the "leaving group"
- C) desulfuration to an oxygen analogue
- D) "aging" of the acetylcholinesterase-inhibitor-complex
- E) hydroxylation to provide water solubility

Piperonyl butoxide is an inhibitor of which of the following?

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- A) pseudocholinesterase
- B) cytochrome P-450
- C) prothrombin synthesis
- D) acetaldehyde dehydrogenase
- E) none of the above



Which of the following is NOT true for metal fume fever?

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- A) it is caused by the freshly formed fumes of the oxides of magnesium, zinc, iron or copper
- B) it is frequently observed in industrial workers on Mondays or after holidays
- C) the toxic effects consist of chills, fever, profuse sweating and weakness that regress after one or two days
- D) the toxic effects are not serious because the causative agents are all essential metals for humans
- E) the pathogenesis is not known but may be due to endogenous pyrogens released from cell lysis

Which of the following is the most frequent route of exposure in cases of nonoccupational poisonings with pesticides?

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- A) dermal
- B) oral
- C) respiratory
- D) subcutaneous
- E) none of the above

Which of the following statements correctly describes the biotransformation of ethanol?

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- A) it is insignificant
- B) it has a half-life of approximately 2 hours in humans
- C) it is initiated primarily by catalase
- D) it is usually a zero-order process
- E) it proceeds from ethanol to acetaldehyde to acetic acid via the microsomal ethanol oxidizing system

The toxicity of the rodenticide fluoracetate is due to its ability to inhibit

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- A) mobilization of calcium from the bone
- B) the TCA (Krebs) cycle
- C) serum cholinesterase
- D) CNS acetylcholinesterase
- E) all of the above

Which of the following correctly pairs the plant and its active principle?

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- A) rhubarb leaves contains oxalic acid
- B) deadly nightshade contains atropine
- C) oleander contains digitalis-like glycosides
- D) nutmeg contains myristicin
- E) all of the above

Which of following solvents have been implicated as causing peripheral neuropathy due to their conversion to a toxic metabolite?

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- A) n-hexane and methyl n-butyl ketone
- B) n-pentane and methyl propyl ketone
- C) 2-methyl pentane and methyl isobutyl ketone
- D) propane and methyl ethyl ketone
- E) Stoddard solvent and toluene

All of the following about animal toxins are correct EXCEPT

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- A) animal venoms tend to be specific for a single organ or function
- B) rattlesnake and other Crotalid venoms are particularly rich in proteolytic enzyme activity
- C) hyaluronidase activity in animal toxins may be particularly important in causing local tissue edema
- D) in addition to specific therapy aimed at decreasing the effects of the envenomation, animal bites should be treated as a potential cause of tetanus
- E) stings caused by marine molluscs (Conus) have been fatal to humans

## The Acceptable Daily Intake (ADI) for pesticide residues

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- A) is based on animal experiments and structure-activity relationships
- B) is based on results of chronic dog studies applying a safety factor of 10
- C) is applied only for fruits and vegetables
- D) is based on animal and human health effects and no-observed-adverse-effect levels
- E) may not be exceeded in drinking water



Organophosphate Induced Delayed Neuropathy is associated with

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- A) triorthocresyl phosphate (TOCP)
- B) mass poisoning in Morocco as a result of cooking with contaminated vegetable oil
- C) paralysis and axonal degeneration
- D) a lesion described as a "dying-back" process
- E) all of the above

Which of the following is considered to be the primary mechanism of action of warfarin, a rodenticide of the coumarin family?

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- A) inhibition of the synthesis of prothrombin
- B) inhibition of microsomal enzymes
- C) free radical formation and lipid peroxidation
- D) inhibition of Vitamin K absorption

The critical organ for exposure to strontium 90 (beta-radiation) is

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- A) spleen
- B) intestine
- C) kidney
- D) liver
- E) bone

Which of the following is the most frequently reported industrial (occupational) disease?

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- A) chemical burns
- B) respiratory conditions caused by toxic materials
- C) dermatitis
- D) pneumoconioses

Drug induced phototoxicity involves which of the following effects?

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- 1) antibody formation
- 2) energy transfer to biological tissues
- 3) conjugation of drug with protein, and recognition as foreign material
- 4) alteration in cell membrane structure

Alkalinization of the urine by administration of sodium bicarbonate will increase the excretion of

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- 1) amphetamine
- 2) acetylcholine
- 3) hexamethonium
- 4) salicylic acid

In which case(s) would you expect to see no changes in the patient when a therapeutic dose of naloxone is administered?

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- 1) in acute methadone overdose
- 2) in a heroin addict with addiction maintained by heroin
- 3) in respiratory depression due to nalorphine overdose
- 4) in the absence of any other drug

## Heavy metal antagonists

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- 1) react with heavy metals to form tightly bound complexes
- 2) react with heavy metal to form chelates
- 3) act by preventing the binding of toxic metals to body ligands
- 4) depend solely upon ring size of the resulting complex for chelate stability



Benzene produces which of the following effects?

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- 1) CNS symptoms with acute oral poisoning
- 2) pancreatic injury in acute or chronic poisoning
- 3) bone marrow damage with chronic poisoning
- 4) toluene as a metabolic product

Which of the following statements correctly describes ethanol?

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- 1) it increases gastric acid secretion
- 2) it is metabolized to acetaldehyde
- 3) it produces more energy per gram than does glucose
- 4) it can produce general anesthesia

Which of the following correctly describes carbon tetrachloride?

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- 1) it is a central nervous system depressant
- 2) it produces liver damage
- 3) its toxicity is enhanced by stimulators of the hepatic microsomal mixed function oxidase
- 4) it produces kidney damage

Which of the following accumulates in bone?

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- 1) mercury
- 2) fluoride
- 3) sulfide
- 4) lead

Which of the following correctly describe cyanide?

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- 1) small amounts of cyanide are normally ingested in certain foods
- 2) cyanide binds to cytochrome oxidase, blocking electron transport
- 3) treatment for cyanide poisoning includes agents to convert hemoglobin to methemoglobin
- 4) in treating cyanide poisoning, thiosulfate administration promotes conversion of cyanide to thiocyanate

## Enzymes that are components of snake venoms

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- 1) proteinases
- 2) hyaluronidase
- 3) phospholipase
- 4) glucuronidase

Which of the following industrial toxicants may produce cancer of the respiratory tract?

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- 1) chlorine
- 2) nickel
- 3) kaolin
- 4) chromium

Lithium salts are often used therapeutically in the treatment of depression. Chronic lithium administration frequently

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- 1) results in toxicity to the blood-forming elements
- 2) is used as a therapeutic modality to treat hypothermia
- 3) is the treatment of choice for individuals with hereditary hypokalemia
- 4) causes toxicity to the cardiovascular, nervous or renal systems if serum levels are not monitored carefully



Which of the following statements regarding the respiratory system is/are true?

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- 1) the major compartments of the respiratory system are nasopharyngeal, tracheobronchial and pulmonary
- 2) impaction of panicles  $> 20\text{mm}$  occurs primarily in the tracheobronchial area
- 3) the pulmonary component is composed of bronchioles, alveolar ducts and alveoli
- 4) the mucociliary escalator is located primarily in the pulmonary area

Which of the following is a pulmonary function test that is useful in quantitating the distribution of ventilation?

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- A) frequency dependence of compliance
- B) maximal voluntary ventilation
- C) multibreath nitrogen washout
- D) peak expiratory flow-volume curve
- E) none of the above

Which of the following is NOT characteristic of teratogenic responses?

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- A) they may change with exposure at different gestation times
- B) they are influenced by the lipid solubility of the agent
- C) they are altered by the dose administered
- D) they may depend on maternal metabolism of an agent
- E) they are dependent entirely on the permeability of the placenta to the agent

Elevations of acid phosphatase are associated with tumors of which tissues?

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- A) chromaffin tissue
- B) breast
- C) testis
- D) bladder
- E) prostate

What is the best protocol for detecting teratogen potential?

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- A) exposure of the pregnant females during the period of major organogenesis; assessment of three dosage levels; and inclusion of a concurrent, negative control group
- B) exposure of the pregnant females during the entire gestational period; exposure of the males prior to mating the females; and inclusion of a concurrent, positive control group
- C) exposure of the pregnant females during the period of major organogenesis; assessment of three dosage levels; and inclusion of historical data from untreated animals
- D) exposure of the females prior to mating and through the period of major organogenesis; assessment of a single dosage level; and inclusion of a concurrent, negative control group

A temperature inversion in a densely populated area is characterized by

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- A) a stalled high-pressure system in which the air becomes stagnant
- B) a decrease in temperature with increasing altitude so that the heavier cold air holds down the polluted warmer air, preventing its dispersion
- C) hotter temperatures at night than in the day-time thereby preventing night breezes which disperse the pollutants
- D) an increase in temperature with increasing altitude so that the cooler polluted air near the ground cannot rise into the upper atmosphere

Which of the following correctly defines a procarcinogen?

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- A) it causes a tumor to develop when applied to a tissue in which initiation has already occurred
- B) it must act in conjunction with a primary carcinogen in order to induce tumor formation
- C) it must be metabolized to a proximate carcinogen in order to induce tumors
- D) it has a much longer latent period than promoting agents
- E) none of the above are correct

Which of the following correctly characterizes carbon tetrachloride?

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- A) it causes azotemia and anuria after respiratory failure
- B) it causes necrosis and fatty infiltration of liver
- C) the lethal dose in humans is approximately 0.5 to 1.0 ml
- D) its toxicity is antagonized by ethanol
- E) none of the above is correct



The incidence of chromosomal anomalies is highest in which group?

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- A) spontaneously aborted embryos
- B) stillbirths
- C) live births
- D) malformed infants
- E) none of the above

Chronic overexposure to which of the following produces organic psychoses?

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- A) carbon monoxide
- B) kerosene
- C) carbon tetrachloride
- D) arsine
- E) tetraethyl lead

Select the agent which is best associated with the acute toxic effect of severe central nervous system depression and ventricular fibrillation.

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- A) arsenic
- B) carbon monoxide
- C) naphthylamines
- D) carbon tetrachloride
- E) warfarin

Ethanol depresses activity within the CNS; however, during the acute phases of ingestion a hyperactivity-hyperexcitement seems to occur in certain individuals. This action of ethanol is due to which mechanism?

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- A) disinhibition - or initial selective depression of inhibitory neurons
- B) initial excitation of higher cortical centers
- C) acute gastritis
- D) spinal stimulation
- E) initial depression of the reticular activating system

A child who systemically absorbs atropine during a routine eye examination might experience all of the following except

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- A) confusion
- B) photophobia
- C) a rash over the face, neck and chest
- D) sialorrhea
- E) tachycardia

A major toxic effect of nitrobenzene is

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- A) mydriasis
- B) pulmonary granulomatosis
- C) methemoglobinemia
- D) Burtonian line formation
- E) hypoprothrombinemia

Methemoglobinemia and hemolytic anemia are characteristic symptoms of acute overdose with

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- A) phenacetin (acetophenetidin)
- B) aminopyrine
- C) aspirin
- D) colchicine
- E) allopurinol

Administration of sodium bicarbonate to treat barbiturate poisoning is an example of

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- A) weak acidic chemicals being excreted better in alkaline urine
- B) weak organic bases are in the non-ionized lipid soluble form in the intestine and tend to be absorbed there
- C) inactivation of a poison for essential receptors
- D) urine buffering
- E) physiologic antagonism



Which of the following does not cause delayed neuropathy in the hen?

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- A) mipafox
- B) leptophos (phosvel)
- C) diisopropylfluorophosphate (DFP)
- D) parathion
- E) tri-o-cresyl phosphate

The greater sensitivity of newborn animals to chemical immunogenesis is related to

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- A) poorly developed granulocytes
- B) incompetence of the newborn immune system compared to that of the adult
- C) the relatively small size of the thymus in the newborn
- D) a poorly developed adrenal medulla
- E) the presence of circulating maternal antibodies in the newborn animal

The primary agent responsible for the development of hematologic malignancies in mice following exposure to immunosuppressive agents is often

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- A) a coliform bacterium
- B) a rickettsial agent
- C) an endogenous retrovirus
- D) pneumocystis carinii
- E) candida albicans

Activated T-lymphocytes have all of the following functions EXCEPT

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- A) local inhibition of macrophage migration
- B) secretion of cytokines
- C) synthesis and secretion of immunoglobulins
- D) recruitment of other T-cells
- E) formation of "memory" cell population

Which of the following is unlikely to increase sensitivity to a contact allergen?

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- A) adjuvants
- B) advanced age
- C) hereditary predisposition
- D) high concentration of the allergen
- E) occlusive patch

The condition known as "wavy ribs" in rats

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- A) is properly classified as a developmental variation
- B) is lethal because the ribs collapse and the neonates suffocate
- C) is a true malformation because it results from direct effects of a test chemical on developing bone
- D) is a true malformation because if one allows neonatal rats with wavy ribs to survive they develop grossly malformed thoracic cavities
- E) develops during day 0 to day 4 of gestation

Abnormal discoloration of the skin or gums is associated with all of the following EXCEPT

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- A) arsenic
- B) cadmium
- C) lead
- D) mercury
- E) silver

Which of the following is the animal species of choice to test chemicals for possible peripheral neuropathy?

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- A) guinea pig
- B) hamster
- C) hen
- D) rabbit
- E) rat



Reduced litter size in rodents could be due to

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- A) increased resorption rate
- B) decreased implantation rate
- C) increased nonlethal malformation rate
- D) A or B
- E) A, B or C

In predicting the irritation potential of a material after accidental human exposure, based on the results of an eye irritation test, it is important to consider

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- A) the results of the test are highly consistent showing very little variation between laboratories
- B) the rabbit eye is more sensitive than the human eye, generally showing more severe and prolonged irritation under the same conditions of exposure
- C) the scores associated with effects on the iris are given the greatest weight in predicting injury to the human eye
- D) significant injury to the eye only occurs with materials which reach the eye by way of the systemic circulation
- E) all of the above

Which of the following characteristics correctly reflects the one-hit model of carcinogenesis?

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- A) one-hit means that one molecule of DNA may be initially affected; such genetic alteration may eventually lead to the expression of a tumor
- B) this model incorporates maximum human exposure
- C) this model assumes linear extrapolation and nonthreshold in the low dose range
- D) this model yields the highest virtually safe dose (VSD) as compared to the other available models
- E) A and C

Which of the following is NOT an advantage of whole-body inhalation systems?

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- A) ease of long-term exposure
- B) lack of restraint of animal subjects
- C) minimize skin contamination
- D) ease of controlling temperature of animal environment
- E) large number of animals can be exposed at once

Which of the following is NOT characteristic of a reticulocyte?

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- A) it can divide
- B) it can synthesize small amounts of hemoglobin
- C) it has functional cytochrome systems
- D) it matures in 24-36 hours in the systemic circulation
- E) it can be easily identified by staining with new methylene blue dye

Genetic toxicology tests designed to detect clastogenesis include

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- A) Ames/Salmonella assay
- B) micronucleus test
- C) fungal aneuploidization
- D) chromosomal observation
- E) B and D

Which of the following neurotoxins inhibits a specific esterase within the nerve fiber?

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- A) methyl n-butyl ketone
- B) carbon disulfide
- C) acrylamide
- D) 2,5-hexanedione
- E) diisopropyl fluorophosphate

The critical period for developmental toxicity from thalidomide in humans is

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- A) weeks 6 to 7
- B) weeks 20 to 25
- C) weeks 30 to 36
- D) weeks 50 to 52
- E) entire gestational period



Which of the following is INCORRECT?

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- A) adactylia = absence of digits on the hand or foot
- B) ectopic kidney = kidney with a dilated pelvis
- C) thoracoschisis = fissure in the thorax due to incomplete juncture or non juncture of the left and right thoracic wall
- D) hemivertebra = a vertebra with incomplete development of one side
- E) patent ductus arteriosus = non closure of the artery connecting the pulmonary artery and aorta

The most common type of reproductive failure in humans is

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- A) testicular atrophy
- B) spontaneous abortion in the first trimester
- C) headaches
- D) cleft palate
- E) none of the above

Primary irritation of the skin is measured by a patch-test technique on the abraded and intact skin of the albino rabbit clipped free of hair. The method consists of introducing under a one-inch patch

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- A) 0.5 ml (in case of liquid) or 0.5 gm (in case of solids or semi-solids) of the test substance
- B) one-half the amount on the abraded site as used on the intact skin
- C) graded doses (generally three) are applied to each test rabbit
- D) one-half the amount and 1/4 the amount found to produce erythema and wheal formation in a range-finding study
- E) enough liquid material to saturate the patch and enough solid material to cover the test site with a thin layer

The liver of the polar bear will induce toxicity in humans when eaten in large quantity. The reason is

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- A) hypervitaminosis K with resultant internal bleeding
- B) hypervitaminosis A and liver necrosis
- C) excess triglyceride content results in vascular occlusion
- D) accumulation of lipid soluble toxicants in polar bear liver induces a variety of toxicoses when consumed

Liver toxicity is associated with exposure to which of the following compounds?

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- 1) carbon tetrachloride
- 2) halothane
- 3) ethanol
- 4) benzene

## Ethylmethanesulfonate (EMS)

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- 1) causes misreading of guanine in DNA
- 2) requires metabolic activation
- 3) causes point mutations
- 4) causes deletion mutations

Which of the following commonly produces kidney damage with acute intoxication?

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- 1) isopropyl alcohol
- 2) propylene glycol
- 3) ethylene glycol
- 4) methylene chloride

Methemoglobinemia may occur on exposure to

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- 1) nitrates
- 2) cyanide
- 3) acetanilid
- 4) carbon monoxide



Cytological examinations are useful in the detection of the following types of mutations:

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- 1) breaks
- 2) base-pair substitution
- 3) deletions
- 4) frameshift

Which of the following is correct for delayed (type IV) hypersensitivity?

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- 1) it is antibody independent
- 2) it is T-lymphocyte mediated
- 3) it is a contact type sensitivity
- 4) it is IgA mediated

Macrophages are involved in which of the following immune functions?

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- 1) antigen processing
- 2) phagocytosis
- 3) release of lymphocyte mitogenic protein
- 4) immunologic memory

Which of the following drugs increase(s) the response to oral anticoagulants?

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- 1) cholestyramine
- 2) acetaminophen
- 3) sodium salicylate
- 4) acetylsalicylic acid

Which of the following statements regarding organophosphate induced delayed neurotoxicity is/are true?

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- 1) attention was drawn to the problem in 1930 by an outbreak of a "ginger Jake" paralysis caused by contamination of ginger extract by cresyl phosphates
- 2) all species of animals are equally affected
- 3) delayed neurotoxicity has been reliably shown to occur in poisoning from DFP, leptofofos and mipafox
- 4) after a single exposure of adequate magnitude to the neurotoxic organophosphate compounds, maximal damages can be demonstrated within 48 hours

Which of the following is/are true integrative processes of the hypothalamo-pituitary-gonadal axis?

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- 1) FSH and LH are glycoproteins synthesized and released from a subpopulation of the basophilic gonadotrophic cells of the pituitary gland
- 2) luteinizing hormone-releasing hormone (LH-RH) acts on gonadotrophic cells, thereby stimulating the release of FSH and LH
- 3) LH-RH and follicle-stimulating hormone-releasing hormone (FSH-RH) appear to be the same substance
- 4) gonadotropins are not important in the female

Which of the following statements is/are true?

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- 1) aneuploidization, clastogenesis and gene locus mutations have all been shown to occur in man as a direct result of exposure to one or more chemical substances in the environment
- 2) the micronucleus, Ames Salmonella and mouse lymphoma assays all test for gene locus mutations
- 3) clastogenesis and gene locus mutagenesis, but not aneuploidization have been shown to be produced by a variety of chemicals
- 4) assaying for clastogenesis consists of examining stained and fixed chromosomes by light microscopy for evidence of chromatid gaps, isochromatid gaps, chromatid breaks, isochromatid breaks, dicentric and exchange chromosomes and translocations

The effect of toxic chemicals on renal function can be assessed by measurement of

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- 1) inulin clearance
- 2) blood urea nitrogen
- 3) urine protein concentration
- 4) plasma creatinine



Which of the following is true regarding hepatotoxicity of a compound?

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- 1) hepatotoxicity need not be correlated with the pharmacokinetics of the parent compound or its major metabolites
- 2) a threshold tissue concentration must be attained before liver injury is elicited
- 3) glutathione plays an essential role in protecting hepatocytes from injury by chemically reactive intermediates
- 4) glutathione transferase and epoxide hydrolase play a role in protecting the hepatocytes

Carbon tetrachloride, ethionine, phosphorus, puromycin and tetracycline have the following common effect regarding fatty liver induction in the rat?

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- 1) lower the level of circulating lipoprotein
- 2) interfere with synthesis of the protein moiety of lipoproteins
- 3) impair phospholipid synthesis
- 4) block the secretion of hepatic triglycerides into the plasma

Which of the following is/are true for the Ames/Salmonella assay?

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- 1) mutant strains require histidine for growth
- 2) mammalian enzymes can be added to test the mutagenicity of chemical metabolites
- 3) reversions are likely to be produced by test chemicals causing base-pair substitution
- 4) mutant strains were first developed in Ames, Iowa

The apparent volume of distribution of a chemical

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- A) is usually equal to the volume of total body
- B) can be larger than the total body volume
- C) can be smaller than the total body water if the chemical is highly tissue bound
- D) is equal to the chemical's water solubility
- E) none of the above

Which of the following is NOT an example of a metabolic oxidation?

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- A) the formation of an alcohol from an aliphatic hydrocarbon
- B) the conversion of an alcohol to an acid
- C) the formation of an epoxide
- D) the methylation of an amino group
- E) the formation of an aldehyde from an alcohol

Lavage is contraindicated after the ingestion of

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- A) concentrated acid
- B) concentrated alkali
- C) kerosene
- D) lethal doses of phenobarbital 12 hours earlier
- E) all of the above

Dioxin (2,3,7,8-tetrachlorodibenzodioxin) contamination may occur in association with the production and/or use of?

---

- A) hexachlorophene
- B) 2,4,5-trichlorophenol
- C) 2,4,5-T (trichlorophenoxyacetic acid)
- D) pentachlorophenol
- E) all of the above

## The blood brain barrier

---

- A) is a less effective barrier than the placental barrier
- B) provides a barrier for charged molecules only
- C) provides a barrier only for water-soluble chemicals
- D) provides a barrier only for lipid-soluble chemicals
- E) none of the above



In first order kinetics, the half-life of a chemical is independent of the dose and of the time of measurement.

---

A) true

B) false

Which of the following groups is the most likely to experience accidental poisoning?

---

- A) adults over eighty years of age
- B) pesticide applicators
- C) children under five years of age
- D) drug manufacturing company employees
- E) chemical manufacturing company employees

Phenobarbital poisoning can be treated by enhancing its excretion into the urine by

---

- A) increasing renal sodium reabsorption
- B) increasing renal potassium reabsorption
- C) increasing pH of the urine
- D) decreasing pH of the urine
- E) none of the above

Which of the following statements about 2,4-dinitrophenol is true?

---

- A) uncouples oxidative phosphorylation
- B) inhibits cytochrome oxidase
- C) inhibits P-450
- D) inhibits aconitase
- E) inhibits vitamin K synthesis

The mechanism by which most agents are absorbed from the gastrointestinal tract is

---

- A) passive diffusion
- B) filtration
- C) active transport
- D) facilitated diffusion
- E) pinocytosis

The passive diffusion of agents across tissue membranes depends primarily on

---

- A) lipid solubility of the ionized molecule
- B) water solubility of the nonionized molecule
- C) membrane pore size
- D) lipid solubility of the nonionized molecule
- E) none of the above

Inorganic mercury deposited in lakes and ponds becomes a toxic hazard

---

- A) as it is consumed by animals or humans drinking the contaminated water
- B) as it is readily absorbed by plants and algae and stored as inorganic mercury in the plant. cellular structure to enter the food chain as plants/algae are consumed by animals higher up the food chain
- C) as microorganisms present in the silt convert inorganic mercury to organic mercury compounds
- D) binds aggressively to the mud and silt and as the ponds/lakes dry up releases inorganic mercury into the breathing air
- E) readily absorbed by fish through the gills and enters the food chain as contaminated fish are consumed

Dermal absorption of a material is influenced most by which of the following chemical properties?

---

- A) pH
- B) molecular weight
- C) density
- D) oil/water partition coefficient
- E) particle size



The most important tissue for a forensic toxicologist to analyze in a death due to suspected as an indicator of exposure to chronic heavy metal poisoning is

---

- A) brain
- B) lung
- C) hair and fingernails
- D) stomach contents
- E) blood

The pharmacokinetic value that most reliably reflects the amount of drug reaching the target tissue after oral administration is the

---

- A) peak blood concentration
- B) time to peak blood concentration
- C) product of volume of distribution and the first-order rate constant
- D) volume of distribution
- E) area under the blood concentration-time curve

Which of the following applies to the conduct and interpretation of epidemiologic studies?

---

- A) explore the origin, nature, pathology and prevention of diseases in a community or large area
- B) are reliable indicators for human risk assessment, but have limitations because disease prevention is usually effective for only later populations
- C) are difficult to utilize when investigating chronic low-level environmental exposure
- D) should be used in conjunction with other toxicologic tests and assays
- E) all of the above are true

Disulfiram (Antabuse) as a drug used in alcoholism blocks

---

- A) monoamine oxidase
- B) alcohol dehydrogenase
- C) the oxidation of acetaldehyde
- D) acetylcholinesterase

## The distinction between genotoxic and epigenetic carcinogens

---

- A) has been useful for regulatory decision making since the early 1960's
- B) is unimportant
- C) is that genotoxic carcinogens produce an observable change in genetic material but epigenetic carcinogens do not
- D) cannot be made on the basis of current knowledge
- E) is crucial because humans are not susceptible to cancers produced in animals by epigenetic carcinogens

Delta-aminolevulinic acid in urine is monitored

---

- A) arsenic
- B) cadmium
- C) chromium
- D) lead
- E) mercury

The federal agency that is required to enforce the Safe Drinking Water Act of 1972 (SDWA) is

---

- A) Environmental Protection Agency
- B) Food and Drug Administration
- C) Consumer Product Safety Commission
- D) Occupational Safety and Health Administration
- E) all of the above

Pesticide petitions submitted to the Environmental Protection Agency by the manufacturer of a pesticide must contain

---

- A) results of crop residue studies
- B) animal and plant metabolism data
- C) description of the manufacturing process for the pesticide
- D) toxicology studies conducted with mammals, birds, fish and invertebrates
- E) all of the above



Which of the following statements correctly describes the blood brain barrier?

---

- A) it allows free passage of circulatory substances into the immediate brain interstitial space
- B) its capillaries do not differ from capillaries found elsewhere in the body in any identifiable manner
- C) it does not exclude methyl mercury to the same extent as it excludes inorganic mercury
- D) its function is independent of age
- E) all of the above

Pyrethroid insecticides appear to affect the nervous system by

---

- A) interfering with sodium conductance in individual neurons
- B) reversibly binding to cholinesterase enzymes
- C) irreversibly binding to brain acetylcholinesterase
- D) inhibiting the TCA (Krebs) cycle
- E) displacing oxygen from receptors in hemoglobin

Which of the following statements characterizes acute organophosphate insecticide poisoning?

---

- A) onset of symptoms are usually delayed due to accumulation of endogenous acetylcholine in nerve tissue
- B) gastrointestinal irritation or discomfort is rare because organophosphates in general are mild irritants
- C) the immediate cause of death in fatal organophosphate poisonings is asphyxia resulting from respiratory failure
- D) in fatal untreated poisonings, deaths usually occur between 48 and 96 hours after exposure
- E) organophosphate insecticides are slowly metabolized and excreted, resulting in slow reversible inhibition of cholinesterase

The most common method used in extrapolating from animal subchronic toxicologic data to estimate an initial human dose for a candidate drug is

---

- A) based on low-dose risk estimation, e.g., Mantel-Bryan procedure
- B) based on the safety factor approach using the type of toxicity, the comparative pharmacokinetic data and benefit consideration
- C) based on the use of the body surface area
- D) to divide the animal toxic dose by 10
- E) none of the above

Clinical signs in a patient include marked respiratory depression (3 to 4 per minute), cyanosis, pinpoint pupils, weak pulse, pulmonary edema, areflexia, and nodding. The person may be suffering from an overdose of

---

- A) atropine
- B) ethylene oxide
- C) phencyclidine
- D) heroin
- E) all of the above

Benzoic acid is a weak acid ( $pK_a$  4) and aniline is a weak base ( $pK_a$  9). Taking the pH of the stomach as 2 and the plasma as 7.4, which one of the following statements is correct?

---

- A) benzoic acid is largely in the non-ionized form and aniline is largely ionized in the stomach (>99%)
- B) benzoic acid is largely ionized and aniline is largely in the non-ionized form in the stomach (>99%)
- C) both benzoic acid and aniline are only partially ionized (about 50%) in the stomach
- D) benzoic acid is largely in the non-ionized form and aniline is largely ionized in the plasma (>99%)
- E) both benzoic acid and aniline are only partially (about 50%) ionized in the plasma

Which of the following statements is/are true?

---

- A) parametric tests assume that the data are continuous, and the population has some known form which can be described with measures of central tendency and dispersion
- B) ANOVA is used for the comparison of three or more groups of continuous data when the variances are homogeneous and the data are independent and normally distributed
- C) non-parametric tests are generally used to analyze data with a normal distribution
- D) A and B

Idiosyncratic reactions to drugs are characterized by

---

- A) an allergic type of response to a chemical
- B) a genetically determined abnormal reactivity to a chemical
- C) an acquired deficiency of the nervous system
- D) an acquired deficiency of the immune system



An easy way to determine the half-life of elimination is to plot the following variables in an appropriate manner. Select the correct one.  $C_p$  is the plasma concentration.  $C_u$  is the urine concentration and  $T$  is the time.

---

- A) plot  $\log C_p$  vs  $T$
- B) plot  $\log C_u$  vs  $\log T$
- C) plot  $C_u$  vs  $T$
- D) plot  $C_u$  vs  $\log T$
- E) A or B can be used

Of the following models currently used for risk extrapolation which one is likely to produce the most conservative estimate (lowest dose) when used to extrapolate data obtained at high doses?

---

- A) logit
- B) linear
- C) multi-hit
- D) probit

One of the most serious concerns impeding the use of domestic sewage sludge as fertilizer for agricultural crops is contamination with

---

- A) cadmium
- B) tributyl tin
- C) chlorine
- D) lead
- E) arsenic

Which of the following air pollutants is not classified as a pulmonary irritant?

---

- A) nitrogen dioxide
- B) carbon monoxide
- C) sulfur dioxide
- D) formaldehyde

Chemicals which are persistent in the environment are characteristically

---

- 1) water soluble
- 2) complexed with metal ions
- 3) palatable to animals
- 4) lipid soluble

Which of the following statements represent(s) valid guidelines for experiments designed to predict the degree of safety of a material for man?

---

- 1) the metabolism and distribution of the material should be as similar as possible in the test species and humans
- 2) a threshold dose for non-neoplastic adverse effects exists for most compounds
- 3) statistical significance may not mean biological significance
- 4) effects demonstrated after excessive unrealistic dosages are of no value in safety studies

The toxic effects of near fatal doses of ionizing radiation almost always include

---

- 1) nausea, vomiting and diarrhea
- 2) peripheral neuritis
- 3) epilation, hemorrhages
- 4) hypertension

The Food and Drug Administration (FDA) is charged with the responsibility for seeing that drugs are

---

- 1) safe
- 2) efficacious
- 3) pure
- 4) properly labeled



Characteristics of a chemical that favors bioaccumulation include

---

- 1) small apparent volume of distribution
- 2) slow rate of biotransformation
- 3) high affinity for plasma proteins
- 4) lipophilicity

Which of the following is/are associated with the cytosol during cell fractionation?

---

- 1) acid phosphatase
- 2) glutathione-s-transferase
- 3) NADPH-cytochrome C reductase
- 4) N-acetyltransferase

Which of the following is/are highly persistent pesticide(s)?

---

- 1) DDT
- 2) carbaryl
- 3) chlordane
- 4) malathion

Pharmacokinetics may be used to

---

- 1) explain species differences in dose response
- 2) predict the steady state concentration of a chemical in the body and tissues after repeated exposure
- 3) explain the plateau of a dose response curve for the adverse affect of a chemical
- 4) determine target organ of toxicity

In calculating the risk of death for a given population from exposure to an environmental chemical it is essential to characterize

---

- 1) the age distribution of the population in question
- 2) potential competing causes of death which may preclude death from the cause in question
- 3) a working hypothesis for a mechanism whereby the chemical may cause death
- 4) exposure parameters to be utilized in making the calculation

Which of the following present(s) air pollution concerns?

---

- 1) lead
- 2) ozone
- 3) oxides of nitrogen and sulfur
- 4) wood smoke

Hyaline droplet nephropathy induced by unleaded gasoline (trimethylpentane) or d-limonene is characterized by which of the following?

---

- 1) activation of these agents by. proximal tubular cytochrome P-450
- 2) a male-specific response in the rat
- 3) binding of activated metabolites to basement proteins in the glomerulus
- 4) binding to alpha-2-microglobulin and subsequent accumulation in proximal tubular cells

Which of the following requires metabolic activation for toxicity to occur?

---

- 1) parathion
- 2) fluoroacetamide
- 3) carbon tetrachloride
- 4) benzo(a)pyrene



Toxicity due to synergism between two drugs can most likely occur if the drugs

---

- 1) have similar pharmacologic activity
- 2) are toxic at high dose levels
- 3) have the same target organs
- 4) interact to mutually inhibit absorption across membranes

Epidemiological studies with immigrant populations are sometimes used to evaluate whether mortality rate differences between the populations are due to environmental factors or to a genetic composition. If observed differences are related to environmental differences, one would expect that

---

- 1) death rates for natives of the country of origin would equal the immigrant death rates
- 2) death rates for natives of the country of origin would differ from immigrant death rates
- 3) death rates for immigrants will differ from death rates for natives of the country of adoption
- 4) death rates for immigrants would equal death rates for natives of the country of adoption

Data from human epidemiologic surveys have shown an association between

---

- 1) hepatic angiosarcoma and vinyl chloride
- 2) bladder cancer and beta naphthylamine
- 3) mesothelioma and asbestos
- 4) dibromochloropropane and male infertility

Succinylcholine is accurately described by which of the following statements?

---

- A) it is a local anesthetic
- B) it is hydrolyzed by plasma cytochrome oxidase
- C) it causes an idiosyncratic response in patients with an atypical pseudocholinesterase
- D) it may cause prolonged apnea in patients previously exposed to anticholinesterase agents
- E) Both C and D are correct

Which of the following statements is true of Arthropod venoms?

---

- A) they often contain serotonin, kinins, haluronidase, phospholipase and a paralyzing factor such as mellitin
- B) they rarely give rise to allergic reactions
- C) they are never treated with antivenins
- D) they are produced by bees, centipedes, assassin bugs, gila monsters and echinoderms
- E) they are usually more complex and contain more enzymes than snake venoms

Alkaline diuresis should be considered in the treatment of poisoning with which of the following

---

- A) phenol
- B) sulfuric acid
- C) phenobarbital
- D) sodium hydroxide
- E) none of the above

Phencyclidine use has been associated with all except which of the following?

---

- A) marked aggressive behavior
- B) disassociative anesthesia
- C) tachycardia
- D) ataxia
- E) colonic spasm

Headache, dizziness, tinnitus, nausea and vomiting, and hyperthermia after overdose are associated with which of the following?

---

- A) aspirin
- B) codeine
- C) acetaminophen
- D) morphine
- E) phenylbutazone



Which of the following is a siderochrome derivative with a strong affinity for iron?

---

- A) dimercaprol
- B) edetate
- C) penicillamine
- D) desferrioxamine
- E) methylene blue

Which of the following statements is an incorrect pairing of metal and toxic effect

---

- A) mercuric ions - toxic to the kidney
- B) methyl mercury - toxic to nervous tissue
- C) ferrous ions - toxic to the gastrointestinal tract
- D) thallium - hirsutism
- E) nickel - contact dermatitis

The signs of ethylenediamine tetraacetic acid (EDTA) poisoning include which of the following?

---

- A) hypercoagulability of the blood with thrombus formation
- B) hypercoagulability of the blood reversible with vitamin K administration
- C) neuromuscular hyperirritability and tetany reversible by neostigmine administration
- D) neuromuscular hyperirritability and tetany reversible by administration of calcium gluconate
- E) somnolence and signs of decreased neuromuscular transmission similar to that seen after intravenous administration of magnesium sulfate

Disulfiram is correctly described by which statement

---

- A) it inhibits alcohol dehydrogenase
- B) it causes acetaldehyde and acetic acid to build up in the blood
- C) it has primary side effects of enhanced sedation when given in the presence of alcohol
- D) its action persists for one week or more after stopping administration
- E) it can be used effectively in treating methanol poisoning

Ingestion of which one of the following plant would be most likely to induce symptoms of severe thirst, nausea and vomiting, dilated pupils, rapid and weak pulse, vertigo, delirium, slow respiration, are convulsions and coma?

---

- A) castor bean (*Ricinus communis*)
- B) foxglove (*Digitalis purpurea*)
- C) rhubarb (*Rheum rhaponticum*)
- D) pokeweed (*Phytolacca americana*)
- E) jimson weed (*Datura stramonium*)

Lead poisoning inhibits heme synthesis leading to an increased urinary excretion of a number of approximately which one of the following heme precursors. Which of the following will appear first with increasingly severe poisoning?

---

- A) uroporphyrinogen 111
- B) delta aminolevulinic acid
- C) protoporphyrin 111
- D) coproporphyrin III
- E) protoporphyrin IX

Chronic manganese poisoning has been most successfully treated with

---

- A) penicillamine
- B) metal sequestering agents
- C) L-Dopa
- D) atropine sulfate
- E) iron salts

Oximes are used to treat human poisoning caused by

---

- A) carbaryl
- B) parathion
- C) carbofuran
- D) aldicarb
- E) aldrin



Ethanol has a specific gravity of 0.79 and a vapor pressure of 50 mm Hg at 25 degrees C. An ethanol saturated atmosphere would contain which of the following amounts of ethanol (on a vol/vol) basis?

---

- A) 4%
- B) 7%
- C) 92000 ppm
- D) 12000 ppm
- E) 7000 ppm

Delayed neurotoxicity is not observed with which of the following?

---

- A) tri-o-cresyl phosphate
- B) mipafox
- C) methoxychlor
- D) leptofos
- E) di- isopropyl fluorophosphate

An infusion of potassium has been shown to correct the potassium displacing capacity of which of the following?

---

- A) aluminum
- B) barium
- C) cadmium
- D) iron
- E) copper

An inhalation toxicity study of chemical "x" is being conducted with exposure concentrations of 1.0 and 5.0 mg/l. "x" has a vapor pressure of 0.038 mm Hg (23 degrees C), a molecular weight of 240 gm/mole, and a density of 0.5. The chambers (4 cubic meters) are operated under dynamic airflow conditions at 800 l/min. A 23 degree C atmospheres saturated with vapors of "x" would contain which one of the following concentrations?

---

- A) 0.5 mg/l
- B) 2.0 mg/l
- C) 5.0 mg/l
- D) 20 mg/l
- E) 200 ppm

Although most mycotoxic fungi invade cereal crops prior to harvest, which of the following is primarily a problem during storage

---

- A) Fusarium
- B) Claviceps
- C) Aspergillus
- D) Rhizoctonia
- E) none of the above

The convulsions caused by picrotoxin result from

---

- A) direct action of picrotoxin on neurons in the CNS, preventing the release of glycine
- B) hyperpolarization of dopaminergic neurons in the CNS
- C) the anticholinesterase activity of picrotoxin
- D) antagonist action of picrotoxin at glycine receptors in the CNS
- E) noncompetitive inhibition of GABA responses

Fat accumulation in liver cells following excessive halogenated hydrocarbon exposure is thought to be

---

- A) due to a reduction in the amount of triglyceride lipoprotein complex resulting in reduced triglyceride removal from the cells
- B) due to mobilization of fatty acids from body stores causing increased triglyceride synthesis
- C) an early stage of cell necrosis due to altered mitochondrial function
- D) caused by an increased level of circulating catecholamines resulting in hepatocellular hypoxia
- E) directly related to increased intracellular availability of glycerol for triglyceride synthesis

Which of the following herbicides is an effective antithyroid agent, and has produced thyroid adenomas and adenocarcinomas in rats after chronic dietary administration?

---

- A) 2,4,5-trichlorophenoxyacetic acid (2,4,5-T)
- B) dinitro-ortho-cresol
- C) paraquat
- D) amitrole
- E) propanil



Which-of the following statements is/are true?

---

- A) all GRAS substances permitted in foods undergo rigorous laboratory tests prior to approval
- B) food ingredients that do not meet GRAS criteria must be eliminated
- C) substances on the GRAS list are subject to review and removal
- D) the Delaney clause states that "no additive shall be deemed to be safe if it is found to be toxic in any species at any dose"
- E) A, C and D

Which of the following rodenticides acts principally by blocking the enzyme aconitase in the Krebs (tricarboxylic acid cycle)?

---

- A) norbormide
- B) warfarin
- C) alpha naphthyl thiourea (ANTU)
- D) red squill (scilliroside)
- E) compound 1080 (sodium fluoroacetate)

Which of the following classes of pesticides would generally be expected to be LEAST acutely lethal to humans by oral ingestion?

---

- A) nicotine
- B) organophosphates
- C) cyclodienes
- D) triazines
- E) bipyridyl compounds

The genetically inherited condition referred to as Wilson's disease is primarily due to toxicity of which of the following?

---

- A) zinc
- B) copper
- C) iron
- D) lead
- E) manganese

Plants containing digitalis-like glycosides include

---

- A) foxglove (*Digitalis purpurea*)
- B) lily of the valley (*Convallaria majalis*)
- C) oleander (*Nerium oleander*)
- D) A and B only
- E) all of the above

Warfarin, a widely used rodenticide, produces toxicity by which mechanism of action?

---

- A) formation of free radicals which bind to DNA
- B) uncoupling of oxidative phosphorylation
- C) anticoagulant action and direct capillary damage
- D) inhibition of acetylcholinesterase
- E) impairment of electron transport

Dioxane is characterized as which of the following?

---

- A) a contaminant in 2,4,5-T
- B) a microsomal enzyme inhibitor
- C) an ethylene glycol dimer
- D) a chlorinated hydrocarbon insecticide
- E) A and B only

Which compound produces toxicity by uncoupling oxidative phosphorylation?

---

- A) dibromochloropropane
- B) hexachlorobenzene
- C) pentachlorophenol
- D) rotenone
- E) toxaphene



Two situations which theoretically provide equal hemoglobin capacity for oxygen transport are (a) 50% blood carboxyhemoglobin concentration, and (b) removal of 50% hemoglobin from the blood. Which of the following comparisons is true?

---

- A) the oxygen content of the blood leaving the lungs is lower in (a) than (b)
- B) the oxygen content of the peripheral venous blood is lower in (a) than (b)
- C) the oxygen content of the blood leaving the lungs is higher in (a) than (b)
- D) the tissue hypoxia is more severe in (a) than (b)
- E) the individual oxyhemoglobin dissociation constants are the same in (a) and (b)

Select the correct toxin and signs/symptoms which may result when certain parts of the puffer fish are consumed.

---

- A) tetrodotoxin and liver necrosis
- B) saxitoxin and convulsion with respiratory arrest
- C) tetrodotoxin and numbness of the skin, salivation, muscle weakness which may lead to respiratory arrest
- D) histamine and rapid decrease in blood pressure

The relative health hazard associated with exposure to radioactive isotopes is greatest for which of the following?

---

- A)  $^{14}\text{C}$
- B)  $^{45}\text{Ca}$
- C)  $^{131}\text{I}$
- D)  $^{90}\text{Sr}$

Which amendment of the Food, Drug and Cosmetic Act added premarket approval and the Delaney amendment to the Act?

---

- A) 1938 amendment
- B) 1958 amendment
- C) 1962 amendment
- D) 1976 amendment

## Inhalation of formaldehyde vapors

---

- 1) has been unequivocally associated with nasal cancer in people
- 2) produces tumors in the nose but not in the lungs in rodents
- 3) induces a greater incidence of nasal tumors in mice than in rats
- 4) causes formation of DNA protein cross links in rodent nasal tissues

Which of the following is/are correct with regard to salicylate poisoning?

---

- 1) oil of wintergreen and aspirin overdose cause similar symptoms
- 2) the poisoned adult is often restless, the poisoned child drowsy
- 3) in therapeutic doses, an adult excreting alkaline urine may put out most of the salicylate metabolically unchanged
- 4) alkaline diuresis can be helpful in salicylate poisoning

Diethylstilbesterol has been used with demonstrated efficacy

---

- 1) to recalcify bone after or during rickets
- 2) as a post-coital (morning after) contraception
- 3) in the treatment of asthma
- 4) as an additive to cattle feed

Toxic effects from oral ingestion of kerosene are described by which of the following?

---

- 1) due to local irritation of the alimentary tract
- 2) not life threatening unless large quantities are ingested or unless it is aspirated
- 3) CNS depression, low grade fever and loss of consciousness with larger doses
- 4) produces bone marrow suppression at low doses



Methoxychlor is described by which of these statements?

---

- 1) it is chemically closely related to DDT
- 2) it has been used as a household and farm insecticide to combat flies and other annoying insects
- 3) it acts by a mechanism that is as poorly defined as that of DDT
- 4) it is much less persistent and less toxic than DDT

Magnesium is characterized by which of the following?

---

- 1) as a form of citrate, oxide, sulfate, hydroxide, or carbonate is widely taken as an, antacid or cathartic
- 2) it is lost daily by excretion into the urine
- 3) the hydroxide is milk of magnesia and is one of the constituents of the universal antidote for poisoning
- 4) its excretion is enhanced by mercurial diuretics

## Recommended Dietary Allowances (RDA) are?

---

- 1) applicable to intentional food additives
- 2) recommended in the USA by the Food and Nutrition Board of the National Academy of Science
- 3) set based on the NOAEL obtained from animal toxicity data and an appropriate safety factor
- 4) set at levels sufficiently high to cover the need of almost all healthy individuals in specified age and sex category with the exception of the recommended intakes of energy

Interstitial emphysema and nonnodular fibrosis of the lung are associated with exposure to

---

- 1) cotton dust
- 2) silica
- 3) mica
- 4) aluminum powder

Acetaminophen (paracetamol), when ingested at high doses, can cause hepatic necrosis. This tissue damage is caused by

---

- 1) oxidation of acetaminophen by a cytochrome P-450-dependent monooxygenase system to an electrophilic metabolite that readily arylates tissue nucleophiles
- 2) oxidation of acetaminophen by NAD with alcohol dehydrogenase and catalysis to the aldehyde, which is hepatotoxic
- 3) oxidation of acetaminophen by a microsomal enzyme system to an electrophilic metabolite that combines with reduced glutathione (GSH) and, when the reduced GSH stores are depleted, initiates hepatotoxicity
- 4) dehydrogenation of the amino group by monoamine oxidase and subsequent dehydration to the aldehyde, which is hepatotoxic

Lithium toxicity is characterized by which of the following?

---

- 1) chronic nephrotoxicity that may occur even when the drug levels are maintained within the therapeutic range
- 2) neuromuscular toxicity that may occur during therapy
- 3) epileptic seizures, slurred speech and blackout spells
- 4) cardiotoxicity

Carbon disulfide exposure causes which of these effects?

---

- 1) peripheral neuropathy
- 2) pulmonary edema
- 3) retinopathy
- 4) proximal tubular damage

Which of the following does NOT induce neurotoxic effects?

---

- 1) black widow spider
- 2) dinoflagellates
- 3) coral snake
- 4) brown recluse spider



Which of the following is/are characteristic of saxitoxin released from dinoflagellates?

---

- 1) gives rise to paralytic shellfish poisoning
- 2) causes mass mortality of marine animals
- 3) produces respiratory and/or gastrointestinal distress in humans
- 4) is associated with red tide

Which of the following statements about asbestosis is true?

---

- 1) chrysotile is the major form of asbestos fibers responsible for asbestosis
- 2) the upper airways are the principal site of development of asbestosis
- 3) chronic effects of asbestos may include fibrosis, pleural calcifications and mesothelioma
- 4) asbestos is both carcinogenic and mutagenic

Which of the following are physical units for radioactivity?

---

- 1) dpm
- 2) Curie
- 3) Becquerel
- 4) rem

Which of the following are true with respect to radiation?

---

- 1) most types of cancer can be increased in frequency by exposure to radiation
- 2) the effective  $1/2$  life of a radioisotope in the body is affected by the physical  $1/2$  life of the isotope and its biological  $1/2$  life
- 3) laboratory animals exposed to single or divided doses of radiation in the sublethal range have demonstrated a shortened life-span
- 4) the health problems in uranium miners are associated with  $^{222}\text{Rn}$  and the attachment of its short-lived daughters to dust particles

Motor sensory polyneuropathy has been observed in industrial workers in Italian shoe and Taipei pressproofing factories after chronic exposure to which of the following?

---

- 1) butane
- 2) n-octane
- 3) turpentine
- 4) n-hexane

The glomerular filtration rate GFR is equal to which of the following?

---

- A) the volume of blood filtered per minute
- B) the volume of plasma filtered per minute
- C) the volume of urine excreted per minute
- D) A divided by C
- E) B divided by C

Which of the following events would most likely lead to alterations in DNA in all cells of the embryo?

---

- A) teratogenic exposure during organogenesis
- B) mutation of parental egg or sperm cells
- C) carcinogenic exposure during the last trimester
- D) embryotoxic exposure two weeks after fertilization
- E) damage to the central nervous system

The most likely outcome of teratogenic exposure during the pre-implantation period is

---

- A) birth defects
- B) transplacental carcinogenesis
- C) spontaneous abortion
- D) intrauterine growth retardation
- E) none of the above



Which of the following types of uteri occurs in of the following the rat?

---

- A) duplex
- B) bicornuate
- C) bipartite
- D) simplex
- E) none of the above

Hyperbaric oxygen utilizes the principle of mass law to convert which of the following blood, pigments to oxyhemoglobin?

---

- A) sulfmethemoglobin
- B) methemoglobin
- C) carboxyhemoglobin
- D) cyanomethemoglobin
- E) none of the above

Which of the following statements regarding carboxyhemoglobin is true?

---

- A) it is bright red in color and is responsible for "cherry red cyanosis" noted in cyanide poisoned patients
- B) it is relatively stable and remains in blood for a long time after death and may be used after death to establish a diagnosis
- C) it is produced by most chemical asphyxiants
- D) all of the above are correct
- E) none of the above are correct

The most frequent type of reproductive failure in humans is which of the following?

---

- A) birth defects
- B) childhood cancer
- C) miscarriage
- D) stillbirth
- E) mental retardation

Kidney damage may alter the formation of hormones involved in

---

- A) erythrocyte formation
- B) phosphate metabolism
- C) calcium metabolism
- D) B and C only
- E) A, B and C

Osmolarity of urine is primarily controlled in the

---

- A) glomerulus
- B) proximal tubule
- C) loop of Henle
- D) distal tubule
- E) collecting duct

Experimentally, renal blood flow can be estimated by measuring which of the following?

---

- A) clearance of inulin
- B) urinary creatinine levels
- C) blood urea nitrogen
- D) urinary glucose levels
- E) clearance of para-aminohippuric acid

The principal site of kidney damage by the analgesic mixture of aspirin and phenacetin is which of the following?

---

- A) proximal tubule
- B) distal tubule
- C) loop of Henle
- D) glomerulus
- E) arterioles



Which of the following is NOT correct?

---

- A) innervation of the pupillary sphincter smooth muscle is by way of the parasympathetic nervous system
- B) the aqueous humor originates from secretion or transudation from the ciliary process and is normally at diastolic blood pressure
- C) Descemet's membrane is a collagenous basement membrane which underlies the corneal epithelium
- D) color sensations are initiated through stimulation of cone cells of the retina, which are less light sensitive than rod cells
- E) accommodation is an active process initiated by contraction of the ciliary muscle

## Consumption of plants containing pyrrolizidine alkaloids

---

- A) results in progressive veno-occlusive disease and centrilobular necrosis of the liver after a single acute dose
- B) cause megalocytosis which is believed to be the result of abnormal cell division in which mitosis ceases but growth continues
- C) probably contributes to the nutritional deficiency disease kwashiorkor due to consumption of traditional herbal medicine commonly called bush tea
- D) normally does not occur with grazing livestock but will occur when pasture grass is in short supply
- E) all of the above

Which one of the following drugs affects fertility by causing a severe drop in spermatogenesis?

---

- A) erythromycin
- B) cyclophosphamide
- C) digoxin
- D) chlorpromazine
- E) lincomycin

The most sensitive diagnostic indicator of chemically induced testicular failure is

---

- A) a decrease in serum testosterone
- B) an increase in serum testosterone
- C) a decrease in serum luteinizing hormone
- D) an increase in serum follicular stimulating hormone
- E) an increase in serum estrogen

Peripheral reticulocytosis is a characteristic feature of

---

- A) pancytopenia
- B) aplastic anemia
- C) thrombocytopenia
- D) increased erythropoiesis
- E) lymphocytopenia

The male sex hormone testosterone is synthesized in and secreted by which of the following?

---

- A) the seminiferous tubules
- B) the leydig interstitial cells
- C) the epididymis
- D) Sertoli cells
- E) the spermatic cord

An acute inhalation toxicity study is conducted by exposing 10 male and 10 female rats for one hour to the test agent. If the airflow is 10 liters/min and the total weight disseminated into the chambers atmosphere is 120 grams, what is the nominal concentration (mg/l) for this exposure?

---

- A) 2
- B) 20
- C) 40
- D) 200
- E) 400

Which of the following have been found to cause immunodeficiency?

---

- A) adrenal glucocorticoids
- B) cyclosporin A
- C) leukemia
- D) viral infections
- E) all of the above



In a traditional subchronic toxicologic evaluation, what would be the primary methodology to ascertain the presence of an immunotoxic response

---

- A) bone marrow progenitor cell assays
- B) histopathology and hematology
- C) a host resistance subgroup
- D) lymphoproliferative responses
- E) qualitative immunoglobulin assays

The phthalate acid ester, di(2-ethylhexyl)phthalate is

---

- A) a hepatic carcinogen in rats and mice
- B) a non-genotoxic carcinogen, (i.e. not known to have a direct effect on DNA)
- C) known to induce peroxisome proliferation which has been postulated to result in the formation of reactive oxygen species
- D) is a testicular toxicant in rodents
- E) all of the above

The altered foci assay in rodent liver utilizes all of the following markers for the histochemical identification of preneoplastic lesions in the liver except

---

- A) gamma-glutamyl transferase activity
- B) glucose-6-phosphatase activity
- C) adenosine triphosphatase activity
- D) resistance to iron accumulation
- E) cytochrome p-450 activity

Thiamine is useful in the treatment of peripheral neuropathies induced by which of the following?

---

- A) arsenic and ethanol
- B) ethanol and carbon dioxide
- C) carbon dioxide and carbon disulfide
- D) ethanol and carbon disulfide

The histological change best associated with the proliferation of smooth endoplasmic reticulum in the liver is which of the following?

---

- A) hepatocellular necrosis
- B) canalicular cholestasis
- C) centrilobular hypertrophy
- D) centrilobular steatosis
- E) toxic hepatitis

Chemicals that cause clastogenesis differ from those that cause aneuploidy because clastogens cause

---

- A) an increase or decrease in the number of intact chromosomes while aneuploidy is a "point" mutation
- B) a "point" mutation while aneuploidy is an increase or decrease in the number of intact chromosomes
- C) a gain, loss, or rearrangement of chromosome pieces while aneuploidy is an increase or decrease in the number of intact chromosomes
- D) a gain, loss, or rearrangement of chromosome pieces while aneuploidy is a "point" mutation
- E) an increase or decrease in the number of intact chromosomes while aneuploidy is a gain, loss, or rearrangement of chromosomes

2,3,7,8-TCDD probably exerts immunosuppressive effects on cell-mediated immunity by which of the following proposed mechanisms?

---

- A) a direct effect on B-cells
- B) a direct effect on T-cells
- C) a direct effect on thymic epithelium
- D) bone marrow depression
- E) all of the above

What is the major fate of the bilirubin excreted into the intestine?

---

- A) conversion to urobilinogen by the flora of the large intestine
- B) reabsorption and recirculation to the liver
- C) conversion to bile salts by. pancreatic enzymes
- D) reabsorption and excretion by the kidney
- E) esterification and reabsorption



Which of the following diseases of the lung is NOT properly paired with its cause?

---

- A) farmer's lung caused by spores of thermophilic actinomyces
- B) bagassosis caused by dust from dried sugar cane
- C) byssinosis caused by cotton flax or hemp dust
- D) asthma-like symptoms caused by toluene diisocyanate
- E) cheese washer's lung caused by exposure to toxin eaten by cattle and concentrated in their milk

Studies of potential mechanisms of chemically induced mouse liver carcinogenesis currently include which of the following approaches?

---

- A) determination of the mutational spectrum of activated Ha-ras oncogenes
- B) determination of the methylation state of Ha-ras oncogenes
- C) evaluation of cell proliferation
- D) A and C only
- E) A, B and C

Tumor suppressor genes show a high degree of evolutionary conservation and were first identified as the transforming genes of retroviruses.

---

A) true

B) false

Which of the following is/are true regarding porphyria cutanea tarda?

---

- 1) characterized by blistering and photosensitivity
- 2) characterized by excessive hirsutism
- 3) caused by hexachlorobenzene
- 4) an example of photoallergy

Which of the following compounds produce(s) cholestasis

---

- 1) amitriptylene
- 2) haloperidol
- 3) chlorpromazine
- 4) aflatoxin

Which of the following properties of an acid is/are relevant to the degree of damage it could produce to the cornea?

---

- 1) solution pH
- 2) heat of hydration
- 3) capacity of associated anion to combine with protein
- 4) specific gravity

Which of the following statements is/are true regarding surfactants and/or detergents and eye injury?

---

- 1) show a reasonable correlation between surface tension lowering capacity and damage produced
- 2) cationic agents are more damaging than anionic ones
- 3) are very likely to cause permanent opacity of the cornea
- 4) nonionic agents tend to be safer than either anionic or cation ones

Which of the following process(es) are important in the pulmonary deposition of insoluble aerosols?

---

- 1) diffusion
- 2) impaction
- 3) sedimentation
- 4) interception



Which of the following industrial chemicals are high risk hazards for type IV allergic reactions (cell-mediated, delayed hypersensitivity) ?

---

- 1) trichlorethylene
- 2) isocyanates
- 3) vinyl acetate
- 4) mercaptobenzothiazole

The use of enzymes from *Bacillus subtilis* to enhance the cleaning power of detergents was associated with occupational asthma in workers. Allergic characteristics of this would include?

---

- 1) positive bronchial provocation tests
- 2) specific IgE in serum
- 3) positive passive transfer tests
- 4) positive patch tests for delayed cutaneous hypersensitivity

Which of the following mechanisms of DNA repair has/have been shown in mammalian cells?

---

- 1) excision repair
- 2) sos repair
- 3) post replication repair
- 4) PCR repair

Consequences of chronic ethanol abuse may include

---

- 1) depression of the central nervous system
- 2) induction of hyperexcitability upon withdrawal
- 3) altered microsomal drug-metabolism
- 4) nervous disorders through vitamin deficiency

Cornification of the vaginal epithelium in the rat is associated with?

---

- 1) the proestrus phase of the reproductive cycle
- 2) estrus
- 3) administration of prolactin
- 4) administration of diethylstilbestrol

T lymphocytes are directly involved in which of the following?

---

- 1) antigen recognition
- 2) delayed hypersensitivity
- 3) cytotoxicity
- 4) production of antibodies

Chloracne in humans has been caused by exposure to

---

- 1) polychlorinated biphenyls
- 2) polybrominated biphenyls
- 3) tetrachlorodibenzo-p-dioxin
- 4) tetrachloroazoxybenzene

Which of the following agent(s) is correctly matched with its primary site of action?

---

- 1) inorganic mercury --- collecting duct
- 2) gentamycin --- proximal tubule
- 3) uranium --- ascending limb of the loop of Henle
- 4) cephaloridine --- proximal tubule



Which of the following are likely to increase the frequency or efficiency of the initiation stage of carcinogenesis?

---

- 1) cell replication
- 2) increased unscheduled DNA synthesis
- 3) induction of metabolic activating enzymes
- 4) increased conjugation of chemical carcinogens

An agent(s) whose principal mode of action is selective damage of the myelin forming cells in the central and peripheral nervous system is which of the following?

---

- 1) triethyl tin
- 2) hexachlorophene
- 3) isonicotinic acid hydrazide
- 4) lead

Ethylenethiourea, a synthetic chemical used in the rubber industry, produces thyroid neoplasms by which of the following mechanisms?

---

- 1) its direct genotoxic interaction
- 2) acting as a procarcinogen
- 3) behaving as a promoter
- 4) acting as a hormone modifier

Which of the following are true of the dominant lethal assay?

---

- 1) both the male and female are exposed to the test chemical prior to mating
- 2) exposure of the female to the test chemical continues throughout gestation
- 3) dominant lethals are calculated as the ratio of dead-born fetuses to total fetuses
- 4) test designed to detect mutagenic effects in the spermatogenic process that are lethal to the embryo or fetus

Which of the following statement(s) is/are true about in vitro chromosomal aberration assays?

---

- 1) it is not necessary to demonstrate cytotoxicity in these assays
- 2) Chinese hamster ovary (CHO) cells or human peripheral blood lymphocytes are routinely used
- 3) metabolic activation is limited to 3-4 hours because of numerous cell cycles and rapid multiplication
- 4) rodent liver-derived S-9 metabolic activation is routinely utilized for metabolism of xenobiotics in CHO and human lymphocyte systems

The micronucleus assay is accurately described by which of the following?

---

- 1) is typically performed on peripheral blood smears of mice
- 2) does not require a selective growth medium
- 3) is not a useful assay for peripheral blood of rats
- 4) counts chromosomes or chromosome fragments that escape incorporation into the nucleus in dividing cells

Which of the following are generally accepted concepts concerning the relationship between mitogenesis or cell proliferation and neoplasia?

---

- 1) necessary for "fixation" of a mutation
- 2) sufficient for tumor induction
- 3) necessary for clonal expansion
- 4) not thought to influence the extent of DNA repair

A volatile chemical was shown to have a first order elimination half life from the blood compartment of 35 minutes. Assuming zero order (constant rate) absorption, how long will it take to reach approximately 90% of the steady state blood level concentration upon exposure to this chemical via inhalation?

---

- A) 0.25 hours
- B) 0.50 hours
- C) 1.0 hours
- D) 2.0 hours
- E) 6.0 hours



In safety evaluation, the general philosophy is to assume that man

---

- A) is less sensitive than the most resistant species
- B) will react to the agent in a manner similar to all test species
- C) is at least as sensitive as the most sensitive species
- D) is less sensitive than species which metabolize the agent like man
- E) none of the above

Which of the following accurately describes hydrogen sulfide?

---

- A) can bind to methemoglobin
- B) inhibits cytochrome P-450 reductase
- C) requires thiosulfate as an antidote
- D) is much less toxic than hydrogen cyanide
- E) all are correct

Which of the following is correct in determining the relative potency of two substances?

---

- A) substances acting by the same mechanisms usually display parallel dose response curves
- B) isobolograms permit one to plot two-dimensional data on a three dimensional graph
- C) synergism occurs if degree of potentiation = 1.0
- D) antagonism occurs if degree of potentiation is more than 1.0
- E) none of the above

## Drug-trapping resins like cholestyramine

---

- A) are rarely given orally due to their high hepatotoxic potential
- B) will effectively reduce concentrations of any drug from the bloodstream if the drug was injected intraperitoneally
- C) enhance excretion of toxicants trapped in the enterohepatic circulation
- D) B and C are correct
- E) all are correct

In testing the toxic effects of 4 different agents on mice, litter mates are randomized to provide blocks of four animals each. The purpose of using such an arrangement in the study is

---

- A) to avoid randomization effects
- B) to enlarge the size of the effects observed
- C) to prevent interaction among the animals in each block
- D) to reduce the residual variation by eliminating the variation between blocks
- E) to reduce the variation between blocks by eliminating variation within each of the blocks

Glutathione S-transferases are involved in the bioactivation of which of the following?

---

- A) aflatoxin B1
- B) ethylene dibromide
- C) n-hexane
- D) ethacrynic acid
- E) none of the above

Toxic effect occurs via asphyxiation from direct depression of medullary respiratory center by the chemical(s)

---

- A) carbon monoxide
- B) cyanide
- C) nitrogen
- D) none of the above

Which of the following is used in medicine in the form of the sulfate for x-ray visualization, but its soluble salts such as the chloride are extremely toxic?

---

- A) calcium
- B) magnesium
- C) barium
- D) strontium
- E) potassium



Indicate the one incorrect statement about DDT

---

- A) It is an extremely nonpolar substance
- B) It is stable against sunlight and can maintain its insecticidal effect on dry surfaces for up to 18 months
- C) It is more toxic to fish than mammals
- D) It has been banned by the U.S. EPA
- E) It uncouples oxidative phosphorylation in insects

A known example of drug action which is dependent upon individual heredity is

---

- A) antibacterial effects of sulfadiazine -- dependent on rates of acetylation of the drug
- B) analgesic effects of meperidine -- dependent on rates of hydrolysis of the drug by liver and plasma enzymes
- C) anticonvulsant effects of phenobarbital -- dependent on rates of penetration of the drug into the CNS
- D) neuromuscular blocking effects of succinylcholine -- dependent on rate of hydrolysis by plasma esterases
- E) antihistaminic effects of chloroquine -- dependent on rates of absorption from the GI tract

Which of the following is NOT a component of the mixed function monooxygenase system

---

- A) NADPH
- B) cytochrome P-450 reductase
- C) cytochrome oxidase
- D) cytochrome P-450
- E) phospholipids

Which of the following is NOT a property of the mixed function monooxygenase system?

---

- A) It is involved in drug oxidation by a hydroxylation mechanism
- B) It is inducible by isoniazid
- C) It is inhibited by carbon monoxide
- D) It requires NADPH
- E) It is located in hepatic microsomes

The acute biological response to intravenously administered inert particles of 10-micron diameter or less is

---

- A) solid state carcinogenesis
- B) clearance by the reticuloendothelial system
- C) granuloma
- D) pulmonary shock
- E) cerebral stroke

The toxicity of a drug that is a weak acid such as phenobarbital can be decreased by?

---

- A) acidification of the urine
- B) alkalization of the urine
- C) administration of sodium bicarbonate
- D) B and C
- E) A and C

An example of the ability of genetic factors to influence the susceptibility to toxic agents is

---

- A) genetic variation of glucuronyl transferase levels in cats
- B) differences in the ability of dogs to acetylate arylamines
- C) the sulfonamides' displacement of bilirubin from its plasma-binding sites in premature infants
- D) hemolytic effects of primaquine in individuals with glucose-6-phosphate dehydrogenase deficiency
- E) nephrotoxic effects of chloroform in mice

Which of the following is NOT required by the Good Laboratory Practice regulations for non-clinical laboratory studies?

---

- A) that the protocol for a study be signed by the study director
- B) that there shall be at least 20 animals/sex/ group in a chronic oral toxicity study
- C) that there shall be an archive for the storage of all raw data, specimens, reports, protocols, and documentation
- D) that there shall be written standard operating procedures for methods used in conducting nonclinical laboratory studies
- E) that equipment shall be adequately inspected, cleaned and maintained



Individuals having an alpha-1-antitrypsin deficiency should especially avoid which of the following?

---

- A) smoking
- B) red wine and cheese
- C) liver and onions
- D) acetylated salicylates
- E) none of the above

Following intravenous dosing, the rate of elimination (mg/hr) of toxicant x from plasma is constant over a period of 24 hours. One could conclude that

---

- A) the plasma half-life of compound x is 12 hours
- B) the toxicant is eliminated by zero-order kinetics
- C) the first-order rate constant of elimination of the toxicant is dose-independent
- D) the toxicant will probably not accumulate on multiple dosing
- E) all of the above

The potentiation of malathion toxicity by TOCP and EPN is mediated by

---

- A) an alteration of malathion protein binding
- B) chelation of the malathion molecule
- C) inhibition of liver aliesterase
- D) induction of cytochrome P-450
- E) none of the above

Various models used in risk extrapolation can predict different potential risks at low doses. Choose the correct rank order (from highest to lowest estimates) of potential risk at low-doses for these models.

---

- A) one-hit > linearized multistage > multi-hit > probit
- B) probit > one-hit > linearized multistage > multi-hit
- C) one-hit > probit > linearized multistage > multi-hit
- D) multi-hit > probit > one-hit > linearized multistage

Nuisance dusts can be described by all of the following statements except

---

- A) they have a long history of little adverse effect on the lungs of exposed workers
- B) they do not produce any significant organic disease or toxic effect when exposures are kept under reasonable control
- C) they are biologically inert dusts which do not provoke any cellular response in the lung when inhaled in any amount
- D) they can produce minimal lung tissue reactions (which are reversible) at high exposure concentrations

Which statement is NOT true regarding regulation of chemicals by the Food and Drug Administration (FDA)?

---

- A) the Food, Drug and Cosmetic Act regulates medical devices as well as foods, drugs and cosmetics
- B) since the enactment of the 1958 Food Additives Amendment, all food ingredients must be demonstrated to be reasonably safe prior to marketing
- C) carcinogenic migrants of packaging materials whose extrapolated risk does not exceed 1 in 1,000,000 are considered "de minimus" and ignored by the FDA
- D) the Food, Drug and Cosmetic Act requires that any marketed drug be proved safe and effective for its labeled indications
- E) the Delaney clause (part of the 1958 Food Additives Amendment) precludes approval by FDA if a food additive induces cancer

Randomizations, stratifications and blinding are used to

---

- A) adjust for confounding factors such as age, smoking and sex
- B) produce a population of sufficient size for statistical analysis
- C) provide a uniform control population
- D) minimize artifactual distortion in selecting a sample population
- E) all of the above

The study mode that defines a group in terms of both disease (dependent) and characteristic (independent) variables is which of the following?

---

- A) cohort
- B) case-control.
- C) cross-sectional
- D) retrospective
- E) cross-over



In a normally distributed population of a quantal dose-response curve, the percent response for a probit value of 7 would be approximately

---

- A) 50
- B) 98
- C) 1
- D) 84
- E) 16

A patient's arterial pH is 7.55 and his arterial pCO<sub>2</sub> is 48 Torr. These findings are consistent with a diagnosis of

---

- A) voluntary hyperventilation
- B) acclimatization to high altitude
- C) asphyxia
- D) ingestion of excessive sodium bicarbonate
- E) severe renal insufficiency (mercury intoxication)

A case-control study of mesothelioma in workers exposed to asbestos is conducted. The "relative risk" figure obtained from this represents

---

- A) an odds ratio of occurrence of asbestos exposure and controls
- B) the incidence of mesothelioma in asbestos-exposed workers compared to the incidence in unexposed workers
- C) the prevalence of mesothelioma in asbestos-exposed workers compared to the prevalence in unexposed workers
- D) an estimate of the disease risk in the general population
- E) none of the above

Which of the following statements is most correct?

---

- A) EPA has no regulatory authority over the production and use of biotechnology-derived products intended for use in animals and/or man
- B) USDA has sole regulatory responsibility over the development, production and use of recombinant DNA-derived animal vaccines
- C) FDA and EPA both have regulatory authority/responsibility over the production and use of recombinant DNA-derived products for human therapeutic use, the former for judging the efficacy/safety to the patient, the latter for possible environmental impact
- D) EPA has sole regulatory authority/responsibility for control of recombinant DNA-derived products intended for use in agriculture crop production

The renal clearance of a toxicant is assessed by determining

---

- A) the rate of elimination in urine/plasma concentration
- B) the urine concentration/blood concentration
- C) the glomerular filtration/blood concentration
- D) none of the above

Persistence in soils for different classes of chemicals is likely to be in the following order. Select from the most to least persistent: 1) chlorinated hydrocarbon insecticides, 2) triazine herbicides, 3) amide herbicides, 4) organophosphate insecticides

---

- A) 1,3,4,2
- B) 1,2,4,3
- C) 1,2,3,4
- D) 4,1,2,3

Acute exposure to high concentrations of the photochemical air pollutant, nitrogen dioxide

---

A) follows Haber's Law

B) results in irritation only in the upper respiratory tract ( cough, sore throat, vertigo, dyspnea )

C) causes immediate pulmonary edema

D) rapidly progresses from pulmonary edema to bronchiolitis obliterans

E) produces a triphasic response pattern (initial mild irritation, latency period, severe effects)

Henry's Law describes which of the following properties of a chemical?

---

- A) soil sorption coefficient
- B) bioconcentration factor in microorganisms
- C) air-water distribution
- D) photodegradation
- E) lipophilic character



If the compound 1,2-bis(2-chloroethyl)-1-nitrosourea (BCNU) is added to hepatocytes which are being subjected to an oxidative stress, what will likely happen to the GSH levels of these cells, compared to an identical hepatocyte incubation without BCNU?

---

- A) GSH levels will be zero in cells with BCNU because BCNU catalytically destroys GSH
- B) GSH levels will be similar in cells with and without BCNU
- C) GSH levels will be higher in cells with added BCNU
- D) GSH levels will be lower in cells with added BCNU

The following statement(s) represent(s) the true principle(s) behind the treatment of cyanide poisoning with intravenous administrations of sodium nitrite and sodium thiosulfate

---

- 1) the affinity of methemoglobin for cyanide exceeds that of the cytochrome
- 2) sodium nitrite converts a fraction of circulating hemoglobin to methemoglobin
- 3) thiosulfate serves as a substrate for the enzyme rhodanese, which mediates the conversion of cyanide to thiocyanate
- 4) cyanomethemoglobin is an easily dissociable complex

According to Good Laboratory Practices regulations, pathologists involved in animal toxicology studies should

---

- 1) read slides in a "blind" manner
- 2) have all the in-life data available when reading slides
- 3) have the slides presented in a randomized manner
- 4) may or may not have the in-life data available depending on the discretion of the study director

The distribution of an agent from the blood to the CNS differs from that for most other tissues because in the CNS

---

- 1) there are tight junctions between capillary and endothelial cells
- 2) strongly ionized agents have great difficulty penetrating the blood-brain barrier
- 3) there is a relative absence of pinocytotic vesicles
- 4) aqueous bulk flow is severely restricted

Major concerns with respect to high levels of inorganic nitrates in the environment include which of the following?

---

- 1) high water solubility and mobility in water supplies
- 2) the formation of nitrites by microorganisms
- 3) the formation of nitrosamines
- 4) formation of acid rain

Which of the following agents have been useful in studying mechanisms associated with oxygen radical induced injury either in vivo or in vitro?

---

- 1) superoxide dismutase
- 2) hydroxyl radical scavengers
- 3) metal chelators
- 4) antioxidants

Overnight food deprivation is commonly used in toxicology studies with rats. This can lead to

---

- 1) reduction in cofactors necessary for conjugation reactions
- 2) stimulation of a number of biotransformation enzymes
- 3) potentiation of the toxicity of such chemicals such acetaminophen and bromobenzene
- 4) nothing, because rats eat during the day

The concept of risk assessment includes

---

- 1) determination of potential adverse health effects
- 2) characterization of uncertainties inherent in the process of inferring risk
- 3) extrapolation from animal data to human exposure
- 4) selecting most appropriate regulatory action



A reference dose (RfD) is most appropriately characterized by which of the following?

---

- 1) it represents the NOAEL divided by an uncertainty factor times a modifying factor
- 2) it is an analogous concept to the Acceptable Daily Intake (ADI)
- 3) it is expressed in mg/kg/day
- 4) it is the dose of a compound producing the same toxicity as a standard, or reference, chemical

## During chronic exposure

---

- 1) the body burden will approach a constant maximum value as long as the excretion mechanisms are not saturated
- 2) the amount accumulated in a particular tissue should be inversely proportional to the rate constant for transfer out of that tissue
- 3) induction of metabolizing enzymes, alterations in plasma flow rates, or damage to excretory organs may cause the kinetics of an agent to be different than predicted from a single dose
- 4) the body burden of accumulative substances will increase without limit

Which of the following describe ambient air quality standards (AQS's) but not threshold limit values (TLV's)?

---

- 1) establish air pollution limits that are designed to protect man and his environment
- 2) are based on an exposure of 40 hours per week
- 3) are designed to protect the general population
- 4) are established to protect the workers

## The Toxic Substances Control Act of 1976

---

- 1) authorizes the EPA administrator to restrict or ban a chemical if it is deemed to present an unreasonable risk to human health or the environment
- 2) calls for an annual review of testing standards
- 3) mentions polychlorinated biphenyls by name and calls for rules to require warning labels for PCB's
- 4) requires health and safety studies on all chemicals in commerce in the U.S. except where such studies would expose proprietary information

Knowing that a drug is a base,  $pK_a = 10$ , and that its nonionized form has a low lipid solubility could enable one to predict that the drug would

---

- 1) not readily cross the placental barrier
- 2) have a small volume of distribution
- 3) penetrate the CNS poorly
- 4) have a long biological half-life ( $t_{1/2}$ )

In the rat, which of the following apply to methanol metabolism?

---

- 1) the kinetics of metabolism and elimination of large doses include the accumulation of damaging amounts of formic acid in the eye
- 2) rapid metabolism to CO<sub>2</sub>
- 3) metabolic alkalosis
- 4) a major route of metabolism involves catalase

The site(s) of action of the insecticide rotenone include which of the following?

---

- 1) the enzyme aconitase in the TCA cycle
- 2) mitochondrial inorganic orthophosphate-adenosine
- 3) acetylcholinesterase at the neuromuscular juncture
- 4) the mitochondrial respiratory chain between NADH and flavin

An analysis of variance was performed on a set of data consisting of 4 groups. The "f" value was found to be greater than suitable tabulated values, hence one or more of the following tests is appropriate

---

- 1) Dunnett's t-test
- 2) Scheffe's multiple comparison test .
- 3) Duncan's multiple range test
- 4) Cochran t-test



Generation of free radicals is thought to be the major mechanism of cell injury in which of the following?

---

- 1) oxygen toxicity
- 2) radiation injury
- 3) carbon tetrachloride poisoning
- 4) ischemia/reperfusion injury

A worker uses an arc-welder on stainless steel under conditions of low ventilation. Hours later the worker complains of chest pains. He is most likely suffering from

---

- A) ozone toxicity with the onset of pulmonary edema
- B) nitrogen dioxide toxicity with incipient lung infection
- C) phosgene toxicity with airway irritation
- D) angina from overexertion and heat stress

Which statement(s) is/are correct?

---

- A) cysteamine or Mucomyst can decrease acetaminophen induced hepatotoxicity
- B) acetaminophen can be directly acted upon by sulfotransferase or UDP-glucuronyl transferase
- C) acetaminophen is activated by cytochrome P450 before binding with glutathione
- D) A and C are correct
- E) A, B and C are correct

Which of the following is/are true concerning phthalate esters?

---

- A) can migrate from plastics into the contents of plastic containers
- B) have a very low LD50
- C) have been found in adipose tissues of humans given blood transfusions
- D) A and C are correct
- E) A, B and C are correct

Desferrioxamine is

---

- A) used in the treatment of poisoning with arsenicals
- B) useful in treatment of hemosiderosis
- C) capable of removing deposits of tetracycline from bone
- D) similar in structure to calcium disodium edetate (EDTA)
- E) effective in the treatment of hepato-lenticular degeneration (Wilson's disease)

Levodopa therapy is useful in the treatment of which of the following?

---

- A) the CNS toxicity of manganese
- B) the CNS toxicity of lead
- C) hypertension
- D) the peripheral neuropathy of arsenic
- E) the CNS toxicity of aluminum effects

Which of the following is true of methanol poisoning in humans?

---

- A) the CNS depressant effects are greater than those observed with the ingestion of an equal amount of ethyl alcohol
- B) a metabolic alkalosis frequently occurs
- C) one treatment is to reduce methanol to formaldehyde using agents such as methylene blue
- D) the toxicity is in part due to formation of metabolites
- E) gastric lavage is ineffective

Which of the statements or phrases is NOT true?

---

- A) red squill - the toxic and pesticidal action against rodents is associated with its cardiac effects
- B) warfarin - a hydroxycoumarin compound which reduces the prothrombin levels in blood
- C) parathion - anticholinesterase insecticide
- D) dieldrin - a chlorinated hydrocarbon insecticide, more toxic than DDT
- E) 2-PAM (pralidoxime) - useful as an antidote for carbaryl (Sevin)



Which of the following therapies is the best choice for treating strychnine poisoning?

---

- A) copper sulfate
- B) dimercaprol
- C) amobarbital
- D) atropine
- E) glycerol monacetin

A 5-week old baby is brought to the emergency room in coma. The baby is blue and cyanotic, respirations deep and labored and heart rate 100. These signs are most likely to be due to poisoning with which of the following chemicals?

---

- A) sodium cyanide
- B) nitrite
- C) carbon monoxide
- D) ferrous sulfate
- E) any of the above

Ingestion of which of the following produces nausea, painful colic and watery diarrhea followed by a symptom-free period of three to five days after which hepatorenal toxicity becomes evident?

---

- A) water hemlock (*Cicuta*)
- B) jimsonweed (*Datura stramonium*)
- C) oleander (*Nerium oleander*)
- D) *Amanita phalloides*
- E) philodendron

The fever which results from excessive absorption of the herbicide dinitroorthocresol is caused by

---

- A) uncoupling of cellular oxidative phosphorylation
- B) suppression of sweating
- C) cutaneous vasoconstriction
- D) disturbance of hypothalamic thermoregulatory centers
- E) all of the above

Chronic inhalation of asbestos particles by humans produces

---

- A) interstitial fibrosis of the lungs
- B) chronic bronchitis
- C) mesenchymal lesions of the pleura
- D) nasal polyps
- E) A and C are correct

Which of the following is/are true about selenium?

---

- A) dietary deficiencies in young calves result in a fatal disease known as white muscle disease
- B) controls lipid peroxidation in conjunction with vitamin E
- C) epidemiological data have indicated a decrease in human cancer death rate (age and sex adjusted) that correlates with selenium content of crops
- D) there is evidence indicating that selenium has an antineoplastic effect in laboratory animal
- E) all of the above

A modern rodenticide with selective toxicity that causes irreversible vasoconstriction in the rat is

---

- A) warfarin
- B) pyriminil
- C) alpha naphthyl thiourea
- D) norbormide

Which of the following pesticides, obtained from plants, has been used against lice in cattle, ear mites in cats and dogs, and to kill fish?

---

- A) nicotine
- B) pyrethrin
- C) rotenone
- D) toxaphene



The 1954 Miller Amendment to the Food, Drug and Cosmetic Act of 1938 provided that

---

- A) teratogenicity testing be carried out in at least two species, with one being nonrodent
- B) the Pharmacopeia of the United States and the National Formulary be designated as the compendia of official standards
- C) sponsors submit a new drug application (NDA) before starting clinical trials in humans
- D) the Food and Drug Administration establish pesticide residue tolerances
- E) sponsors must wait 30 days to begin clinical trials in humans after receipt by the Food and Drug Administration of an investigational new drug application (IND)

## Batrachotoxin

---

- A) is a toxin of a coelenterate
- B) produces massive hemolysis
- C) is a protein
- D) has an effect on neuronal conduction
- E) none of the above

Which of the following herbicides alters thyroid function and increases the incidence of thyroid tumors in laboratory animals?

---

- A) 2,4-D
- B) amitrole
- C) paraquat
- D) pentachlorophenol
- E) dinitro-orthocresol (DNOC)

Lead-induced porphyria is due to

---

- A) inhibition of ferrochelatase activity in liver
- B) stimulation of delta-aminolevulinic acid dehydratase activity in liver
- C) induction of uroporphyrinogen decarboxylase activity in liver
- D) stimulation of delta-aminolevulinic acid synthetase and consequent destruction of heme
- E) all of the above

Mushrooms have been shown to cause all of the following EXCEPT

---

- A) gastroenteric symptoms
- B) sweating
- C) a digitalis-like effect on the cardiovascular system
- D) inebriation or hallucinations without drowsiness or sleep
- E) irreversible hepatorenal toxicity

Almost all fatalities caused by mushroom poisoning are due to the *Amanita phalloides*. Its mechanism of action is

---

- A) inhibition of acetylcholinesterase
- B) binding to RNA polymerase
- C) competitive muscarinic antagonism
- D) inhibition of choline acetyltransferase
- E) preventing release of acetylcholine from nerve terminals

Inhibition of the citric acid cycle is thought to be the mechanism by which this compound produces its toxic effect

---

- A) alpha naphthyl thiourea (ANTU)
- B) strychnine
- C) kepone
- D) dinitroorthocresol (DNOC)
- E) fluoroacetate

What is the most common adverse effect of pyrethrins in humans?

---

- A) excitation
- B) diarrhea
- C) muscular fibrillation
- D) contact dermatitis
- E) anaphylaxis



The most commonly used species in screening for delayed neurotoxic effects of organophosphates is

---

- A) dog
- B) chicken
- C) cat
- D) mouse
- E) pig

Which of the following is NOT true for amphetamine toxicity?

---

- A) diazepam is an effective treatment for CNS symptoms
- B) excretion can be increased significantly by alkalinization of the urine
- C) chronic ingestion of amphetamine can produce a psychotic state that resembles schizophrenia
- D) amphetamine stimulates the central nervous system
- E) cardiovascular effects may include headache, angina and palpitations

Exposure to  $^{125}\text{I}$ -iodine or  $^{131}\text{I}$ -iodine is most toxic to which of the following tissues?

---

- A) bone
- B) fat
- C) lung
- D) thyroid
- E) heart

In children with lead encephalopathy a proximal tubule reabsorption defect is often found. This defect is characterized by which of the following?

---

- A) aminoaciduria, phosphaturia and glucosuria
- B) phosphaturia, proteinuria and excess potassium loss
- C) glucosuria and renal stone formation
- D) glucosuria, calciuria and excess potassium loss
- E) proteinuria and calciuria

Tricothecene toxins such as T-2 are produced by a species of

---

- A) the fungi *Fusarium*
- B) the bacteria *Clostridium*
- C) the virus *Tricothormus*
- D) the dinoflagellate *Tricornus*

Which of the following is true regarding the poisonous toxin in the castor bean (*Ricinus communis*)?

---

- A) the active principal is a lectin
- B) it consists of 2 polypeptide chains connected by a disulfide bridge
- C) one chain is essential for binding to an intestinal cell's membrane, and this binding leads to the entry of the other chain into the cell
- D) a potent toxin to intestinal cells
- E) all of the above

Which of the following statements regarding inhalation toxicology is/are true?

---

- 1) dusts defined as homogeneous and polydispersed are of one chemical substance of non-uniform size
- 2) gases are chemicals that tend to be in the gaseous state under normal conditions of temperature and pressure
- 3) vapors are chemicals that tend to coexist in equilibrium between liquid or solid state and the gaseous state
- 4) dusts defined as heterogeneous and polydispersed are of one chemical substance of non-uniform size

## Workers exposed to toluene diisocyanate (TDI) experience

---

- 1) blurred vision
- 2) bronchoconstriction
- 3) oliguria
- 4) dermatitis



The treatment of a patient suffering from an overdose of aspirin may include which of the following?

---

- 1) gastric lavage or induction of emesis
- 2) administration of a respiratory stimulant
- 3) maintenance of plasma volume and electrolyte balance
- 4) administration of  $\text{NH}_4\text{Cl}$  to decrease urinary pH

Which of the following is true in the treatment of organophosphate insecticide poisoning?

---

- 1) caution must be exercised in the use of atropine since victims of OP poisoning are more sensitive to atropine than normal individuals
- 2) a major therapeutic aim is to reactivate acetylcholine esterase with pralidoxime (2-PAM)
- 3) atropine effectively antagonizes the peripheral neuromuscular paralysis
- 4) atropine is used to block the action of excess acetylcholine

## Succinylcholine

---

- 1) is rapidly hydrolyzed by plasma pseudocholinesterase
- 2) is ineffective in patients with low dibucaine numbers
- 3) may- cause prolonged apnea due to an atypical plasma pseudocholinesterase
- 4) should be used with extreme caution in patients of Mediterranean origin

Kernicterus may be produced in neonates by

---

- 1) phototherapy
- 2) phenobarbital
- 3) protamine sulfate
- 4) sulfonamides

## The toxic effects of acute exposure to carbaryl (Sevin)

---

- 1) usually include flaccid paralysis
- 2) are related to excessive accumulation of acetylcholine at cholinergic synapses
- 3) could be blocked by physostigmine
- 4) can be accompanied by marked pupillary constriction

Which of the following is/are true concerning arsenic?

---

- 1) it has a high affinity for sulfhydryl enzymes
- 2) it causes pyruvate accumulation in blood
- 3) it binds to lipoic acid
- 4) it inhibits pseudocholinesterase

Which of the following has/have tested positive for mutagenic activity in the Ames test?

---

- 1) coffee extracts
- 2) safrole
- 3) tea extracts
- 4) broiled meat extracts

Scombroid poisoning has occurred after the ingestion of

---

- 1) mackeral
- 2) mahi mahi
- 3) sardines
- 4) rainbow trout



Tetracyclines can cause which of the following as adverse effects?

---

- 1) gastrointestinal irritation
- 2) hepatotoxicity
- 3) photosensitivity
- 4) discoloration of dental enamel of offspring following maternal treatment after day 120 of gestation (human)

BAL (dimercaprol) is the preferred chelating agent for which of the following?

---

- 1) iron
- 2) mercury
- 3) selenium
- 4) arsenic

Which of the following statements is/are true?

---

- 1) the roentgen (Coulomb per Kg of air) is a unit of radiation exposure
- 2) the Gray (rad) is a measurement of energy deposition in any medium by any type of ionizing radiation
- 3) the Sievert (rem) takes into consideration such modifying factors of dose as the quality of the radiation
- 4) the quality factor for various types of ionizing radiation is based on the linear energy transfer of the type of radiation

Which of the following is/are true of radon and radon daughters?

---

- 1) levels may be increased indoors in situations where the ventilation rate is poor
- 2) inhalation of radon daughters attached to dust particles results in exposure of the lungs to alpha radiation
- 3) they are associated with an increased incidence of small cell carcinoma in the lungs of uranium miners
- 4) epidemiology studies have conclusively shown that exposure to radon daughters in the indoor environment results in an increased incidence of lung tumors

## Nitrogen dioxide

---

- 1) is associated with pulmonary damage known as silo-filler's disease
- 2) is a deep lung irritant producing pulmonary edema
- 3) exposure may increase susceptibility to bacterial infections
- 4) induces tracheobronchial cancer after long term inhalation exposure to rats and mice

The major sites of toxic manifestations of gold include

---

- 1) the skin and mucous membranes
- 2) the central nervous system
- 3) the kidneys
- 4) the reproductive system

Which of the following groups of drugs is/are, correctly paired with the predominant toxic side effect?

---

- 1) beta adrenergic blocking agents (e.g., propranolol) - hypotension and bradycardia
- 2) classical antihistamines (e.g., diphenhydramine) - sedation
- 3) aminoglycoside antibiotics (e.g., gentamicin) . - ototoxicity and renal toxicity
- 4) nonsteroidal anti-inflammatory drugs (e.g., ibuprofen) - gastrointestinal irritation

## The Delaney Amendment to the Federal Food, Drug and Cosmetic Act

---

- 1) specifies that no additive shall be deemed to be safe if it is found, after tests which are appropriate for the evaluation of the safety of food additives, to induce cancer in animals or man
- 2) applies to unintentional food additives (e.g. pesticides) found in raw (unprocessed) agricultural commodities
- 3) applies to color additives in processed food
- 4) has a de minimus clause that allows trace amounts of materials which pose at most a negligible risk of cancer when used as a food additive



The current radiobiologic principle commonly accepted for the carcinogenic potential of ionizing radiation is

---

- 1) linear cancer induction
- 2) threshold cancer induction
- 3) non-threshold cancer induction
- 4) the Compton effect

Incidents of solvent intoxication are common because

---

- 1) billions of pounds are used annually as solubilizers, dispersants or diluents
- 2) occupational and home/recreational exposures occur frequently
- 3) users often fail to take adequate precautions
- 4) these chemicals are extremely toxic

The toxicity of methyl-n-butyl ketone (MBK) in humans is

---

- 1) expressed as a peripheral neuropathy
- 2) less than that of methyl ethyl ketone (MEK)
- 3) related to metabolites
- 4) has not been produced in experimental animals

Which of the following chemicals are excreted most readily in the urine?

---

- A) readily filtered, highly water-soluble, not actively secreted
- B) readily filtered, highly water-soluble, actively secreted
- C) readily filtered, highly lipid-soluble, not actively secreted
- D) readily filtered, highly lipid-soluble, actively secreted
- E) not readily filtered, highly water-soluble, not actively secreted

Damage to proximal convoluted tubules in the nephrons following exposure to inorganic mercury or uranium might be expected to be reflected in which of the following?

---

- A) aminoaciduria
- B) elevated aspartate aminotransferase (AST)
- C) decreased glomerular filtration rate (GFR)
- D) proteinuria
- E) elevated blood glucose

Dermal delayed hypersensitivity reactions are mediated by

---

- A) mast cells
- B) platelets
- C) T-lymphocytes
- D) B-lymphocytes
- E) type I pneumocytes

Which of the following is NOT used to monitor alterations in renal function?

---

- A) clearance of phenolsulfonphthalein
- B) urine osmolality
- C) urine volume and pH
- D) clearance of sulfobromophthalein
- E) excretion of sodium and potassium

Itai-Itai, also known as "ouch-ouch" disease, was diagnosed in Japan shortly after World War II. Those affected complained of severe rheumatic and myalgic pain as well as exhibiting proteinuria and glycosuria. Environmental exposure of the following was found to be the cause?

---

- A) mercury
- B) lead
- C) cadmium
- D) arsenic
- E) none of the above



Which of the following is true concerning metallothioneins?

---

- A) these proteins have about 1/3 of their amino acids as cysteine
- B) these proteins are selectively induced by prior exposure to and are selectively bound to a single heavy metal (cadmium)
- C) these proteins can only be isolated and obtained from kidney tissue
- D) A, B and C
- E) none of the above

## In teratologic testing in rodents

---

- A) both male and female gametes should be exposed to the compound
- B) treatment should be confined to the first week of the pregnancy when the embryos are small and sensitive
- C) the compound should be given to the pregnant mother for about 1 week prior to delivery and the neonates should be examined within a few hours after birth
- D) the animals should be treated for 2 weeks prior to delivery and 1 week following delivery
- E) none of the above

Eye defects can be induced in rats by administering a teratogenic agent on gestation days

---

- A) 2-4
- B) 8-10
- C) 16-19
- D) 17-21
- E) none of the above

In the alveolar region of the lung, dusts are deposited by

---

- A) sedimentation
- B) impaction
- C) diffusion
- D) A and B
- E) A and C

Phagocytosis in the lung is involved in

---

- A) interstitial clearance into the lymph nodes
- B) bronchiolar clearance through transport by the mucociliary escalator
- C) pulmonary fibrosis
- D) A, B and C
- E) none of the above

The voiding of a diminished amount of urine in the presence of a normal fluid intake is called

---

- A) anuria
- B) oliguria
- C) polyuria
- D) uremia
- E) enuresis

Which of the following tests is most commonly used for detecting chromosomal aberrations?

---

- A) mammalian cytogenetic analysis
- B) unscheduled DNA synthesis in mouse germ cells
- C) morphological abnormalities in sperm
- D) sister chromatid exchange (SCE) in mammalian cells in culture
- E) none of the above

The immunoglobulin which is important in allergic responses (type I hypersensitivity) in humans is

---

- A) IgM
- B) IgG
- C) IgA
- D) IgE
- E) IgD



Which of the following poisons has NOT been reported to induce acute liver failure?

---

- A) acetaminophen
- B) carbon tetrachloride
- C) carbon monoxide
- D) halothane mercuric chloride
- E) mercuric chloride

3,4-Dimethyl-2,5-hexanedione produces axonopathy more rapidly and more potently than 2,5-hexanedione. This is probably due to

---

- A) electron-withdrawal by the methyl groups causing slower reactivity with amino groups to form pyrroles
- B) electron-donation by the methyl groups causing faster reactivity with amino groups to form pyrroles
- C) steric hindrance by the methyl groups blocking pyrrole formation
- D) greater specificity of the dimethyl compound in inhibiting  $\text{Na}^+/\text{K}^+$ -ATPases
- E) formation of toxic adducts between the dimethyl compound and glutathione

Which of the following has NOT been associated with the development of aplastic anemia?

---

- A) X irradiation
- B) chloramphenicol
- C) radon
- D) benzene
- E) cyclophosphamide

Drug-induced megaloblastic anemia caused by

---

- A) blockade of the synthesis of pyrimidine nucleotides
- B) reduction of cytochrome b5
- C) glucose-6-phosphate dehydrogenase (G6PD) inhibition
- D) A and B
- E) none of the above

A cell is said to be aneuploid when, compared to the number of chromosomes it should have, it has which of the following?

---

- A) too many chromosomes
- B) a reciprocal translocation
- C) too few chromosomes
- D) either too many or too few chromosomes
- E) none of the above

The unscheduled DNA synthesis assay is classified as which of the following?

---

- A) a gene mutation assay
- B) a chromosome aberration assay
- C) a dominant lethal assay
- D) a reverse mutation assay
- E) none of the above

Neuropsychiatric effects underlying the "mad hatter" syndrome in manufacturers of felt hats result from exposure to

---

- A) volatile mercuric salts
- B) solvents
- C) lead
- D) organophosphorous compounds
- E) formaldehyde

Male rats are more sensitive than female rats to chemicals that induce hepatotoxicity by the formation of free superoxide anion radicals because in the male

---

- A) the level of superoxide dismutase is higher and produces a higher number of free radicals
- B) the level of superoxide dismutase is lower and produces a lower number of free radicals
- C) the level of superoxide dismutase is higher and produces a higher rate of detoxification
- D) the level of superoxide dismutase is lower and produces a lower rate of detoxification



The excitotoxic concept of neuronal death embraces a set of assumptions. Which of the following statements is NOT a part of the excitotoxic concept of neuronal death?

---

- A) amino acid excitants, when present in excessive concentrations at the dendritic or somal surfaces of a neuron, produce a continuous state of depolarization
- B) this state of depolarization draws heavily on the cell's energy stores in an attempt to restore ionic balance
- C) by virtue of their direct inhibitory action on sodium-potassium-ATPase, these agents further directly block the cell's efforts to restore ionic balance
- D) cell death ensues when the energy stores are exhausted

Which of the following best characterizes chloracne?

---

- A) a sebaceous gland that becomes a keratinous cyst
- B) a sweat gland that becomes plugged with sebum
- C) an inflamed hair follicle
- D) a cutaneous granuloma
- E) allergic contact dermatitis caused by chlorinated dibenzodioxins

Which of the following blood cells is/are derived from the megakaryocyte?

---

- A) macrophage
- B) thrombocyte
- C) lymphocyte
- D) erythrocyte
- E) all of the above

Which of the following is NOT a clotting factor?

---

- A) fibrinogen
- B) prothrombin
- C) calcium
- D) ferritin
- E) kallikrein

Alpha-chloroacetophenone and ethyliodoacetate are examples of

---

- A) caustic bases
- B) detergents
- C) lacrimators
- D) nitrogen mustards
- E) quinones

Epoxides generally are

---

- A) chemically too unstable to cause cancer
- B) neutrophiles
- C) alkylating agents
- D) epigenetic carcinogens
- E) not mutagens

In which of the following conditions would the half saturation concentration for oxygen be lowest?

---

- A) 50% decrease in hemoglobin
- B) 25% carboxyhemoglobinemia
- C) 50% methemoglobinemia
- D) 50% carboxyhemoglobinemia
- E) 25% methemoglobinemia

Which of the following is/are true in regard to ocular toxicity?

---

- A) the acute damage observed with an alkali burn is an accurate indicator of long term damage
- B) the normal course of treatment with acid burns is prolonged irrigation with water even though with some acids such as  $H_2SO_4$  considerable amounts of heat are generated which may damage the cornea
- C) lime burns are like acid burns in that the material generates heat and tends to precipitate thereby making it difficult to remove from the eye
- D) with alkali burns the use of steroids, prostaglandins and collagenase inhibitors are contraindicated
- E) all of the above



Many inflammatory mediators in the lung are released from cells participating in the inflammatory process. Which of the following mediators is NOT released by the macrophage/ monocyte?

---

- A) complement factors
- B) interleukin-1
- C) toxic oxygen species
- D) histamine
- E) tumor necrosis factor (INF)

Southern analysis of DNA isolated from animals treated with chemical "X" as compared to controls can provide data regarding

---

- A) the capacity of chemical "X" to increase the transcriptional activity of DNA
- B) restriction fragment length polymorphism (RFLP) indicating a capacity of chemical "X" to cause mutations
- C) RFLP indicating a capacity of chemical "X" to increase the half life of RNA
- D) the capacity of chemical "X" to inhibit protein synthesis
- E) the capacity of chemical "X" to stimulate protein synthesis

A nonsense mutation can result in which of the following?

---

- A) a fully functional gene product
- B) the gain or loss of one or more base pairs in a gene
- C) a nonfunctional gene product because of premature termination of protein synthesis
- D) a fully functional gene with altered substrate specificity
- E) none of the above

## Aflatoxin B1

---

- 1) frequently occurs as a contaminant of stored grains
- 2) is subject to metabolic activation by cytochrome P450
- 3) is synergized by hepatitis in causing carcinogenicity
- 4) is synthesized by corn, peanuts and legumes

Blood is carried to the liver by the

---

- 1) hepatic vein
- 2) portal vein
- 3) lymphatics
- 4) hepatic artery

When an agent is eliminated by first-order kinetics, the plasma half-life may be

---

- 1) related to genetic factors
- 2) shortened by an agent that interferes with binding of the agent to plasma protein
- 3) altered by changes in urine pH
- 4) prolonged by induction of hepatic microsomal enzymes

Which of the following is/are true concerning Aroclor 1254?

---

- 1) it is an inducer of microsomal drug metabolizing enzymes
- 2) it is a mutagen
- 3) it is a mixture of polychlorinated biphenyls
- 4) it is a contaminant of 2,4,5-T

Which of the following functions are performed by T lymphocytes?

---

- 1) secrete cytotoxic factors
- 2) secrete immunoglobulins
- 3) produce lymphokines
- 4) differentiate into plasma cells



Which of the following typically cross the glomerular membranes in a healthy kidney?

---

- 1) erythrocytes
- 2) glucose
- 3) hemoglobin
- 4) polynuclear hydrocarbons

Which of the following are true for antimetabolites such as 6-mercaptopurine and thioguanine?

---

- 1) they are used for the treatment of acute leukemia
- 2) they have immunosuppressive activity
- 3) overdosage can cause hepatic necrosis or jaundice
- 4) prolonged treatment can cause cancer

Which of the following hepatotoxic substances produce cholestasis?

---

- 1) chlorpromazine
- 2) carbon tetrachloride
- 3) erythromycin
- 4) aflatoxin

Which compound(s) produce(s) neuropathy in chronically exposed workers, presumably through a common metabolite?

---

- 1) n-hexane
- 2) methyl isobutylketone
- 3) methyl n-butylketone
- 4) methyl ethylketone

Which of the following are nephrotoxins?

---

- 1) halogenated hydrocarbons (e.g. carbon tetrachloride, chloroform)
- 2) analgesics (e.g. aspirin, phenacetin)
- 3) antibiotics (e.g. streptomycin, gentamicin)
- 4) heavy metals (e.g. mercury, chromium)

Which of the following tests are routinely used to measure the effect of immunotoxicants on B-cell activity?

---

- 1) plaque-forming cell response to sheep red blood cells
- 2) complement depletion
- 3) immunoglobulin quantitation
- 4) delayed hypersensitivity reactions

Which statement(s) is/are true for the Ames/Salmonella typhimurium assay?

---

- 1) strains TA 1535 and TA 100 respond to mutagens that induce base-pair mutations
- 2) tester strains TA 98 and TA 1537 contain the PKM 101 plasmid which increases the capacity of the bacteria to carry out mixed function oxidation reactions
- 3) strains TA 1537, TA 1538 and TA 98 respond to mutagens that induce frame-shift mutations
- 4) all strains have the normal DNA excision repair system

Sodium saccharin ingestion causes bladder cancer in rats when consumed in high concentrations over a lifetime, including in utero exposure. The mechanism of action of sodium saccharin is believed to include

---

- 1) genotoxic damage through frame shift mutations
- 2) increased urothelial cell proliferation resulting from formation of abrasive silicate microcrystals in the urine
- 3) formation of a highly electrophilic metabolite that readily traverses the placental barrier of rats
- 4) high levels of protein in rat urine



Promoters of carcinogenesis have which of the following characteristics?

---

- 1) they need to be given repeatedly by exposure after initiation
- 2) they appear to be nongenotoxic
- 3) they show evidence for reversibility of effect if exposure ceases
- 4) they always result in polyploid cells in the target tissue

Phototoxicity may result from

---

- 1) dermal exposure to furocoumarins, (e.g. 8-methoxypsoralen)
- 2) normal doses of certain antibiotics, (e.g. lomefloxacin or tetracycline)
- 3) exposure to petroleum products, (e.g. coal tar or pitch vapors)
- 4) dermal exposure to certain cationic detergents, (e.g. lauryl dimethyl benzyl ammonium)

In an attempt to reduce animal usage, many labs today would not proceed with a primary eye irritation test if the test substance

---

- 1) is a strong acid (pH 2.0 or less) or strong alkali (pH 11.0 or greater)
- 2) has an octanol -buffer partition coefficient greater than 10
- 3) is a severe dermal irritant (primary dermal irritation index of 5 to 8) or causes corrosion of the skin
- 4) has a particle size greater than 50 microns in diameter

A neurotoxicant that damages the hippocampus would be expected to produce functional changes that include

---

- 1) weakness
- 2) blurred vision
- 3) ataxia
- 4) memory dysfunction

Which of the following is NOT true for the TLV (threshold limit value)?

---

- A) is a time-weighted concentration for a 7-8 hour workday or a 40 hour workweek
- B) refers to airborne concentrations of substances
- C) represents conditions under which nearly all workers may be repeatedly exposed day after day without adverse effects
- D) might be inappropriate for a small fraction of workers because of biological variability
- E) is determined only on the basis of experimental animal studies

Which of the following are true for saturable plasma clearance described by Michaelis-Menten kinetics?

---

- A) first order when plasma concentration,  $C$ , is less than the Michaelis constant
- B) first order when  $C$  is much larger than the Michaelis constant
- C) zero order when  $C$  is less than the Michaelis constant
- D) zero order when  $C$  is equal to the Michaelis constant
- E) the Michaelis constant is not useful for describing concentration dependent kinetics

The theoretical % of "test material" in a diet designed to maintain rats weighing 250 grams and ingesting 20 grams/rat/day at a dose level of 100 mg/kg/day is?

---

- A) 0.125%
- B) 0.025%
- C) 0.0125%
- D) 0.0025%
- E) 0.00125%

The most common fatal acute poisoning in adults in the U.S. follows overdoses of

---

- A) alcohol
- B) aspirin
- C) barbiturates
- D) carbon monoxide
- E) heroin



## In safety evaluation

---

- A) subchronic studies are shorter than acute studies
- B) chronic rodent studies last for most of the life span of the test species
- C) acute poisoning never causes chronic effects
- D) acute and chronic effects of the test substance tend to produce the same effect in different target organs
- E) none are correct

Which of the following is/are true regarding absorption of a weak organic acid ( $pK_a$  4.0) following ingestion?

---

- A) the rate of absorption will be greater in the stomach (pH 2.0) than in the intestine (pH 6.0)
- B) the rate of absorption will be greater in the intestine (pH 6.0) than in the stomach (pH 2.0)
- C) the extent of total absorption may be greater in the intestine than in the stomach
- D) the rate of absorption will be the same in both stomach and intestine
- E) both A and C are true

A probit plot of mortality data would provide the most reliable estimate of the

---

- A) no observed effect level in the test species
- B) dose that would kill 30% of the test species
- C) carcinogenic threshold dose for the test species
- D) dose that would kill 0.5% of the test species
- E) threshold limit value for the agent in man

Which species is the most commonly used surrogate for freshwater fish in warm waters in laboratory aquatic toxicity testing?

---

- A) daphnia
- B) rainbow trout
- C) bluegill sunfish
- D) fathead minnow
- E) goldfish

In direct age-adjustment of mortality in population x, one applies

---

- A) the standard crude death rate to population x
- B) the standard death rates for each age group to age groups in population x
- C) the population x crude death rate to a standard population
- D) the population x death rates for each age group to age groups in a standard population
- E) no adjustments

An individual drank radiator antifreeze in a suicide attempt. The person exhibited symptoms of epigastric pain, nausea and vomiting; weakness, delirium, restlessness, convulsive spasms and coma for two days. Blood studies at this point revealed the presence of formic acid and a subsequent autopsy disclosed multiple tissue hemorrhages and edema of the brain and retina. Which of the following solvents is most likely responsible?

---

- A) methyl alcohol
- B) ethyl alcohol
- C) ethylene glycol
- D) propylene glycol
- E) glycerol

Cells which exclude vital stains such as trypan blue are?

---

- A) unable to actively transport organic cations
- B) normal or in an early stage of cell membrane injury
- C) abnormal since cytoplasm of normal cells stains with trypan blue
- D) abnormal since trypan blue stains nuclear chromatin and is not excluded from normal cell
- E) necrotic

Overdose symptoms of vasoconstriction, local anesthesia, tachycardia, mood elevation, alertness, increase in mean arterial pressure, and death due to lethal arrhythmias or respiratory arrest are characteristic of

---

- A) phencyclidine
- B) marijuana
- C) hallucinogenic mushrooms
- D) cocaine
- E) heroin



Bioaccumulation of chemicals in the food is most likely to occur for compounds which display

---

- A) high lipophilicity
- B) high chemical and metabolic stability
- C) high tendency to bind to humic acids
- D) A and B
- E) A, B and C

The Consumer Product Safety Act applies to all of the following  
EXCEPT

---

- A) flammable fabrics
- B) refrigerators
- C) motor vehicles
- D) packaging of poisons
- E) lawn-mowers

Which of the following has NOT been associated with the production of human cancer?

---

- A) nickel
- B) asbestos
- C) arsenic
- D) barium
- E) chromium

Benzene exposure may lead to urinary excretion of

---

- A) aniline
- B) biphenyls
- C) phenol
- D) toluene
- E) hexane

Which of the following chemicals induces hemolysis and has been alleged (1982) to have been used in chemical warfare in Southeast Asia (yellow rain)

---

- A) trichothecene
- B) 2,3,7,8-tetrachlorodibenzo-p-dioxin
- C) polychlorinated biphenyls
- D) tetraethyl pyrophosphate
- E) nitrogen mustard

## Parathion

---

- A) lacks significant direct anticholinesterase activity
- B) is a phosphorothionate
- C) is nonselective with respect to target versus nontarget species
- D) displays potent anticholinesterase action following metabolic activation
- E) all of the above

In order to properly select three dose levels for a subchronic toxicity study, which of the following factors is least important to know prior to the study

---

- A) potential of the chemical to bioaccumulate
- B) shape of the dose-response curve from acute studies
- C) target organs
- D) linearity of pharmacokinetics with dose

The membrane permeability coefficient may depend on all of the following EXCEPT

---

- A) membrane/aqueous medium partition coefficient
- B) membrane surface area
- C) membrane constituents
- D) thickness of the membrane
- E) diffusivity of the molecule through the membrane



Which of the following would be an argument against the existence of "thresholds" for carcinogens?

---

- A) it is possible for a single molecule of a chemical to cause cancer
- B) some carcinogens exhibit a dose-response relationship which appears to be non-linear (convex) in the low-dose region
- C) efficient DNA repair mechanisms exist in mammalian species
- D) a carcinogen may produce a tumor through "physical" factors rather than by direct damage to the genome
- E) it has been established that as the dose of a carcinogen decreases, the time required for, tumor formation increases

The initial oxidant species formed when an electron is transferred to molecular oxygen is

---

- A) singlet oxygen
- B) superoxide anion
- C) hydroxyl radical
- D) hydrogen peroxide
- E) hydronium radical

An odds ratio obtained in a case-control study serves to estimate

---

- A) case fatality rate
- B) p value
- C) f statistic
- D) relative risk
- E) disease prevalence

SKF-525A is an example of a microsomal enzyme inhibitor. Which of the following statements is true about SKF-525A?

---

- A) decreases the duration of hexobarbital-induced anesthesia
- B) increases the duration of hexobarbital-induced anesthesia
- C) completely blocks the toxicity of phosphorothionates
- D) is effective in vivo but not in vitro

In a subchronic dog oral toxicity study, the low dose (1 mg/kg) and intermediate (3 mg/kg) dose levels cause mild and moderate toxicity, respectively. Toxicity at the high dose (10 mg/kg) was similar to that observed at the intermediate dose level. The most likely biological explanation for this effect is

---

- A) dose levels were poorly chosen
- B) the compound enhances its own absorption at high dose levels
- C) the response is typical for a flat dose response curve
- D) emesis and/or diarrhea may have occurred following dosing at the high dose level limiting absorption

As a consultant toxicologist you are called to a farm at which a number of cattle have recently died. The cattle had shown a failing appetite, thirst and appeared to be depressed. On necropsy there was generalized icterus, a swollen yellow liver, bluish swollen kidneys and frequent hemoglobinuria. The farmer admitted to the extensive use of pyrethrin based insecticides. A rodenticide was used in the feed storage area. The pond in the field was repeatedly treated with a bluish green compound to prevent overgrowth of algae. An adjacent swamp area had been recently sprayed with a foul smelling compound to control mosquitoes. Would you suspect

---

- A) pyrethrin toxicity
- B) thallium poisoning
- C) copper poisoning
- D) malathion toxicity

Which of the following factors contribute to individual differences in human metabolism?

---

- A) diverse genetic pool
- B) diverse diet
- C) diverse exposure to xenobiotics
- D) A, B and C

An estimate of the daily exposure to the human population (including sensitive subgroups) that is likely to be without appreciable risk of deleterious effects during the lifetime is the

---

- A) uncertainty factor
- B) TLV
- C) MTD
- D) NOAEL
- E) RfD



In ecotoxicology, a sentinel species

---

- A) is most abundant in the location of interest
- B) is at the top of the food chain
- C) contains a large, consistent amount of biomarkers
- D) is exceptionally sensitive to a pollutant's effects
- E) is opportunistic in its choice of habitats

The main reason DDT was banned in the United States was

---

- A) its extremely high toxicity to mammals and birds
- B) its teratogenic potential in humans
- C) it causes eggshell thinning in raptors and other high trophic level birds
- D) its antiestrogenic potential in mammals and birds
- E) all of the above

Which of the following function as receptor-mediated intracellular "amplifiers"?

---

- 1) adenylate cyclase
- 2) phospholipase C
- 3) tyrosine kinase
- 4) acetylcholine esterase

Phosgene can be a metabolite of

---

- 1) phosphoric acid
- 2) parathion
- 3) diisopropyl fluorophosphate
- 4) chloroform

A "stress protein response" has been identified in which cells increase synthesis of a family of proteins at the expense of proteins synthesized under normal conditions. Which of the following types of "toxic stress" have been demonstrated to induce this effect?

---

- 1) radiation
- 2) heavy metals
- 3) oxidative stress
- 4) heat

Which of the following is/are true regarding TCDD?

---

- 1) it can be a contaminant of chemical processes using chlorophenols as reactants
- 2) it can readily bind with the Ah receptor
- 3) it can bioaccumulate
- 4) it can be a contaminant of pulp bleaching processes

Which of the following produce BOTH psychological and physical dependence?

---

- 1) LSD
- 2) morphine
- 3) marihuana
- 4) barbiturates

## Eutrophication

---

- 1) describes the effects of pollution on natural waters
- 2) is another measure of biological oxygen demand
- 3) leads to overpopulation of waters with undesirable fish
- 4) results from enrichment with nutrients



Known to be highly persistent in the environment

---

- 1) DDT
- 2) chlorinated cyclodienes
- 3) polychlorinated biphenyls
- 4) chlorophenoxyacetic acids (2,4-D)

Which of the following acts are regulated by the U.S. Consumer Product Safety Commission (CPSC)?

---

- 1) Occupational Safety and Health Act
- 2) Consumer Product Safety Act
- 3) Food, Drug and Cosmetic Act
- 4) Federal Hazardous Substances Act

Test standards, definitions or guidelines relating to acute oral toxicity studies have been published by

---

- 1) EPA under the Toxic Substances Control Act (TSCA)
- 2) EPA under the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)
- 3) Consumer Product Safety Commission (CPSC) under the Federal Hazardous Substances Act (to give specificity to the definitions of "toxic" and "highly toxic")
- 4) Department of Transportation (DOT) (class b poison )

The term ppm means

---

- 1) parts per million
- 2) mg of x/1,000,000 mg
- 3) mg of x/kg
- 4) 0.0001% of x

Which of the following pairings is/are correct?

---

- 1) amanita phalloides - diarrhea and delayed jaundice
- 2) cadmium - anosmia and renal damage
- 3) kerosene - cough and rales
- 4) phosphorous - "phossy jaw"

## Bioavailability

---

- 1) quantifies the extent of drug absorption
- 2) is the fraction of a dose reaching the general circulation
- 3) is 1 for intravenously administered substances
- 4) can be greater than 1 if accumulation occurs

In reviewing a typical oral dose-response curve for a chemical agent, which of the following is/are commonly assumed to be true?

---

- 1) the sigmoid curve has a linear portion between 16 and 84 percent
- 2) the response follows a normal gaussian distribution
- 3) if the distribution is normal then the mean  $\pm$  2 standard deviations represents 95% of the population
- 4) a very steep slope of the dose-response curve is indicative of a heterogeneous population

Which of the following statement(s) is/are correct with respect to occupational skin diseases?

---

- 1) they account for approximately 50% of all reported occupational diseases in the U.S.
- 2) contact dermatitis is the most frequent manifestation of occupational skin disease
- 3) workers in manufacturing and agriculture, forestry and fishing industries have a substantially greater risk of occupational skin disease than workers in other sectors of industry
- 4) distinguishing occupational from nonoccupational dermatoses is difficult because of preexistent skin disorders and nonoccupational aggravating factors



Which of the following physiological functions is/are NOT constant over a 24 hour period?

---

- 1) body temperature
- 2) heart rate
- 3) urine production
- 4) hepatic glutathione content

1,4-Dioxane is an industrial solvent that has which of the following characteristics?

---

- 1) metabolized to beta-hydroxyethoxyacetic acid in rat and human
- 2) metabolized to hippuric acid in the rat and human
- 3) causes cancer in laboratory animals
- 4) causes chloracne and liver damage following chronic exposure

Which of the following has/have been shown to produce bronchoconstriction and pulmonary disease wholly or in part through an allergic mechanism?

---

- 1) spores of thermophilic actinomycetes (farmer's lung)
- 2) kaolin
- 3) toluene diisocyanate vapors
- 4) benzo(a)pyrene in coke oven emissions

## Quantitative cancer risk estimation, based on results of animal toxicity tests

---

- 1) is based on the results obtained at the maximum tolerated dose
- 2) is extrapolated only using linear dose-response relationships
- 3) is based on application of multiple safety factors
- 4) is based on both statistical procedures and mechanistic models of chemical carcinogenesis

Important components of a risk characterization for workplace exposure to a chemical include

---

- 1) hazard identification
- 2) dose-response assessment
- 3) exposure assessment
- 4) risk management

Which of the following is NOT true for asbestos exposure by inhalation?

---

- A) all persons with exposure to more than 10 fibers per cc will develop X-ray changes compatible with the disease
- B) fibers less than 3 microns in length are unlikely to produce asbestosis
- C) at the-same concentration of fibers (i e. more than 10/cc) the longer the exposure time in years, the-greater number of cases
- D) asbestosis occurs more frequently than does mesothelioma among exposed workers
- E) none of the above

Which of the following statements about lead is/are correct?

---

- A) lead causes potassium influx into the red blood cell
- B) lead inhibits both mitochondrial and cytoplasmic enzymes in erythrocyte precursors
- C) skeletal storage of lead is stabilized by metabolic acidosis
- D) a decrease in blood pressure is the most common effect of lead exposure to the adult population
- E) all of the above

Of the following, which is most likely to cause contact dermatitis?

---

- A) isopropyl myristate
- B) coumarin
- C) formaldehyde
- D) quaternary ammonium salts
- E) benzyl alcohol



## Characteristic signs of acute morphine poisoning

---

- A) diarrhea
- B) pinpoint pupils
- C) depressed respiration
- D) A, B and C
- E) B and C

Atropine poisoning may produce all of the following symptoms  
EXCEPT

---

- A) mydriasis
- B) hypothermia
- C) tachycardia
- D) hallucinations
- E) dry mouth and skin

If sodium bicarbonate is given following phenobarbital ingestion, the excretion of phenobarbital is affected because the

---

- A) acid urine decreases amount of ionized drug
- B) acid urine increases amount of ionized drug
- C) alkaline urine increases amount of ionized drug
- D) alkaline urine decreases amount of ionized drug
- E) alkaline urine increases glomerular filtration rate

In a hard rock miner with polyneuritis, exfoliative dermatitis and jaundice, the most likely diagnosis is chronic poisoning by

---

- A) silica
- B) beryllium
- C) fluorides
- D) arsenic
- E) asbestos

A 17 year old boy was drenched with a 45% parathion solution. Which of the following effects would be LEAST likely to occur as a consequence of the exposure?

---

- A) muscle fasciculations within 5 hours
- B) nausea and vomiting within 3 hours
- C) pinpoint pupils within 4 hours
- D) delayed neurotoxicity several weeks after exposure
- E) respiratory depression within 8 hours

The major life-threatening effect following the suicidal use of phenol is

---

- A) optic neuritis with permanent blindness
- B) severe gastrointestinal irritation
- C) esophageal stricture
- D) fatty degeneration of the liver
- E) respiratory depression and other CNS effects

Which of the following effects is NOT produced by botulinum toxin?

---

- A) paralysis of skeletal muscles
- B) involvement of the third cranial nerve with resultant diplopia and mydriasis
- C) dysarthria and dysphagia
- D) blockade of acetylcholine synthesis
- E) blockade of transmission of nerve impulses to the motor end plate

## Tetrodotoxin

---

- A) reduces the entry of  $\text{Na}^+$  that produces the action potential in electrically excitable tissues
- B) exerts a curare-like blockade at the neuromuscular junction
- C) destroys prejunctional nerve endings after an initial phase of transmitter release
- D) increases the  $\text{Na}^+$  permeability of electrically excitable membranes
- E) prolongs repolarization by extending the duration of  $\text{Na}^+$  permeability during the nerve action potential



Which of the following is NOT true for allergic reactions to chemical agents?

---

- A) a genetic predisposition may be demonstrable
- B) the allergic response has an erratic (not directly proportional) relationship to the dose of the eliciting agent
- C) prior contact with the agent (or an agent of very similar chemical structure) is necessary before an individual will show this type of response
- D) only a few percent of the individuals who are exposed to a specific agent will become allergic to it
- E) the different clinical signs of allergy will depend on the particular agent which elicited the reaction

Passive diffusion of drugs across tissue membranes depends primarily on

---

- A) water solubility of the ionized drug molecule
- B) water solubility of the unionized drug molecule
- C) pH of the blood
- D) membrane pore size
- E) lipid solubility of the unionized form

Pretreatment with sublethal doses of carbon tetrachloride protects the liver against subsequent exposures because the

---

- A) metabolism of carbon tetrachloride is induced
- B) metabolism of carbon tetrachloride is reduced
- C) body temperature is lowered
- D) renal excretion of carbon tetrachloride is increased
- E) bioavailability of carbon tetrachloride is reduced

Anaphylaxis can be a major problem caused by the bite or sting of which of the following arthropods?

---

- A) fire ant (*Solenopsis richteri*)
- B) brown recluse spider (*Loxosceles reclusa*)
- C) blister beetle - Centipedes (*Scolopendra* sp.)
- D) black widow spider (*Latrodectus* sp.)
- E) honey bee (*Apis mellifera*)

Paralytic shell fish poisoning is caused by ingestion of seafood contaminated with

---

- A) mercury
- B) saxitoxin
- C) tetrodotoxin
- D) botulinum toxin
- E) domoic acid

Which of the following common garden plants contains solanine?

---

- A) potato
- B) carrot
- C) turnip
- D) red beet
- E) radish

Acute poisoning has been most often caused by which class of pesticides?

---

- A) organochlorines
- B) carbamates
- C) phenoxy acids
- D) organophosphates
- E) pyrethroids

The toxicity from Dieffenbachia (dumb cane) is generally attributed to

---

- A) pilocarpine
- B) a toxic albumin called abrin
- C) calcium oxalate
- D) belladonna alkaloid
- E) ibotenic acid



Which of the following slurry tank gases is the most toxic?

---

- A) carbon dioxide
- B) nitrogen tetroxide
- C) hydrogen sulfide
- D) methane
- E) ozone

The primary source of food contamination with phthalic acid esters is

---

- A) pesticide residues
- B) food processing procedures
- C) food packaging
- D) microorganisms and their metabolites
- E) environmental pollutants

## Polychlorinated biphenyls

---

- A) are generated by burning dioxins
- B) are found in the blood only after acute exposure
- C) exhibit marked differences in toxicity among isomers
- D) accumulate in fruit
- E) are not biodegraded

Which of the following statements is/are NOT true about 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD)?

---

- A) TCDD is an unwanted byproduct of the synthesis of trichlorophenol
- B) TCDD has been shown to be teratogenic in experimental animals
- C) TCDD is rapidly metabolized and excreted by mammals
- D) TCDD is a potent inducer of aryl hydrocarbon hydroxylase activity
- E) A and C

The pesticide DDT exerts its toxic effects on the nervous system primarily by which of the following mechanisms?

---

- A) blocking acetylcholinesterase at cholinergic synapses
- B) inhibiting the release of GABA from nerve endings
- C) blocking the efflux of calcium from nerve axons
- D) blocking postsynaptic acetylcholine receptors
- E) delaying the inactivation of sodium channels

The half-life of  $^{14}\text{C}$  is approximately

---

- A) 5.6 years
- B) 56 years
- C) 560 years
- D) 5600 years
- E) 56000 years

Renal tubular necrosis and renal and bladder stone formation are characteristic effects of

---

- A) aromatic hydrocarbons
- B) glycols
- C) aliphatic hydrocarbons
- D) carbon disulfide
- E) aliphatic alcohols

The major mode of energy transfer leading to biological damage from ionizing radiation is

---

- A) spallation of  $^{12}\text{C}$  nuclei
- B) excitation of DNA atomic electrons
- C) electron-positron pair production
- D) electron capture
- E) ion pair production



Fractional deposition of dust particles in the respiratory tract is LEAST affected by

---

- A) the concentration of the dust particles in the inhaled air
- B) the mass median aerodynamic diameter of the dust particles
- C) the standard geometric deviation of the dust particles
- D) the breathing pattern of the subject
- E) the geometry of the respiratory tract

The initial interaction between ionizing radiation and a cell that results in cell damage or death is between the radiation and

---

- A) water, resulting in the generation of free radicals
- B) DNA, resulting in the generation of DNA adducts
- C) DNA, resulting in the induction of thymine dimers
- D) the cell membrane, resulting in changes in ion fluxes
- E) polysomes, resulting in disruption of protein synthesis

The primary neoplastic change observed in radium dial painters occupationally exposed to radium via ingestion was

---

- A) leukemia
- B) intestinal carcinoma
- C) osteogenic sarcoma
- D) neuroblastoma
- E) hepatocellular carcinoma

Silica is thought to exert its pulmonary toxicity

---

- A) inhibiting the secretion of surfactant
- B) disrupting mucociliary clearance leading to increased retention of inhaled dusts
- C) rupturing the lysosomes of macrophages leading to cell lysis
- D) producing hypersensitivity pneumonitis
- E) producing byssinosis

Intravenous administration of an oil (vegetable oil, mineral oil, silicone oil, etc.) may produce

---

- A) complement activation
- B) coagulation
- C) carcinogenicity
- D) pulmonary embolism
- E) anaphylaxis

Intravenous administration of a high concentration of human serum albumin to a dog may cause

---

- A) increased osmotic pressure in the vascular system
- B) serum sickness
- C) nephropathy
- D) A, B and C
- E) none of the above

Which of the following statements concerning tin is correct?

---

- A) most soluble and insoluble tin salts are well absorbed following oral administration
- B) tin is an essential element in humans
- C) trialkyl tins are the least toxic organotin compounds
- D) the major target site for triethyl tin is the CNS
- E) dietary tin increases gastrointestinal absorption of calcium and iron

A seven-year-old boy came to an emergency room with high fever and hallucinations. While being examined by a physician, the boy lost consciousness and started tonic and clonic convulsions. The boy's mother told the physician that he was picking mushrooms from the nearby woods and may have eaten some. Based on the symptoms, the mushrooms could have contained which of the following toxins?

---

- A) amatoxins
- B) psilocybin
- C) muscimol
- D) ibotenic acid
- E) monomethylhydrazine



## 2-PAM

---

- 1) has the ability of making a nucleophilic attack on the covalent O-P bond in phosphorylated acetylcholinesterase
- 2) shows its most striking effect in the improvement of neuromuscular function
- 3) is a quaternary compound with little beneficial effect on inhibited acetylcholinesterase in an animal's CNS
- 4) is indicated as the drug of choice in any case of cholinesterase inhibition

Hepatolenticular degeneration (Wilson's disease) is a genetically determined abnormality of copper metabolism. An individual affected with this trait would have

---

- 1) an increase in urinary copper concentrations
- 2) improvement following treatment with penicillamine
- 3) a decrease in plasma ceruloplasmin concentrations
- 4) a decrease in liver copper concentrations

Digitalis-like cardiotoxicity can be caused by

---

- 1) brown recluse spiders (*Loxosceles reclusa*)
- 2) coral snakes (*micruras* sp.)
- 3) puffer fish (*Tetrodonor Eugu* sp.)
- 4) Bufo toads (*Bufo* sp.)

Nickel has been shown to be a respiratory tract carcinogen following chronic occupational exposure. Which of the following is true?

---

- 1) the carcinogenicity of chronic nickel exposure has been exclusively linked to nickel carbonyl
- 2) the mechanism of nickel's carcinogenicity has been shown to be due to its clastogenic effect
- 3) soluble forms are considered to be more potent carcinogens
- 4) in animals, nickel carcinogenicity has also been shown when given subcutaneously

## Nitrogen dioxide

---

- 1) is a deep lung irritant
- 2) produces obstructive bronchiolitis
- 3) can be liberated from silage
- 4) is carcinogenic

A geometric standard deviation ( $\sigma_g$ ) of less than 1.5 for particles within an aerosol means that the particles

---

- 1) are similar in size
- 2) will be deposited in the alveolar region
- 3) were probably generated in a laboratory
- 4) have a small aerodynamic diameter

Deep lung irritation and pulmonary edema are observed in animals exposed experimentally to high doses of

---

- 1) ozone
- 2) sulfur dioxide
- 3) nitrogen dioxide
- 4) formaldehyde

Insoluble particles such as  $^{239}\text{Pu-02}$  and carbon black will accumulate in

---

- 1) Type II alveolar cells
- 2) lymph nodes
- 3) goblet cells
- 4) lung macrophages



Which of the following statements regarding DDT is/are correct?

---

- 1) DDT has a longer biological half-life than methoxychlor in rats and man
- 2) blood DDE/DDT ratios greater than 1.0 indicate recent DDT exposure
- 3) DDT concentrates at the top of ecological food chains
- 4) DDT has the same effect on the nervous system as does nicotine

## The hexachlorocyclohexanes

---

- 1) are used primarily as agricultural defoliants
- 2) form toxic epoxides
- 3) are chemically unstable
- 4) show important steric differences in insecticidal activity

With reference to poisonings with cyanide or carbon monoxide

---

- 1) 50% methemoglobin is more detrimental than 50% carboxyhemoglobin
- 2) artificial respiration with hyperbaric oxygen is very effective for both poisonings
- 3) sodium nitrite IV is indicated in the treatment of both poisonings
- 4) fatalities generally occur more rapidly with cyanide than with carbon monoxide

Inhalation exposure to which of the following has been associated with bronchial asthma?

---

- 1) chlorine
- 2) sulfur dioxides
- 3) toluene diisocyanate
- 4) western red cedar dust

## Mercury

---

- A) is not widespread in our environment
- B) once absorbed, is entirely excreted within a few hours
- C) produces basophilic stippling as an early clinical symptom
- D) is absorbed via the lungs as metallic mercury vapor
- E) has never been shown to produce adverse effects

An epileptic taking phenobarbital and diphenylhydantoin takes 16 gm of acetaminophen. You would expect the patient to

---

- A) recover relatively fast due to increased detoxification of acetaminophen
- B) show enhanced acetaminophen-induced hepatotoxicity
- C) show elevated plasma acetaminophen levels, but no liver failure, due to protection by the other two drugs
- D) excrete acetaminophen in the urine predominantly in the sulfate form
- E) die

Which of the following species would be most likely to accumulate the highest levels of polychlorinated biphenyls?

---

- A) polychaete worms
- B) daphnids
- C) pelicans
- D) minnows
- E) catfish

For a chemical that can elicit a toxic response in an animal species, the risk of a particular exposure is determined from

---

- A) the total exposure that the animal will experience
- B) experimental studies leading to estimation of a precise LD50 for that species
- C) a combination of measures of intrinsic toxicity and intensity of chemical exposure
- D) a toxicokinetic study of how the animal metabolizes, stores and excretes the chemical
- E) a combination of the information in B and D



In a toxicity study, the animals under test were housed in room 1 while the controls for the study were housed in room 2 at the other end of the hall. This is an example of

---

- A) good experimental design
- B) confounding of effects
- C) blinding a study
- D) a valid confidence interval
- E) intergroup variance

Which of the following is the most common target organ in cases of systemic poisoning following suicide attempts?

---

- A) blood and hematopoietic system
- B) liver
- C) kidneys
- D) lung
- E) central nervous system

Gastric lavage is indicated for narcotic overdose because of

---

- A) anticipated delayed gastric emptying
- B) a need to counter side effects of naloxone therapy
- C) gastric irritation of narcotic overdose
- D) the solubility of the drugs
- E) none of the above

Disulfiram is used to block

---

- A) l-aromatic amino acid decarboxylase
- B) the oxidation of acetaldehyde
- C) the oxidation of formaldehyde
- D) alcohol dehydrogenase
- E) acetylcholinesterase

Which of the following is a specific antidote for formaldehyde?

---

- A) methylene blue
- B) ethanol
- C) dimercaprol
- D) oxygen
- E) none of the above

The hazard of a radioactive isotope in the body is diminished if it

---

- A) has a long half-life
- B) is homogeneously distributed throughout the body
- C) emits beta rather than gamma radiation
- D) emits neutrons rather than beta radiation
- E) localizes in a particular organ of the body

Which of the following may be helpful in the treatment of phosphorus burns?

---

- A) 2% copper sulfate solution
- B) 5% dimercaprol solution
- C) 5% amobarbital solution
- D) 1% atropine solution
- E) 10% glycerol monoacetic acid solution

In a carefully designed clinical trial, control groups which are necessary for valid observations

---

- A) can include only those patients who have not been treated for the disease under study
- B) can include placebo controls or the standard treatment for the disease
- C) will have their data normalized to remove potential bias
- D) require different criteria for evaluation than the treatment groups
- E) are unethical because patients must be treated with drugs when they consult a physician



Which of the following is NOT true for methemoglobinemia?

---

- A) may be endogenous or exogenous in origin
- B) under normal conditions, normal humans will have about 2% of their total hemoglobin in the methemoglobin form
- C) congenital erythrocyte abnormalities may result in steady-state levels of methemoglobin as high as 30 - 50%
- D) the condition results when conversion of ferrous to ferric hemoglobin occurs at a rate which exceeds the methemoglobin reducing capacity of the cell
- E) the enzyme system responsible for reducing > 90% of the methemoglobin produced naturally is the NADPH methemoglobin reductase/methylene blue system

Iatrogenic disease refers to disease resulting from

---

- A) food additives
- B) air pollution
- C) one's occupation
- D) water pollution
- E) physician's activities

Sodium phosphate pollution of water results in

---

- A) adverse effects on fish schooling behavior
- B) in increased occurrence of fish tumors
- C) loss of fecundity in fish population
- D) teratogenicity in fish
- E) eutrophication

The most likely source of interindividual variability of the same species is

---

- A) genetic factors
- B) body weight and surface area
- C) geographical and environmental factors
- D) immunologic competence
- E) diet

Threshold limit values (TLV) are

---

- A) normally expressed both as ppm and mg/m<sup>3</sup>
- B) confined to single chemical entities
- C) based on continuous exposure for a lifetime
- D) always based on toxicity
- E) all of the above

Which of the choices below is represented by the expression:  
number of new cases during a specified time interval divided by the  
base population

---

- A) incidence
- B) coefficient of variation
- C) probit
- D) prevalence
- E) sensitivity

If the same contaminated diet were eaten by the following species, which species would a larger dose per unit body weight?

---

- A) mice
- B) rats
- C) rhesus monkeys
- D) dogs
- E) man

Intravenous-administration of 1200 ml/kg of a new chemical entity (m.w. = 40000 Daltons) to be used as a plasma expander resulted in death of mice. The most likely mechanism of toxicity is

---

- A) hypoxemia
- B) hypervolemia
- C) hyperosmolality
- D) nephrotoxicity
- E) solubility



Which of the following is true concerning the effects of phenobarbital on chloroform toxicity in rodent kidney?

---

- A) pretreatment with phenobarbital has been shown to decrease toxicity
- B) toxic chloroform metabolites are not formed in the kidney; therefore phenobarbital pretreatment has no effect
- C) pretreatment with phenobarbital has no effect because neither chloroform nor its metabolites are toxic
- D) pretreatment with phenobarbital has been shown to increase toxicity
- E) the production of toxic chloroform metabolites does not involve cytochrome P450 and thus phenobarbital pretreatment has no effect

Which one of the following are NOT characteristic to both active transport and facilitated diffusion?

---

- A) carrier-mediated transport
- B) substrate specificity
- C) saturability of the transport system
- D) transport at a rate greater than by simple diffusion
- E) expenditure of metabolic energy for transport

Encephalopathy occurring during treatment of tuberculosis with isoniazid is attributed to

---

- A) drug-induced sensitization
- B) disease-induced cytotoxic anoxia
- C) drug-induced vitamin B6 deficiency
- D) drug-induced cytotoxic anoxia
- E) breakdown of the blood-brain barrier

Features characteristic of lead intoxication may include all of the following EXCEPT

---

- A) intranuclear inclusion bodies in renal tubular cells
- B) increased erythrocyte protoporphyrin
- C) increased urinary excretion of delta-aminolevulinic acid
- D) increased ferrochelatase activity
- E) decreased activity of delta-aminolevulinic acid dehydratase

In defining an adverse effect of a chemical the dose-response relationship

---

- A) is always a curve-with a sharp break point when represented graphically
- B) is always of less concern than is the nature of the adverse finding
- C) is always quite easy to establish
- D) can be established by using two dose levels and the Kruskal-Wallis transformation equation
- E) none of the above

Which of the following is true for both the U.S. Toxic Substances Control Act and the sixth amendment of the European Economic Community?

---

- A) risk determination is included
- B) both are applicable to imports
- C) an inventory of existing chemicals is involved
- D) mandatory toxicity and other testing is involved
- E) B and C

The volume of distribution is

---

- A) large for agents, that are highly bound to plasma proteins
- B) the ratio of half-life to total dose
- C) small for most toxic agents
- D) not applicable to compounds with nonlinear toxicokinetics
- E) large for agents that are actively sequestered in a specific organ

The use of British Anti-Lewisite (BAL) to treat poisoning by arsenic is an example of

---

- A) functional antagonism
- B) receptor antagonism
- C) chemical antagonism
- D) dispositional antagonism



In deriving a chronic oral reference dose (RfD) in accordance with EPA guidelines, which of the following extrapolations does NOT require an uncertainty factor?

---

- A) LOAEL to NOAEL extrapolation
- B) subchronic to chronic extrapolation
- C) animal to human extrapolation
- D) male to female extrapolation

In risk assessment "loss of life expectancy" calculations attempt to estimate

---

- A) the number of people expected to die due to an adverse event
- B) the probability that a given individual will die due to an adverse event
- C) the average reduction of lifetime for those involved in an adverse event
- D) the probability that an adverse event will occur which will result in the loss of lives

Cohort studies are

---

- A) designed to determine the effect of a specific chemical of interest on a population with a known exposure to the chemical
- B) designed to determine the causative agent for a specific disease in a population afflicted with the disease
- C) often useful in proving that a chemical is not a human carcinogen
- D) a valuable tool for extrapolating animal data to humans
- E) another name for case-control studies

Which of the following insecticides is NOT primarily a neurotoxicant?

---

- A) DDT
- B) parathion
- C) pyrethrum
- D) dieldrin
- E) rotenone

A major difference between cytochrome P450 dependent monooxygenase (P450) and flavin-dependent monooxygenase (FMO) is

---

- A) P450 enzymes exhibit no substrate overlap
- B) the FMO enzyme system does not require molecular oxygen
- C) P450 enzymes contain a heme prosthetic
- D) FMO enzymes are not involved in the oxidation of amines
- E) P450 enzymes are not membrane associated

The cofactor used by sulfotransferases is

---

- A) acetyl coenzyme A
- B) nicotinamide adenine dinucleotide phosphate (NADPH)
- C) phosphoadenosine-phosphosulfate (PAPS)
- D) S-adenosylmethionine
- E) uridine diphosphate glucuronic acid (UDPGA)

Match the specific forms of rat cytochrome P450 (i-iv) with the appropriate substrate for oxidation (1-4). (i) P450 1A1 (1) benzene (ii) P450 2B1 (2) benzo(a)pyrene (iii) P450 2E1 (3) lauric acid (iv) P450 4A1 (4) phenobarbital

---

A) i-3; ii-4; iii-2; iv-1

B) i-2; ii-3; iii-1; iv-4

C) i-1; ii-2; iii-3; iv-4

D) i-2; ii-4; iii-1; iv-3

E) i-1; ii-3; iii-4; iv-2

The blood:air partition coefficient for a compound is 40. In an inhalation exposure in rats the blood to air concentration ratio after 24 hours is 5. What is the most likely explanation for the lower than expected blood:air concentration ratio?

---

- A) the time of exposure is so short that the blood is not expected to equilibrate with the inhaled concentration
- B) most gases and vapors never really achieve equilibrium because of diffusion-limited uptake of these chemicals across the alveolar membranes
- C) there is extensive metabolism at the inhaled concentration used in the experiment
- D) there is very rapid exhalation of the inhaled chemical, limiting the total uptake that can be achieved by the inhalation route



Which of the following reactions should NOT be inhibited by piperonyl butoxide?

---

- A) aminopyrine N-demethylation
- B) parathion desulfuration
- C) aflatoxin B1 epoxidation
- D) naphthalene hydroxylation
- E) malathion hydrolysis

Which of the following is NOT primarily a metabolic inhibitor?

---

- A) rotenone
- B) cyanide
- C) carbon monoxide
- D) fluoroacetate
- E) pyrethrum

Increased enzyme activity of which of the following cytochrome P450 isoforms is increased by starvation or by ethanol administration

---

- A) 1A2
- B) 4A2
- C) 2E1
- D) 2D6
- E) 21A1

Which of the following is/are active at neuronal sodium channels?

---

- 1) saxitoxin
- 2) tetrodotoxin
- 3) batrachotoxin
- 4) botulinum toxin

As a general rule cytochrome P450 related drug metabolism can be demonstrated in

---

- 1) isolated perfused organs without addition of cofactor
- 2) boiled cell-free tissue homogenates
- 3) supernatant from centrifugation at  $9000 \times g$  for 20 minutes; sufficient to sediment mitochondria
- 4) supernatant from centrifugation at  $10,000 \times g$  for one hour; sufficient to sediment the endoplasmic reticulum

Definition of a "case" is an important element of an epidemiologic study because

---

- 1) the "spectrum of disease" implies a range of clinical manifestations
- 2) our disease classifications are largely symptom based and may not adequately represent the disease process under study
- 3) the definition chosen may affect comparability of the results with those of another study using a different definition of cases
- 4) the definition chosen affects the validity of results extrapolated from animal studies

Which of the following statements about warfarin is/are true?

---

- 1) is a rodenticide in the coumarin family
- 2) is used clinically as an anticoagulant
- 3) acts as a vitamin K antagonist
- 4) inhibits synthesis of prothrombin

To evaluate the environmental fate and transport of a chemical, which of the following is/are important considerations?

---

- 1) water solubility
- 2) biodegradation
- 3) absorption to soil
- 4) toxicity to aquatic invertebrates



Which of the following has/have been associated with reproductive and/or developmental dysfunction after environmental or workplace exposure?

---

- 1) ethylene oxide
- 2) anesthetic gases
- 3) polychlorinated biphenyls
- 4) methyl mercury

Which of the following is/are true regarding the Maximum Tolerated Dose in carcinogenicity studies?

---

- 1) has been estimated from subchronic studies
- 2) may overwhelm normal physiology, homeostasis and detoxication or repair mechanisms and thereby induce or promote cancer
- 3) provides maximum opportunity for detection of a neoplastic response
- 4) is a controversial issue concerning carcinogen bioassays

The essential steps required in the risk assessment are

---

- 1) hazard identification
- 2) dose-response assessment
- 3) exposure assessment
- 4) risk characterization

Contamination of aquatic environments by polycyclic aromatic hydrocarbons is of concern because

---

- 1) they are bioconcentrated by aquatic organisms from water, food and sediments
- 2) they are direct-acting carcinogens in several aquatic organisms
- 3) they have relatively long environmental half-lives
- 4) they are readily soluble in water

## Selenium

---

- 1) is a nonessential element
- 2) protects against some heavy metal toxicity
- 3) deficiency is associated with kidney disease
- 4) concentration in water depends on the occurrence of seleniferous soils

Which of the following can be acted upon by hydrolases?

---

- 1) epoxides
- 2) amides
- 3) carboxylic acid esters
- 4) phosphoric acid esters

Which of the following is/are true concerning dominant lethal mutations?

---

- A) they result from a mutation of somatic cells
- B) they are not compatible with survival of offspring
- C) they cause tumors to develop in surviving offspring
- D) they follow a Mendelian inheritance pattern
- E) A and C

The state in which most chemicals cross the placenta is

---

- A) in the protein bound form
- B) as a free chemical
- C) as a metabolite
- D) in the ionized form
- E) in the conjugated form



Cough, fever and blood stained sputum is an expected consequence of industrial exposure to high concentrations of

---

- A) nitrogen dioxide
- B) zinc fumes
- C) cotton dust
- D) all of the above

"Ginger jake" paralysis resulting from ingestion of a contaminated ginger extract was caused by

---

- A) deltamethrin
- B) methanol
- C) triorthocresyl phosphate (TOCP)
- D) triethyltin
- E) acrylamide

14.7 ml of pure CO gas was injected into an evacuated 440 liter tank and brought back to atmospheric pressure at 760mm Hg with CO-free air. Calculate the concentration of CO in the tank in ppm.

---

- A) 1 ppm
- B) 0.3 ppm
- C) 3.3 ppm
- D) 33.3 ppm
- E) 500 ppm

Which of the following is best absorbed from the intestine (pH 8.0)?

---

- A) the weak base aniline (pK<sub>a</sub> 5.0)
- B) the weak acid salicylic acid (pK<sub>a</sub> 3.0)
- C) the weak base ephedrine (pK<sub>a</sub> 9.0)
- D) a strong acid
- E) a strong base

Treatment of a pregnant rat with a competitive inhibitor of testosterone would be expected to produce

---

- A) masculinized female offspring
- B) sterile female offspring
- C) feminized male offspring
- D) no effect on the offspring since embryos do not make testosterone
- E) no effect on the offspring since pregnant female rats do not make testosterone

Which of the following is true for gametogenesis

---

- A) the primitive male germ cells are primary spermatocytes
- B) the chromosome and cytoplasm of secondary oocytes are evenly divided among the ovum and the three polar bodies
- C) the production of oocytes takes place before birth
- D) A and B
- E) B and C

A majority of congenital malformations in humans is due to

---

- A) drugs
- B) emotional stress
- C) environmental pollutants
- D) infectious agents
- E) unknown causes

Which of the following toxicant exposure-treatment pairs is NOT properly matched?

---

- A) parathion - atropine
- B) curare - artificial respiration
- C) carbaryl - 2-PAM
- D) DDT- barbiturates
- E) C. botulinum - antitoxin



Measurement of which of the following should NOT be used to assess glomerular function?

---

- A) para-aminohippurate
- B) inulin clearance
- C) blood urea nitrogen
- D) urinary albumin
- E) plasma creatinine

The mechanism by which a sperm contains both members of a chromosomal pair is

---

- A) nondisjunction
- B) clastogenesis
- C) mitosis
- D) mutagenesis
- E) apoptosis

## Respiratory distress following exposure to paraquat

---

- A) is usually rapid in onset, occurring within minutes to hours after exposure
- B) occurs only after inhalation of the herbicide
- C) is the result of acute bronchoconstriction
- D) may occur several days after oral exposure
- E) none of the above

The skin is an organ

---

- A) incapable of metabolism
- B) with little metabolic capacity
- C) capable of substantial metabolism
- D) whose metabolic capacity is fully known
- E) none of the above

Alanine aminotransferase (ALT) is

---

- A) another name for SGOT
- B) an essential amino acid
- C) a good marker enzyme for hepatotoxicity
- D) increased in GGT-positive foci in the liver
- E) none of the above

A basic requirement for the classification as a mutagen is that

---

- A) the parent compound or its metabolite(s) must covalently link to DNA
- B) it must be positive in the Ames test
- C) it must undergo metabolic activation
- D) all of the above
- E) none of the above

The criteria that are used for identifying a new teratogenic agent in humans include

---

- A) an abrupt increase in the frequency of a particular defect or association of defects (syndrome)
- B) coincidence of a defect with known exposure to an environmental change
- C) known exposure to an environmental change early in pregnancies yielding characteristically defective infants
- D) absence of other factors common to all pregnancies yielding infants with the characteristic defect(s)
- E) all of the above

Select the original definition of "promotor" in the context of carcinogenesis.

---

- A) the phenomenon that two carcinogens are more powerful than each one of them alone
- B) the observation that a substance, by itself not carcinogenic, increases the potency of a carcinogen when that substance is applied before exposure to the carcinogen
- C) the enhancement of carcinogenicity of an agent by a second agent, not carcinogenic by itself, acting after exposure to the first agent has ended
- D) the effect of an agent, not carcinogenic by itself, enhancing the carcinogenicity of a chemical simultaneously administered
- E) any factor that will transform a latent tumor cell into an actively proliferating one



Which of the following cell types of the immune system can be identified by a distinct pattern of cell surface-associated markers and receptors?

---

- A) T-cells
- B) B-cells
- C) macrophages
- D) A, B and C
- E) none of the above

The presence of stem cells in the bone marrow is indicative of which of the following?

---

- A) benzene intoxication
- B) chronic heavy metal intoxication
- C) aniline intoxication
- D) hexachlorophene intoxication
- E) none of the above

Which of the following substances is/are associated with the formation of cataracts?

---

- A) steroids
- B) chlorpromazine
- C) 2,4-dinitrophenol
- D) hyperbaric oxygen
- E) A, B and C

A type I immediate hypersensitivity reaction is initiated by the binding of an antigen to which of the following?

---

- A) IgG
- B) IgE
- C) IgA
- D) IgM
- E) none of the above

The most sensitive determinations to quantify the hematotoxic effects of aniline in a 4-week study are

---

- A) erythrocyte count, hematocrit, MCH, MCHC, MCV
- B) erythrocyte count, differential blood count, histopathology of bone marrow, Heinz body count, weight of spleen
- C) hemosiderin stain of lung tissue, reticulocyte count, methemoglobin determination
- D) erythrocyte count, reticulocyte count, methemoglobin determination, Heinz body count, weight of spleen

Increases in which of the following serum enzymes is most indicative of biliary tract damage rather than hepatocyte toxicity?

---

- A) aspartate aminotransferase
- B) alanine aminotransferase
- C) lactic dehydrogenase
- D) alkaline phosphatase
- E) sorbitol dehydrogenase

Which of the following is a human syndrome characterized by monosomy?

---

- A) Down syndrome
- B) Fanconi's syndrome
- C) xeroderma pigmentosum
- D) Turner's syndrome
- E) none of the above

Polycythemia has been observed in drinkers of cobalt-contaminated beer due to

---

- A) cobalt salts increasing production of renal erythropoietic factor by the kidneys
- B) cobalt stimulation of adrenal medulla
- C) bivalent cobalt binding with an inhibiting erythrolysin
- D) B and C
- E) none of the above



Alopecia is an important clinical diagnostic clue in

---

- A) thallium toxicity
- B) botulinum toxicity
- C) arsenic toxicity
- D) triorthocresyl phosphate toxicity
- E) methylmercury toxicity

Which of the following types of immunologic responses is NOT mediated by immunoglobulins?

---

- A) type I (anaphylactic)
- B) type II (cytolytic)
- C) type III (arthrus)
- D) type IV (delayed-hypersensitivity)
- E) type V (mixed)

Which of the following is NOT a clotting factor?

---

- A) fibrinogen
- B) prothrombin
- C) calcium
- D) ferritin
- E) kallikrein

What are the most likely adverse reproductive outcomes of human exposure to environmental toxicants

---

- A) male infertility and abortion in the female
- B) male Sertoli cell tumors and premature labor in the female
- C) male Leydig cell tumors and intrauterine fetal growth retardation
- D) altered libido in both males and females
- E) abnormal sperm production and dysmenorrhea

Corneal damage by direct contact with acids results from which of the following?

---

- A) pH alone
- B) the affinity of the anion for corneal tissue
- C) the affinity of the cation for corneal tissue
- D) pH and the capacity of the anion to combine with protein
- E) none of the above

The toxicity of hyperbaric oxygen to premature infant photoreceptor cells may be explained by

---

- A) oxygen-induced retinal vasoconstriction
- B) pupillary constriction
- C) inhibition of glycolysis
- D) degeneration of ganglion cells
- E) accumulation of minute bubbles of oxygen in the vitreous

Hepatic peroxisome proliferation in the rat has been produced by exposure to

---

- A) clofibrate
- B) diethylhexylphthalate
- C) trichloroethylene
- D) all of the above
- E) none of the above

Which of the following types of neurons is the most sensitive target for 1-methyl-4-phenyltetrahydropyridine (MPTP)?

---

- A) adrenergic
- B) cholinergic
- C) GABAergic
- D) dopaminergic
- E) glutamatergic



## Certain chemical carcinogens

---

- 1) are not carcinogenic per se and consequently are classified as procarcinogens
- 2) are metabolized to electrophilic reactants
- 3) in addition to being carcinogenic may also be mutagenic
- 4) have the potential for indirect action via activation of a latent virus

Type IV allergic reactions may be

---

- 1) cytotoxic
- 2) cell mediated
- 3) anaphylactic
- 4) delayed hypersensitivity

Which of the following is/are true?

---

- 1) alkaline phosphatase in urine is a highly specific indicator of renal damage
- 2) proteinuria may result from effects of a toxic agent on the glomerular membrane
- 3) the glomerular filtration rate is conveniently estimated by the renal clearance of para-aminohippurate
- 4) urine osmolality changes may reflect effects of a toxic agent on renal medullary function

Which of the following statements concerning the kidney is/are true?

---

- 1) in a single pass through the kidney, most chemicals will have a greater propensity to influence cortical rather than medullary function
- 2) the nephron represents a functional unit of the kidney
- 3) the countercurrent mechanism within the medulla can trap a compound, leading to establishment of a high concentration of a compound in this region
- 4) many nephrotoxics act on (in) the proximal tubule

Which of the following statements are true concerning nitrosamines and nitrosamides?

---

- 1) nitrosamines under some conditions exhibit specific organotropism
- 2) dimethylnitrosamine causes primarily renal toxicity and is not highly hepatotoxic
- 3) nitrosamines as a class are converted to electrophilic reactants through oxidation
- 4) the biological significance of nitrosamine concentrations in the parts per billion range is now well established

The estimation of a toxicant's ability to impair the humoral antibody response can be made by measuring its effect upon

---

- 1) the splenic population of plaque-forming cells in animals sensitized with sheep red blood cells
- 2) the sheep red blood cell hemagglutinin titer in animals sensitized with sheep blood cells
- 3) serum immunoglobulin levels in unimmunized animals by means of radial immunodiffusion
- 4) the in vitro lysis of radioactive-chromium labelled tumor cells by spleen cells of animals sensitized specifically against the tumor cells

Thymus-dependent lymphocytes provide which of the following functions within the immune system?

---

- 1) immunological memory and immune amplification
- 2) recognition of foreign antigens
- 3) host resistance against persistent intracellular pathogens
- 4) cytotoxicity for tumor cells and cell mediated immunity

Which of the following suggest(s) that DNA is a critical target in carcinogenicity?

---

- 1) many carcinogens are also mutagens
- 2) most, if not all, cancers display chromosomal abnormalities
- 3) many carcinogens can be metabolized to electrophiles that react covalently with DNA
- 4) nuclear DNA levels are markedly decreased in cancerous tissues



Characteristics of IgG include which of the following?

---

- 1) can cross the placenta
- 2) can mediate antibody dependent cellular cytotoxicity
- 3) can fix complement
- 4) found in high amounts in tears and saliva

Which of the following is/are correct for taxol?

---

- 1) is a plant alkaloid extracted from the bark of yew trees
- 2) can stabilize the polymerized form of microtubules
- 3) is used in the treatment of certain cancers
- 4) causes an axonopathy at high doses

Chromosomal aberrations that are stable and can be transmitted repeatedly through cell divisions include

---

- 1) deletions
- 2) acentric fragments
- 3) reciprocal translocations
- 4) dicentric fragments

Which of the following pairings are correct regarding the life span of the erythrocyte in the peripheral circulation?

---

- 1) mouse - 20 to 45 days
- 2) rat - 50 to 65 days
- 3) guinea pig - 80 to 90 days
- 4) human - 180 days

Which of the following statements concerning the kidney is/are true?

---

- 1) phenobarbital and salicylates will be cleared more rapidly from the plasma after IV administration of sodium bicarbonate
- 2) amphetamines will be cleared more rapidly from the plasma after IV administration of ammonium chloride
- 3) the mechanism for secretion of organic compounds is localized in the proximal tubule
- 4) because of high blood flow to the medulla, it is very sensitive to toxic insult

Which of these animal carcinogens has/have not been demonstrated to be genotoxic?

---

- 1) butylated hydroxytoluene (BHT)
- 2) di-(2-ethylhexyl)phthalate
- 3) saccharin
- 4) d-limonene

Chemical teratogens may

---

- 1) be mutagenic
- 2) alter fetal development at doses which do not alter placental function
- 3) alter maternal health at dose levels below those altering fetal development
- 4) also cause tumors

Which of the following is the most appropriate antidote for organophosphate poisoning?

---

- A) dimercaprol
- B) d-penicillamine
- C) desferrioxamine
- D) pralidoxime
- E) naloxone



Which of the following is the major organ affected by acute DDT poisoning?

---

- A) kidney
- B) pancreas
- C) lung
- D) nervous system
- E) bone marrow

Chlorophenothane (DDT), dieldrin, chlordane and methoxychlor have in common which of the following characteristics?

---

- A) cholinesterase inhibitors
- B) organophosphate compounds
- C) bone marrow depression
- D) chlorinated hydrocarbons
- E) reversed by 2-PAM (pralidoxime)

What is the most common cause of food poisoning?

---

- A) botulinum toxin
- B) oxalic acid
- C) Staphylococcus or Salmonella toxin
- D) nitrates
- E) cyclamates

Which of the following is characteristic of chlorophenothane (DDT)?

---

- A) can accumulate through food chains
- B) is rapidly detoxified by o-demethylation
- C) has a shorter biological half-life than malathion
- D) is rapidly excreted in the urine
- E) has a very narrow margin of safety

Which of the following classes of pesticides have the greatest potential for long-term effects on ecosystems?

---

- A) chlorinated hydrocarbon insecticides
- B) pyrethrins
- C) organophosphates
- D) triazine herbicides
- E) carbamates

All but one of the following atmospheric pollutants is matched with its probable environmental effect. Which agent is matched to an unlikely environmental consequence?

---

- A) chlorofluorocarbons - stratospheric ozone depletion
- B) carbon monoxide - spruce forest die-off
- C) nitrogen oxides - photochemical smog
- D) carbon dioxide - global warming
- E) sulfur dioxide - acid precipitation

A clinical sign(s) associated with chronic benzene exposure in humans is/are?

---

- A) bone marrow depression
- B) leukemia
- C) aplastic anemia
- D) decreased white blood cell count
- E) all of the above

The average daily air concentration of a chemical which should not be exceeded if harm to an exposed worker is to be avoided, is referred to as

---

- A) NOEL
- B) ED50
- C) LD50
- D) threshold limit value (TLV)
- E) LC50



Which of the following individuals is the best candidate for IPECAC emesis?

---

- A) a 60 year-old person who is in a deep coma following suicidal ingestion of a barbiturate
- B) a middle-aged alcoholic with a history of atherosclerotic heart disease who has ingested methyl alcohol
- C) a woman in her third trimester of pregnancy who has tried to commit suicide with carbon tetrachloride
- D) a 25-year-old person who has no history of previous disease who has chewed jimson weed
- E) a college student who has no history of previous disease who tried to commit suicide by drinking lye

Dimercaprol protects against the toxic action of certain metals because it

---

- A) combines with the blood-forming elements
- B) combines with sulfhydryl groups in proteins, displacing the metal
- C) forms a stable complex with the metal
- D) increases the excretion of the metal by a direct action on the kidneys
- E) supplies sulfhydryl groups to replace those inactivated by the metal

An ideal chelating agent used in the treatment of heavy metal poisoning would be the one that

---

- A) has a greater affinity for the metal than do endogenous ligands
- B) readily forms a lipid soluble complex with the metal
- C) has an equal affinity for both the metal and the endogenous ligands
- D) readily forms an unstable, large molecular complex with the metal for fecal excretion
- E) none of the above

A worker in a munitions plant developed symptoms of nausea and vomiting, hypertension, progressive liver damage and mandibular necrosis. The most likely cause of these symptoms is

---

- A) potassium chlorate
- B) phosphorus (white)
- C) nitroglycerine
- D) sodium nitrate
- E) picric acid

A 16-month-old child ingested an unknown quantity of pills which had been prescribed for his mother. Several hours later he was brought to the emergency room because he could not be aroused. On arrival the child was semicomatose and had labored respiration. the pulse was 160, respiration 30, temperature 38.6 C and blood pressure was 80/50. Gavage produced bright red blood and there was blood in the stool. From these symptoms it is likely that the prescription was

---

- A) headache pills
- B) birth control pills
- C) iron pills
- D) diet pills
- E) sleeping pills

Which of the following is not true of arsenic poisoning?

---

- A) once arsenic is accumulated by keratinized tissue it is not removable
- B) most arsenic poisonings in man and domestic animals result from the accidental ingestion of arsenical pesticides
- C) the clinical manifestations of poisoning with arsine and arsenate salts are identical
- D) behavioral changes eventually becoming dementia are typical of chronic arsenic poisoning
- E) the lesion of arsenic poisoning common to many tissues involves capillary damage

Ingestion of about 200 ml of an inexpensive radiator antifreeze resulted in symptoms of epigastric pain, nausea and vomiting, weakness, restlessness, delirium, convulsive spasms and coma. the patient died two days later and autopsy revealed severe GI irritation, hyperemia and edema of the brain and retina, degenerative changes in the spinal cord, and hemorrhages in the brain, heart, lungs and other tissues. Which of the following was most likely to be the cause of these symptoms?

---

- A) methanol
- B) ethanol
- C) propylene glycol
- D) ethylene glycol
- E) glycerol

For which one of the following pesticides is dimercaprol (BAL) not an effective antidote?

---

- A) sodium fluoride
- B) corrosive sublimate (mercuric chloride)
- C) arsenic trioxide
- D) Paris green (copper acetoarsenite)
- E) tartar emetic (antimony potassium tartrate)



Which of the following is the molecular basis for the acute effects of cyanide poisoning?

---

- A) formation of methemoglobin
- B) formation of carboxyhemoglobin
- C) inhibition of sulfydryl enzymes
- D) inhibition of cytochrome oxidase
- E) inhibition of hydrolytic enzymes

The therapeutic efficacy of sodium nitrite in the treatment of patients with cyanide poisoning depends upon

---

- A) the formation of methemoglobin
- B) an increase in oxidative phosphorylation
- C) an increase in coronary arterial blood flow
- D) a decrease in intracranial pressure
- E) relaxation of the non-vascular smooth muscle

A 23-year-old person is brought to the emergency room after attempting suicide with ink remover (oxalic acid). Which of the following antidotal treatments is indicated for this patient?

---

- A) administration of a chelating agent
- B) atropine and 2-PAM
- C) calcium gluconate
- D) methylene blue
- E) vitamin K

The toxins of both tetanus and botulism are produced by species of which genus?

---

- A) Pneumococcus
- B) Clostridium
- C) Pseudomonas
- D) Aerobacter
- E) Klebsiella

Agitation of sludge in tanneries, paper mills or liquid manure systems may have lethal consequences for humans because of the release of which of the following?

---

- A) chlorine
- B) hydrogen sulfide
- C) methane
- D) nitrogen
- E) skatole

Which of the following is an agent which is likely to cause injury to the liver when its vapors are inhaled?

---

- A) gasoline
- B) kerosene
- C) aniline
- D) carbon tetrachloride
- E) benzene

A basic drug with a  $pK_a$  of 3 is effective orally. Assume that the pH of the stomach is 1, the pH of the small intestine is 6 and the pH of the plasma is 7.4. Under these conditions, the drug would be expected to

---

- A) be absorbed into the blood stream more rapidly from the stomach than from the small intestine
- B) be absorbed into the blood stream more rapidly from the small intestine than from the stomach
- C) be absorbed into the blood stream equally rapidly from the stomach and the small intestine
- D) not be absorbed from either the stomach or the small intestine
- E) pass principally from plasma into the stomach

Proteinaceous drugs are not given orally because

---

- A) they are more toxic by this route
- B) they are destroyed by digestive enzymes
- C) they tend to induce nausea and vomiting
- D) they are destroyed by passage through the liver
- E) none of the above



The renal clearance of a drug with a half-life of 78 minutes and an apparent volume of distribution of 14 liters is approximately

---

- A) 1 ml/min
- B) 10 ml/min
- C) 100 ml/min
- D) 700 ml/min
- E) 7000 ml/min

Toxins of animal origin are found in which of the following animal phyla?

---

- A) protozoa
- B) annelida
- C) arthropoda
- D) coelenterata
- E) all of the above

Inorganic nitrates in well water may cause which of the following effects?

---

- A) methemoglobinemia
- B) metabolic acidosis
- C) porphyria
- D) kernicterus
- E) cardiac arrhythmias

In humans, chronic ingestion of aflatoxins has been associated with an increased incidence of which of the following?

---

- A) breast cancer
- B) stomach/intestinal cancer
- C) bladder tumors
- D) liver cancer
- E) abortion

Chronic poisoning from which of the following is characterized by a Parkinsonian-like syndrome?

---

- A) manganese
- B) zinc
- C) beryllium
- D) antimony
- E) silver

Which of the following is accurately matched with its toxic effects?

---

- A) *Papaver somniferum* - narcosis
- B) squill - anorexia, nausea, vomiting
- C) *Digitalis purpurea* - blurred vision
- D) *Datura stramonium* - bizarre mental and neurological symptoms
- E) all of the above

Which of the following statements is not true about fluorine?

---

- A) it is mainly found in nature in the fluoride form
- B) chronic intoxication is called fluorosis
- C) it is added in the form of sodium fluoride to water supplies at 0.8 to 1 ppm, to prevent dental caries
- D) it adds to existing bone to form a softer type skeletal matrix
- E) in the form of fluoride ions, it exchanges with hydroxyl ions in the hydroxyapatite lattice of bone

Ingestion of improperly prepared pufferfish can lead to intoxication by which of the following?

---

- A) tetrodotoxin
- B) ciguatoxin
- C) saxitoxin
- D) alpha - bungarotoxin
- E) botulism toxin



Poisoning with methanol or ethylene glycol are alike in that both

---

- A) may cause blindness
- B) may be treated with ethanol
- C) uncouple oxidative phosphorylation
- D) vaporize and cause toxicity after inhalation
- E) undergo primary enzymatic reduction

Once the oxalate from oxalate-rich plants such as halogeton has been absorbed into the blood stream it may

---

- A) produce acute hypocalcemia leading to death
- B) cause kidney damage owing to blocking of the tubules by crystals of calcium oxalate
- C) crystallize in brain tissue and cause symptoms of paralysis and other disorders of the CNS
- D) A and B
- E) A, B and C

Which of the following common garden plants is/are poisonous?

---

- A) lily-of-the-valley
- B) rhododendron
- C) yew
- D) A and C
- E) A, B and C

Which of the following is not true regarding the anatomy/ physiology of the mammalian respiratory system?

---

- A) the exchange of oxygen and carbon dioxide occurs only in the terminal portions of the respiratory system
- B) the uptake of toxic gases occurs only in the terminal portions of the respiratory system
- C) the thickness of the liquid layer lining the airways decreases from the upper airway to the terminal alveoli
- D) mucous secretions are thickest in the upper airway and decrease toward the alveoli
- E) all of the above

Exposure to carbon tetrachloride causes which of the following symptoms?

---

- A) CNS depression, fatty liver, myocardial sensitization to catecholamines, kidney failure
- B) CNS depression, elevated carboxyhemoglobin levels
- C) CNS depression, acidosis, retinal damage
- D) CNS depression, fatty liver, physical dependence, cirrhosis
- E) CNS dependence, gastritis

A glass bottle containing one curie of ( $^3\text{H}$ ) cholesterol is left on the table in the center of a laboratory of 1000 square feet in which several people work. What would most likely be the effect on the workers after two weeks?

---

- A) no effect
- B) moderate depression of the lymphocyte count with increased incidence of infections
- C) moderate gastrointestinal upset by the end of the two-week period
- D) death of at least half of the workers due to acute radiation sickness
- E) an increased incidence of malignant tumors after a latent period of about 10 years

The site of aerosol deposition in the pulmonary system is highly dependent upon particle size. Particles of 2 to 5  $\mu\text{m}$  are predominantly deposited in which of the following regions?

---

- A) nasopharyngeal
- B) tracheobronchial
- C) alveolar
- D) all of the above
- E) none of the above

Indicate the correct combination of agent and effect

---

- A) ethylene glycol ethers - reproductive toxicity
- B) wood alcohol - retinopathy
- C) haloalkanes - hepatotoxicity
- D) n-hexane - peripheral neuropathy
- E) all of the above



Amphetamines are correctly characterized by which of the following?

---

- 1) CNS stimulants
- 2) not orally active
- 3) appetite suppressants
- 4) alpha and beta adrenergic agonists

Which of the following statements is/are correct?

---

- 1) infectious hepatitis virus in food is ordinarily destroyed by thorough cooking
- 2) the spores of *Clostridium botulinum* are quite heat resistant, but the toxins are readily destroyed by normal cooking
- 3) aflatoxins produced by *Aspergillus flavus* are commonly found in peanut meals
- 4) *Escherichia coli* is generally considered to be an indication of fecal contamination of food

Which of the following toxins act by selectively blocking the sodium channels along the axon?

---

- 1) tetrodotoxin
- 2) botulinum toxin
- 3) saxitoxin
- 4) batrachotoxin

Which of the following statements regarding fluoride are true?

---

- 1) it is recognized as a human carcinogen
- 2) it acts as a metabolic poison via inhibition of key enzymes in oxidative phosphorylation
- 3) it is allowed as an ingredient in dentifrices and mouthwashes, but not as a component of foods or products to be ingested
- 4) high intakes are associated with fluorosis, a condition affecting both the skeletal and the dental systems

A produce stand operator who grows his own fruits/vegetables developed a persistent, non-pruritic, blotchy, blistering rash on his hand and forearms over a period of several days. Several of his field hands developed similar symptoms. The most likely cause(s) of these symptoms is/are

---

- 1) pinenes in the vegetable crates
- 2) tricothecenes on corn husks
- 3) a reaction to peach fuzz amygdalin
- 4) phototoxicity from furocoumarins and psoralens found in some vegetables

Tolerance limits for a food additive are determined on the basis of which of the following?

---

- 1) a one-hundred-fold safety factor for additives which are weak carcinogens
- 2) levels improve the taste and texture of the food most significantly
- 3) levels which result in an ADI (acceptable daily intake) in man equal to a no-effect dose in animal toxicity studies
- 4) the lowest level which is still effective in good technological practice

Which of the following is characteristic of cyanide poisoning?

---

- 1) sodium nitrite is administered to convert methemoglobin to oxyhemoglobin
- 2) methemoglobinemia is produced in treatment because ferric ion can effectively bind the cyanide ion
- 3) sodium thiosulfate is administered because this compound competes effectively with cyanide ion
- 4) oxygen levels in venous blood tend to be higher than normal

Which of the following are considered to be risk factors bronchogenic carcinoma?

---

- 1) exposure to asbestos
- 2) exposure to beryllium
- 3) atmospheric pollution in cities
- 4) smoking cigarettes



Toxic symptoms following ingestion of which of the following include vomiting and alterations in cardiac conduction?

---

- 1) foxglove
- 2) lily-of-the-valley
- 3) oleander
- 4) jimsonweed

Which of the following describes the purpose of expression time experiments in establishing a mammalian cell culture mutagenicity assay?

---

- A) to provide information as to how long the cells must be subjected to the mutagen in order to obtain a response
- B) to demonstrate at which cell density there is interference with expression of the mutation due to cross-feeding from cell to cell
- C) to demonstrate phenotypic lag and indicate the time after treatment that must elapse before the stable mutation is expressed and can be scored
- D) to describe the time of focus formation in cell transformation assays
- E) none of the above

Which of the following causes hepatic angiosarcoma in humans?

---

- A) lead
- B) diethylstilbestrol
- C) soot
- D) vinyl chloride
- E) asbestos

Which of the following tests can best detect point mutations?

---

- A) cytogenetics
- B) micronucleus
- C) specific locus
- D) dominant lethal
- E) heritable translocation

Which of the following is a primary cutaneous irritant which may also be associated with an increased risk of bronchogenic carcinoma?

---

- A) chromate
- B) hydrogen sulfide
- C) zinc
- D) toluene
- E) formaldehyde

Which of the following is generally considered to be a carcinogenic response to a chemical?

---

- A) increased latency of tumor appearance compared to controls
- B) an increase in the number of preneoplastic lesions per animal
- C) a decrease in the incidence of a tumor which is usually seen in the controls
- D) an increase in the number of animals with tumors
- E) all of the above

Nitrosamines may be characterized by which of the following?

---

- A) they are used as preservatives in hot dogs
- B) they are formed from precursors in human stomach acid
- C) they result from reactions of nitrites and secondary amines
- D) A and B
- E) B and C

Damage in which of the following tissues results in high Draize scores in the eye irritation test in rabbits?

---

- A) conjunctiva
- B) cornea
- C) iris
- D) lens
- E) retina



Which of the following is correct regarding inhalation exposure?

---

- A) deposition by sedimentation occurs mainly with particles having a mass median diameter of less than 1 micron
- B) deposition by diffusion occurs principally in the deep lung
- C) impaction at bifurcations occurs where flow rates are highest
- D) B and C are correct
- E) none are correct

Allergic responses to chemicals result from damage to target cells that release which of the following?

---

- A) histamine
- B) slow reacting substance (anaphylactoid SRS-A)
- C) serotonin
- D) kinins
- E) all of the above

Which of the compounds listed below is most damaging to ocular tissues?

---

- A) cationic surfactants
- B) anionic surfactants
- C) nonionic surfactants
- D) soaps
- E) sodium bicarbonate

Which human genetic disorders show reduced DNA repair capabilities?

---

- A) Xeroderma pigmentosum
- B) ataxia telangelectasia
- C) Fanconi's anemia
- D) all of the above
- E) none of the above

The term LCT50 is

---

- A) that dose which will kill 50 percent of a group of animals when a compound is administered orally in a suitable vehicle
- B) equivalent to the threshold limit (or MAC) value
- C) inversely related to the LD50 in the same species
- D) useful in evaluating inhalation hazard
- E) the minimum lethal dose of a compound in one species

Chronic poisoning with which of the following agents usually causes peripheral neuropathy in adult humans but encephalopathy in children?

---

- A) arsenic
- B) lead
- C) mercury
- D) chromium
- E) beryllium

Which of the following has a mechanism of toxic action related to hepatic microsomal free radical formation?

---

- A) carbon monoxide
- B) methane
- C) kerosene
- D) carbon tetrachloride
- E) carbon disulfide

Which of the following is a potent inhibitor of cytochrome oxidase?

---

- A) acetazolamide
- B) hydrocyanic acid
- C) iproniazid
- D) physostigmine
- E) para-aminosalicylic acid



100% oxygen therapy is effective when hemoglobin is in which of the following forms?

---

- A) oxyhemoglobin
- B) methemoglobin
- C) carboxyhemoglobin
- D) all of the above
- E) A and C above

What is the time-weighted exposure to dust for a limestone quarry worker if he was exposed to 2.0 mg/m<sup>3</sup> for 30 minutes, 8.0 mg/m<sup>3</sup> for 1-1/2 hours, 2.0 mg/m<sup>3</sup> for 2 hours, and 18 mg/m<sup>3</sup> for 1 hour?

---

- A) 1.1 mg/m<sup>3</sup>
- B) 3.2 mg/m<sup>3</sup>
- C) 7.0 mg/m<sup>3</sup>
- D) 8.0 mg/m<sup>3</sup>
- E) > 8 mg/m<sup>3</sup>

In assessing reproductive function, the fertility index is calculated as the number of pregnancies (x 100) divided by the number of

---

- A) resorptions
- B) viable pups born
- C) successful matings
- D) pups born
- E) estrus cycles

A compound is given, to male mice on day 1. Using the serial mating technique, male fertility is severely depressed for the first 7 days but returns rapidly to normal and remains high for the rest of the 60 day test. The compound has most likely affected the

---

- A) spermatogonia
- B) spermatocytes
- C) spermatids
- D) spermatozoa
- E) all of the above

The main route for the penetration of chemicals through the skin is

---

- A) sebaceous glands
- B) hair follicles
- C) epidermis
- D) sweat ducts
- E) none of the above

Bile acids are characterized as

---

- A) procarcinogens
- B) co-carcinogens
- C) ultimate carcinogens
- D) promoters
- E) immunosuppressors

Which of the following is most effective for the treatment of narcotic opiate poisoning?

---

- A) levallorphan
- B) cyclazocine
- C) nalorphine
- D) meperidine
- E) naloxone

Which of the following is present in minute amounts in normal human plasma, and attaches to basophils and mast cells resulting in the release of mediators (histamine and slow reacting substance)?

---

- A) IgM
- B) IgG
- C) IgA
- D) IgE
- E) IgD



Rodents treated experimentally with immunosuppressive drugs or sera have been shown to develop

---

- A) immunity to adenoviruses
- B) hyperimmune responses to most antibodies
- C) allergies to certain prepared animal diets
- D) allergic rhinitis
- E) leukemia, lymphomas or reticulum cell sarcomas

Select the cell type that is capable of antigen processing, phagocytosis and lysosomal enzyme release

---

- A) T-lymphocytes
- B) B-lymphocytes
- C) macrophages
- D) platelets
- E) basophils

Damage to the glomerulus by toxic substances usually results in

---

- A) glucose in the urine
- B) protein in the urine
- C) absence of urea in the urine
- D) absence of phosphates in the urine
- E) excess amino acids in the urine

Which one of the following is not considered to be a teratogenic event in experimental teratology?

---

- A) cleft palate
- B) delayed ossification
- C) missing rib
- D) syndactyly
- E) incomplete neural canal closure

Which of the following effects is/are associated with oral exposure of bulls to ethylene dibromide?

---

- A) reduced sperm motility
- B) abnormal sperm structure
- C) reduced sperm production
- D) all of the above
- E) A and B only

Basal cell and squamous cell carcinomas in human skin are usually associated with repeated exposure to

---

- A) agents causing urticaria
- B) all mutagenic agents
- C) cold weather
- D) solar UV radiation
- E) cutaneous irritants

Chemicals pass through the skin primarily by

---

- A) passive diffusion
- B) active transport
- C) filtration
- D) facilitated diffusion
- E) facultative diffusion

During the course of a segment III preclinical reproduction study in the rat, anophthalmia is noted in 1/280 controls and 1/230 low dose, 1/190 mid dose and 2/260 high dose rats. You should conclude that

---

- A) the test compound is a weak teratogen
- B) the incidence of malformation, although low, appears to be dose related. the study should be repeated at higher dosages to see whether a higher malformation rate could be obtained
- C) detailed and appropriate statistical analyses should be performed before a definitive conclusion can be drawn
- D) the findings are spontaneous and not treatment related because the type of defect noted cannot be induced by treatment at this stage of development



Hemolytic anemia is produced by

---

- A) ferris oxide
- B) arsine
- C) desferrioxamine
- D) mercuric chloride
- E) lead acetate

An association between maternal treatment during the first trimester of pregnancy and vaginal adenocarcinoma in female offspring has been made for which of the following chemical(s)?

---

- A) thalidomide
- B) diethylstilbestrol
- C) phenytoin
- D) aspirin
- E) none of the above

All of the following are true about methemoglobin except

---

- A) it interferes with oxygen delivery to the tissues
- B) it is the ferrous form of hemoglobin
- C) it is restored to normal hemoglobin through the action of an NADH-dependent enzyme
- D) cyanosis is a common clinical manifestation of significant methemoglobinemia
- E) methemoglobinemia may be produced by aniline dyes and a variety of organic and inorganic nitrites and nitrates

Which of the following is the primary hematotoxic effect of aspirin overdose?

---

- A) thrombocytopenia
- B) granulocytopenia
- C) prevention of platelet aggregation
- D) inhibition of clotting factor K in the liver
- E) hemolysis

An increase in gamma glutamyltranspeptidase-positive foci in the liver suggests

---

- A) an induction of smooth endoplasmic reticulum
- B) hepatobiliary disease
- C) a potential for neoplasia
- D) intrahepatic cholestasis
- E) extrahepatic cholestasis

Secretory function in the liver is performed by which of the following cell types?

---

- A) bile duct cells
- B) hepatocytes
- C) Kupffer cells
- D) Ito cells
- E) macrophages

Which of the following developmental toxicant/malformation pairs is incorrect?

---

- A) phenytoin / digital hypoplasia
- B) vitamin A / malformations of the, eye, palate and genitourinary tract
- C) thalidomide / phocomelia
- D) ethanol / mental retardation and facial malformations
- E) aspirin / hypoplasia of the thymus

Which of the following represents the correct sequence of development?

---

- A) zygote-blastula-gastrula-neurula-embryo-fetus
- B) zygote-gastrula-blastula-neurula-embryo-fetus
- C) zygote-neurula-blastula-gastrula-embryo-fetus
- D) blastula-gastrula-neurula-zygote-embryo-fetus
- E) blastula-gastrula-zygote-neurula-embryo-fetus



Administration of sodium bicarbonate to treat barbiturate poisoning is based on which of the following principles?

---

- A) weak acidic chemicals are excreted better in alkaline urine
- B) weak organic bases are in the nonionized lipid soluble form in the intestine and tend to be absorbed there
- C) inactivation of a poison for essential receptors
- D) physiological antagonism
- E) all of the above

Which of the following metabolic reactions occur in the skin?

---

- 1) cytochrome P450 mediated oxidation
- 2) glucuronide conjugation
- 3) sulfate conjugation
- 4) methylation

The blood-brain barrier is a functional concept for which the following can be stated:

---

- 1) highly polar compounds readily pass the barrier and enter the brain
- 2) the barrier is most likely due to the combination of a glial barrier and unique properties of brain capillaries
- 3) the barrier has been proven not to exist by definitive EM studies
- 4) lipid-soluble compounds usually penetrate the barrier

Which of the following is/are true about methemoglobinemia?

---

- 1) methemoglobin is unable to bind oxygen
- 2) the oxidation of ferrous to ferric iron in one or more of the hemes affects the ability of other hemes to release oxygen
- 3) the red blood cell can reduce methemoglobin to hemoglobin
- 4) methemoglobin can form spontaneously

Which area(s) in the central nervous system are thought to have incomplete protection from blood-borne toxicants?

---

- 1) choroid plexus
- 2) frontal cortex
- 3) area postrema
- 4) cerebellum

Which of the following is/are characteristic of bay region epoxides of polycyclic aromatic hydrocarbons?

---

- 1) synonymous with "K" region epoxides
- 2) potent mutagens
- 3) formed by the action of epoxide hydrolase enzymes
- 4) formed by the action of cytochrome P450

Mutagens are characterized by which of the following statements?

---

- 1) they may induce base-pair transformations by altering the chemical nature of the base guanine in DNA
- 2) they may produce changes in phenotype
- 3) they may induce changes in genotype
- 4) they are always cell cycle independent

Which of the following commonly produces kidney damage with acute intoxication?

---

- 1) isopropyl alcohol
- 2) propylene glycol
- 3) ethylene glycol
- 4) methylene chloride



Which of the following is/are known to produce skin cancer humans?

---

- 1) arsenic
- 2) ultraviolet light
- 3) mixtures containing polyaromatic hydrocarbons
- 4) tetraethyl lead

Which of the following statements is/are true?

---

- 1) the proximal tubule is the primary site of action for many nephrotoxins because it is often exposed to higher concentrations of chemicals than are other cells in the kidney
- 2) the glomerulus is often susceptible to immunologic injury
- 3) the collecting duct appears to be relatively insensitive to most nephrotoxins
- 4) because of the high blood flow to this region, the renal medulla is a common target of nephrotoxins

## The potential of skin irritation by chemicals

---

- 1) has been predicted during the performance of guinea pig sensitization patch tests
- 2) is now tested with great accuracy by using non-animal models and in vivo tests
- 3) can be evaluated with studies in humans in case of mild irritants
- 4) is predictable solely on the basis of chemical structure

Which of the intracellular organelles is most involved with the autolytic removal of cellular debris?

---

- A) ribosomes
- B) Golgi complex
- C) endoplasmic reticulum
- D) lysosomes
- E) mitochondria

Which of the following defines the threshold limit value for a chemical?

---

- A) the metabolic threshold in humans
- B) the upper limit for the time-weighted average exposure for an 8-hour workshift
- C) the concentration at which toxicity is seen in humans
- D) the concentration at which a chemical can be smelled
- E) the detection limit (analytically) of the chemical in the workplace

A chemical is administered daily by oral gavage to a test animal at a dose of 15 mg/kg/day. It takes 10 days to reach a steady state level in the body. Assuming first order elimination, if the dose were increased to 30 mg/kg/day, how long will it take to reach the new steady state level?

---

- A) 5 days
- B) 10 days
- C) 20 days
- D) 40 days
- E) 100 days

The dose response curve is critical for which of the following?

---

- A) estimation of risk
- B) choice of species for study
- C) selection of an antidote
- D) determination of side effect frequency
- E) establishment of the frequency of dosing

The LD50 of a mixture containing 25 parts of A (LD50 = 250 mg/kg) and 75 parts of B (LD50 = 750 mg/kg) was determined to be 625 mg/kg. The dose-response curves for A and B, each given alone, are parallel. Which statement is correct?

---

- A) the predicted LD50 of this mixture is 1000 mg/kg. This represents antagonism
- B) the predicted LD50 equals the observed LD50. This represents summation
- C) the predicted LD50 equals the observed LD50. This is synergism
- D) the predicted LD50 is 1000 mg/kg. This is synergism
- E) the predicted LD50 = 500 mg/kg. This is antagonism



The effect seen when the administration of an agent with no appreciable effect on a biological system enhances the response to another agent is defined as

---

- A) competition
- B) summation
- C) cumulative effect
- D) potentiation
- E) antagonism

In doing a linear regression analysis on the dose of an agent you conduct a statistical test on the regression slope and find that it is significant. The interpretation of this result is that

---

- A) the intercept is large
- B) the slope is large
- C) the intercept is not zero
- D) the slope is not zero
- E) both the intercept and slope are not zero

High molybdenum content in the soil may lead to defective myelination in young ruminants by

---

- A) a direct destructive effect on myelin
- B) damaging the cells that produce and maintain myelin
- C) interfering with copper absorption to produce a copper deficiency which causes demyelination
- D) activation of a virus which produces demyelination
- E) stimulation of the immune system to produce antibodies against the basic myelin protein

Which of the following statements is not correct in regard to nitrite poisoning in wild and domestic ruminants?

---

- A) nitrates present in tissues and plants may be reduced to nitrites in the presence of moisture and bacteria
- B) methemoglobinemia is the principal toxic effect of nitrite poisoning in ruminants and in humans
- C) stomach cancer is the principal toxic effect of nitrites in ruminants but not in humans
- D) nitrites transform the iron in hemoglobin from the ferrous to the ferric state
- E) common clinical signs of nitrite poisoning in ruminants include cyanosis, dyspnea, tachycardia, weakness, coma and diarrhea

Facilitated diffusion and active transport share all but which of the following characteristics?

---

- A) substrate specificity
- B) transport at a rate greater than that by diffusion
- C) transport against a concentration gradient
- D) saturability of the transport system
- E) all of the above

Which of the following is the most important first step in addressing an epidemiologic question?

---

- A) read extensively in the literature
- B) decide how you might carry out a study
- C) decide what you really want to know
- D) draft a study protocol and solicit comments on it
- E) check the regulations

Which of the following is characteristic of ethylene glycol?

---

- A) it uncouples oxidative phosphorylation
- B) it is metabolized to formic acid
- C) it is metabolized to oxalic acid
- D) it inhibits aconitase
- E) it inhibits the effect of vitamin K

In antagonizing methemoglobinemia, methylene blue

---

- A) serves as an artificial oxygen transporting system in blood
- B) is converted from its oxidized form to its reduced form by NADPH
- C) facilitates the reduction of iron in hemoglobin from  $\text{Fe}^{+++}$  to  $\text{Fe}^{++}$
- D) displaces the toxicant from hemoglobin
- E) is oxidized by methemoglobin reductase



Which of the following is the most reliable method of identifying a poisonous agent?

---

- A) locating a toxic agent near the victim
- B) observation of the symptoms
- C) pathologic changes observed grossly or microscopically
- D) chemical analysis of the blood or tissues
- E) effectiveness of a specific antidote

## Barium sulfate as used in radiology

---

- A) is nearly inert and is associated with few side effects
- B) may cause barium intoxication if used more than once or twice
- C) has barium sulfate kidney stone formation as a hazard to use
- D) can be used orally or intravenously
- E) is used only occasionally because most people are sensitive to barium

Microsomes are defined as

---

- A) small liposomes
- B) isolated mitochondria
- C) fragments of nuclei
- D) small bodies present in living liver cells
- E) pieces of endoplasmic reticulum

A weak organic acid is 50% ionized at a pH of 3.9. What is the pKa of this acid?

---

- A) about 1.95
- B) 3.9
- C) 7.8
- D) over 7.8
- E) insufficient information given to determine pKa

An epidemiologist analyzes the last 100 evening rush hour automobile accidents and finds that 23 of the involved drivers had potential occupational exposures to solvents. To conclude that solvent exposure may be an important factor in auto accidents, he needs to know

---

- A) the number of automobile accidents in the morning rush hour
- B) how many passengers in each car had consumed alcohol
- C) how many pedestrians were involved and whether they had been exposed to solvents
- D) how many evening rush hour drivers not involved in accidents had potential occupational exposure to solvents
- E) all of the above

It has been estimated that, if the average 60 kilogram person consumes one diet drink per day (about 150 milligrams of saccharin), the lifetime risk of bladder cancer is

---

- A) greater than 24%
- B) 10% to 24%
- C) between 1% and 10%
- D) equal to 1%
- E) less than 1%

Which of the following is/are standard study designs in epidemiology?

---

- A) cohort
- B) cross-sectional
- C) case-control
- D) all of the above
- E) none of the above

Differential susceptibility of males and females to toxicants is likely to result from sex differences in

---

- A) rate of absorption
- B) urinary excretion
- C) nutritional requirements
- D) hormones affecting biotransformations
- E) body weight and surface area



Which of the following substances prevents the release of acetylcholine from the axon terminal?

---

- A) curare
- B) parathion
- C) carbaryl
- D) botulinum toxin
- E) DDT

Which of the following is the single greatest contributor degradation of surface water quality?

---

- A) contaminated industrial waste discharged directly into waterways
- B) burial of solid waste
- C) burial of liquid waste
- D) soil erosion
- E) acid rain

Which one of the following governmental bodies does not have guidelines for conducting toxicology studies?

---

- A) Environmental Protection Agency
- B) Department of Transportation
- C) Food and Drug Administration
- D) Consumer Product Safety Commission
- E) National Institute of Occupational Safety and Health

Extrapolation of toxicity data from animal to man can best be done by

---

- A) structure-activity relationships
- B) physiological, anatomical and kinetic models
- C) genetic studies of animals and man
- D) comparisons based on organ weight to body weight ratio
- E) computer models of biochemical schemes

A wide spectrum of lung changes varying from a clinically insignificant focal pigmentation to crippling fibrosis is most likely to be encountered among which of the following?

---

- A) distillery workers
- B) coal miners
- C) farmers
- D) histologists
- E) city traffic directors

Which of the following metals has been associated with an increased incidence of cardiomyopathy in heavy beer drinkers?

---

- A) chromium
- B) cobalt
- C) nickel
- D) cadmium
- E) copper

Pseudocholinesterase activity in the serum is a useful indicator of occupational exposure to

---

- A) organophosphate insecticides
- B) nitrites
- C) vinyl chloride
- D) DDT
- E) chlordane

Opisthotonus, which may occur following a toxic dose of strychnine, refers to

---

- A) paralysis of involuntary muscles
- B) labored breathing
- C) back arched in hyperextension with only the head and feet touching the ground
- D) muscles of the face and neck relaxed
- E) all of the above



The toxic substances control act delineated the concept of unreasonable risk in greater detail than any previous regulatory enabling legislation. However, several real constraints are factored into decision making. These include

---

- A) technical feasibility
- B) cost of control
- C) availability of chemical
- D) A and B are correct
- E) all of the above

Which of the following soil properties affect the persistence of organic insecticides in soils?

---

- A) organic carbon content
- B) moisture content
- C) pH
- D) temperature
- E) all of the above

Which of the following statements is/are true about a 60-day toxicity study of a proposed new food additive to be conducted in ferrets using GLP guidelines

---

- A) the balances used to weigh out the amount of test compound must be calibrated at prescribed intervals
- B) the study cannot be conducted according to GLP because only dogs, rats, mice, hamsters and monkeys may be used in toxicity studies
- C) the stability of the test compound in the dosing solution needs to be determined and the concentration of the test compound monitored in the dosing solution
- D) the study cannot be conducted in accordance with GLP because the guidelines specify that subchronic studies must be at least 90 days long

E) A and C

Occupational exposure to asbestos may lead to

---

- A) pulmonary fibrosis
- B) an increased risk of be onchogenic carcinoma which may be potentiated by cigarette smoke
- C) pleural mesothelioma
- D) peritoneal mesothelioma
- E) all of the above

If the rate of excretion of a chemical is 0.3 gm per hour, which of the following is true?

---

- A) its biological half-life is 2.31 hours
- B) it follows first order kinetics
- C) it follows zero order kinetics
- D) 30% of the chemical is excreted in an hour
- E) none of the above

Under the current FDA regulations, the sponsor of an IND trial is required to notify FDA and all participating investigators of any adverse experience associated with use of a drug. Such notification shall be made within \_\_\_\_\_ after the sponsor's initial receipt of the information?

---

- A) 1 working day
- B) 10 working days
- C) 1 month
- D) 6 months
- E) 1 year

Which of the following correctly describes the "Delaney clause"?

---

- A) prohibits marketing of a new human prescription drug that has been shown to cause cancer in rats
- B) allows the marketing of a new food-animal feed additive that increased the number of tumors (over control) in a 2-year rat carcinogenicity/toxicity study, but the mechanism of the occurrence of increased tumors was shown to be non-genotoxic but rather hormonal
- C) allows the marketing of a new food-animal drug that has been shown to cause cancer in mice but produces residues of the drug in meat, milk or eggs below the NOAEL
- D) applies to herbicides applied to human food crops
- E) none of the above



Licensing, distribution and tolerance limits of pesticides are regulated by which governmental body?

---

- A) Department of Agriculture
- B) National Institute of Occupational Safety and Health (NIOSH)
- C) Environmental Protection Agency (EPA)
- D) Department of Transportation
- E) none of the above

In the extrapolation of animal data to man it has been suggested that mg/surface area/day provides an equivalent dose between species. To a close approximation, the surface area is proportional to the:

---

- A)  $2/3$  power of the weight
- B) square of the body weight expressed in grams
- C) square root of the weight expressed in kilograms
- D) reciprocal of the square root of the weight in kilograms
- E)  $3/4$  power of the weight expressed in grams

Which of the following chemical pair interactions are appropriate?

---

- 1) propanol - carbon tetrachloride: potentiation
- 2) morphine - naloxone: synergism
- 3) mercury - BAL: chemical antagonism
- 4) propoxyphene - amytal: dispositional antagonism

Emesis is contraindicated in which of the following cases?

---

- 1) child who just swallowed an estimated number of 100 mg/kg salicylate tablets
- 2) young adult in deep coma following narcotic ingestion
- 3) child who ingested two mothballs 20 minutes before
- 4) child who ingested a strong acid 20 minutes before

Hyalin droplet nephropathy induced by unleaded gasoline (trimethylpentane) or d-limonene is characterized by which of the following?

---

- 1) activation of these agents by proximal tubular (P-2 segment) cytochrome P450
- 2) a male-specific response in the rat
- 3) binding of activated metabolites to basement proteins in the glomerulus
- 4) binding of the agents to alpha-2-microglobulin and subsequent accumulation in P-2 segment proximal tubular cells

Which of the following are factors that may influence the rate of excretion of a chemical by the kidneys?

---

- 1) renal blood flow
- 2) protein binding of the chemical
- 3) urine production
- 4) pKa of the chemical

Which of the following is/are true of salicylate poisoning?

---

- 1) causes disturbance in acid base balance
- 2) causes metabolic acidosis
- 3) is treated by administration of sodium bicarbonate
- 4) is treated by administration of sodium lactate

Which of the following are characteristic of first-order kinetics?

---

- 1) the toxicant is eliminated at a fixed amount per day
- 2) a constant fraction of the toxicant present is eliminated per unit time
- 3) it can occur when the toxicant elimination process is saturated and a fixed number of mg or gm are excreted per unit time
- 4) when toxicant concentration is high such as occurs after acute exposure a large amount is initially eliminated per unit time



A conservative estimate of carcinogenic risk may involve which of the following?

---

- 1) extrapolation into a region below the lowest observed adverse effect level (LOAEL)
- 2) the use of a linearized, non-threshold multistage model
- 3) the use of animal data from the most sensitive species
- 4) assumption of a threshold in the dose-response

The reason that the epidemiological approach for discovering the relationship between chemicals and human disease is not more successful lies in the

---

- 1) complexity of human genetic make-up
- 2) long latent periods required to detect changes
- 3) diversity of human social and economic conditions
- 4) difficulty of accurate exposure assessments

Induction of vomiting is recommended for poisonings involving which of the following?

---

- 1) ingested anticonvulsants
- 2) inhaled poisons with systemic action
- 3) ingested anticoagulants
- 4) sodium hydroxide ingestion

The principle of "selective toxicity" implies

---

- 1) the selection of the drug with the least toxicity
- 2) the toxicity of drugs dictates the drug that must be selected
- 3) the selection of one toxic effect of a drug that can be monitored to determine the risk to benefit ratio of therapy
- 4) the exploitation of differences between the biochemistry of the infecting organism and the host

For a toxic effect to occur, it is necessary that the toxic agent or its metabolites

---

- 1) reach an appropriate receptor
- 2) be present at sufficiently high concentrations
- 3) be present for enough time to produce an effect
- 4) exhibit bioaccumulation

If saturation of a sulfate conjugation pathway occurred with increased dosage of a compound and if urinary excretion of the sulfate conjugate was a major means of elimination, at higher dose levels one would expect:

---

- 1) decreased blood levels of the sulfate conjugate
- 2) increased blood levels of the parent compound
- 3) increased percentage of the dose being eliminated in the urine
- 4) decreased percentage of the dose being eliminated in the urine

Exposure to  $^{90}\text{Sr}$  by inhalation or orally induces bone neoplasia because strontium competes with calcium for incorporation into the bone.

---

Choose A if the statement and reason are true and related as to cause and effect

Choose B if the statement and reason are true but not related as to cause and effect

Choose C if the statement is true but the reason is false

Choose D if the statement is false but the reason is true

Choose E if the statement and reason are both false

Toxic effects from thallium salts can be treated effectively with calcium EDTA BECAUSE thallium replaces calcium and is chelated by EDTA

---

Choose A if the statement and reason are true and related as to cause and effect

Choose B if the statement and reason are true but not related as to cause and effect

Choose C if the statement is true but the reason is false

Choose D if the statement is false but the reason is true

Choose E if the statement and reason are both false



Chronic administration of silver salts leads to a permanent bluish darkening of the skin BECAUSE some of the silver compound is deposited in the skin where it is reduced by light to metallic silver.

---

Choose A if the statement and reason are true and related as to cause and effect

Choose B if the statement and reason are true but not related as to cause and effect

Choose C if the statement is true but the reason is false

Choose D if the statement is false but the reason is true

Choose E if the statement and reason are both false

Alkylating agents can induce neoplasms BECAUSE they form a reactive cationic intermediate, a carbonium ion, which reacts with an electron donor group in DNA to form a covalent bond.

---

Choose A if the statement and reason are true and related as to cause and effect

Choose B if the statement and reason are true but not related as to cause and effect

Choose C if the statement is true but the reason is false

Choose D if the statement is false but the reason is true

Choose E if the statement and reason are both false

Acetaminophen can cause severe hepatic damage BECAUSE its activated metabolite is detoxified by conjugation with glutathione.

---

Choose A if the statement and reason are true and related as to cause and effect

Choose B if the statement and reason are true but not related as to cause and effect

Choose C if the statement is true but the reason is false

Choose D if the statement is false but the reason is true

Choose E if the statement and reason are both false

Pralidoxime (2-PAM) should be used for treatment only when organophosphate poisoning has been positively identified BECAUSE it can potentiate the toxic effects of carbamate pesticide poisoning.

---

Choose A if the statement and reason are true and related as to cause and effect

Choose B if the statement and reason are true but not related as to cause and effect

Choose C if the statement is true but the reason is false

Choose D if the statement is false but the reason is true

Choose E if the statement and reason are both false

The most appropriate emergency antidote for methanol is

---

- A) black coffee
- B) Gatorade
- C) whiskey
- D) burned toast
- E) milk of magnesia

Dioxin ( 2, 3, 7, 8-TCDD) has occurred as a contaminant of

---

- A) 2,4-D
- B) 2,4,5-T
- C) paraquat
- D) sodium cacodylate
- E) picloram

Which one of the following is not a toxic manifestation of poisoning by mushrooms of the genus *amanita*?

---

- A) convulsions
- B) jaundice
- C) vomiting
- D) increased systolic blood pressure
- E) salivation

Which of the following is false?

---

- A) pufferfish is a poisonous animal while the stingray is a venomous animal
- B) in the USA there are more deaths caused by arthropod envenomations than there are due to bites of venomous snakes
- C) beta-bungarotoxin has its primary action at the adrenergic terminal to decrease the release of norepinephrine
- D) a snake can control the amount of venom release during a strike (bite)
- E) death after snakebite is usually due to respiratory failure



Which of the following has the highest oxidant value in "photochemical smog"?

---

- A) carbon monoxide
- B) ozone
- C) sulfur dioxide
- D) peroxyacetyl nitrate (PAN)
- E) 3, 4-benzpyrene

Halogenated aliphatic hydrocarbons exhibit which of the following toxic symptoms?

---

- A) CNS depression
- B) liver and kidney toxicity
- C) cardiac sensitization to epinephrine
- D) all of the above
- E) none of the above

Organochlorine insecticides have become less frequently used because they

---

- A) are persistent in the soil
- B) tend to accumulate in biologic media
- C) tend to accumulate in nonbiologic media
- D) have resulted in development of resistant strains of insects
- E) all of the above

Atropine toxicity is characterized by all of the following except

---

- A) tachycardia
- B) CNS stimulation
- C) hypothermia
- D) dry mouth and skin
- E) flushing of the skin

Which of the following refers to increased bone mineral density which may be fluoride-induced?

---

- A) osteoporosis
- B) osteosclerosis
- C) mottled enamel
- D) osteomalacia
- E) hypocalcemia

Clinical laboratory studies of a patient with lead poisoning might reveal

---

- A) basophilic stippling of the erythrocytes
- B) elevated urinary protoporphyrin
- C) decreased urinary coproporphyrin
- D) renal casts and hematuria
- E) none of the above

Complications accompanying chronic and excessive use of ethanol include

---

- A) severe mental disturbances
- B) thiamine deficiency
- C) the development of physical dependence on ethanol
- D) convulsions during withdrawal of ethanol
- E) all of the above

All of these cause interstitial pneumonia except

---

- A) Q fever
- B) influenza
- C) cytomegalic inclusion disease
- D) Pseudomonas
- E) berylliosis



Hyperpyrexia, dry skin, dry mouth, blurred vision and confusion are all symptoms of

---

- A) poisoning with cholinesterase inhibitors
- B) treatment with a beta-adrenergic blocker
- C) overdose of nicotine
- D) ingestion of nightshade berries
- E) none of these

As the dose of ingested fluoride increases, a chronic adverse effect appears first in

---

- A) kidneys
- B) ligaments
- C) muscles
- D) arteries
- E) dental enamel

The concentration gradient for passive diffusion between blood and tissues is determined by the concentration difference between

---

- A) protein bound drug in the blood and free drug in the tissue
- B) protein bound drug plus free drug in the blood and free drug in the tissue
- C) free drug in the blood and free drug plus protein bound drug in the tissue
- D) free drug in the blood and free drug in the tissue
- E) protein bound drug plus free drug in the blood and protein bound drug plus free drug in the tissue

Which of the following is/are correct statements regarding inhalation of dusts?

---

- A) chronic inhalation of tin in the form of dust or fumes leads to benign pneumoconiosis
- B) aluminum dusts or mists were formerly used medically for workers with pulmonary fibrosis due to silica
- C) chrysotile is a common air pollutant in most large urban areas in the united states
- D) inhaled beryllium in ore treatment rooms may cause both acute chemical pneumonitis and chronic granulomatous pulmonary disease
- E) all of the above

Polychlorinated biphenyls cause

---

- A) acneiform dermatitis
- B) eye irritation
- C) liver toxicity
- D) epileptiform convulsions
- E) all of the above

An inhibitor of dihydrofolate reductase resulting in megaloblastic anemia is

---

- A) methotrexate
- B) aspirin
- C) phenacetin
- D) chloramphenicol
- E) warfarin

In the 1970's, the James river in Virginia was polluted with the following insecticide that is still abundant in the river's sediment

---

- A) mirex
- B) kepone
- C) methoxychlor
- D) lindane
- E) DDT

Paralytic shellfish poisoning in humans is caused by a toxin produced by

---

- A) benthic algae
- B) *Vibrio parahaemolyticus*
- C) dinoflagellates
- D) *Vibrio cholerae*
- E) the roe of marine game fish



## Venom from Hymenoptera

---

- A) cannot result in an anaphylactic reaction in humans
- B) is the major toxic moiety in the black widow spider bite
- C) has been shown to release histamine from leukocytes of hymenoptera-allergic humans
- D) is a simple chemical rather than a mixture
- E) has not been implicated in any human death in the United States

The "beneficial protective effect" of various dietary fibers against colon carcinogenesis is attributed to

---

- A) increased fecal output
- B) decreased intestinal transit time
- C) dilution of toxic chemicals
- D) binding of toxic chemicals
- E) all of the above

Minamata disease involved contamination of

---

- A) peanuts with aflatoxins
- B) fish with methylmercury
- C) cereal grain with arsenic
- D) vegetable oil with TOCP
- E) whiskey with methanol

Itai-Itai disease was produced by exposure to

---

- A) cadmium
- B) lead
- C) mercury
- D) zinc
- E) arsenic

A toxic effect not associated with iron poisoning is

---

- A) gastrointestinal bleeding
- B) cyanosis
- C) siderosis
- D) hemochromatosis
- E) osteoporosis

Thiocyanates are the primary toxic agent in cabbage and the major toxic effects occur in the thyroid

---

A) true

B) false

In addition to carcinoma of the pleura, exposure to high levels of asbestos has also been associated with carcinoma of the

---

- A) peritoneum
- B) liver
- C) kidney
- D) brain
- E) thyroid

Which one of the known or suspected carcinogens below is not considered an occupational or environmental pollutant?

---

- A) cadmium
- B) benzene
- C) benzdine
- D) cyclophosphamide
- E) arsenic



Which of the following industrial solvents has been associated with neuropsychiatric disorders ranging from irritability to manic-depressive psychosis with clinical manifestations of nerve damage and signs of Parkinsonism?

---

- A) benzene
- B) carbon tetrachloride
- C) carbon disulfide
- D) propylene glycol
- E) toluene

Which of the following statements correctly compares neonatal and adult animals?

---

- A) oxidative metabolism of foreign compounds is markedly reduced in newborn mice and rats when compared to the respective adults
- B) conjugation reactions are well developed at birth and decrease significantly by adulthood
- C) adult rats develop more ocular damage from hyperoxia than do neonates
- D) neonates are particularly resistant to the induction of carcinogenesis compared to exposure as adults
- E) all of the above

Fava beans or broad beans are a major source of dietary protein in many parts of the world. However, these beans contain the pyrimidine glucosides, vicine and convicine. Which of the following is/are true?

---

- A) favism results from an inability of some to convert the glucosides to the non-toxic aglycones
- B) victims have a genetic erythrocytic deficiency of glucose-6-phosphate dehydrogenase
- C) the incidence of favism outside the Mediterranean area is extremely low
- D) all of the above
- E) B and C

## Amphibian toxins

---

- A) serve as a defense against predators
- B) are produced in secretory glands in the skin
- C) include bufotalin, tetrodotoxin, and batrachotoxin
- D) all of the above
- E) none of the above

Toxic mycotoxins associated with food contamination include which of the following?

---

- A) saxitoxin
- B) zearalenones
- C) saffrole
- D) tetrodotoxin
- E) none of the above are mycotoxins

Which of the following toxicants has not been reported to cause cancer of the respiratory system?

---

- A) asbestos
- B) iron oxide
- C) nickel
- D) arsenic
- E) chromium

Which of the following pairs is incorrect?

---

- A) nickel - nasal cancer
- B) cadmium - emphysema
- C) ozone - emphysema
- D) ammonia - lung cancer

The aminoglycoside antibiotics such as gentamicin and tobramycin produce toxic effects which are clinically manifested as

---

- A) mild and reversible renal impairment in 8-26% of the patients treated for several days
- B) the initial damage of the proximal tubule cells is usually characterized by the excretion of enzymes such as alkaline phosphatase
- C) a mild rise in plasma creatinine level
- D) all of the above
- E) none of the above



BAL (British anti lewisite) is used in the treatment of toxicity by

---

- 1) cadmium
- 2) selenium
- 3) tellurium
- 4) arsenic

Inhalation or ingestion of a toxic amount of gasoline may lead to the development of

---

- 1) ventricular fibrillation
- 2) pneumonia
- 3) respiratory depression
- 4) convulsions

Phenacetin and aspirin are about equally effective in producing

---

- 1) analgesia
- 2) methemoglobinemia
- 3) antipyresis
- 4) an anti-inflammatory effect

Wilson's disease involves a defect in the body mechanism for metabolizing which of the following?

---

- 1) magnesium
- 2) calcium
- 3) mercury
- 4) copper

Which of the following is/are true for coal worker's pneumoconiosis (CWP)?

---

- 1) the characteristic lesion of CWP is a focal macule associated with perifocal emphysema
- 2) the lungs of workers with CWP may be intensely black, but measurements of pulmonary function usually show minimal effect
- 3) exposure to coal dust may be complicated by simultaneous exposure to silicon dioxide leading to progressive massive fibrosis
- 4) CWP has not been associated with lung cancer because smoking is not allowed in mines

Treatment of cyanide poisoning relies on

---

- 1) production of methemoglobin
- 2) maintenance of an adequate concentration of carboxyhemoglobin
- 3) supply of a source of sulfur to form thiocyanate
- 4) induction of an increased activity of rhodanese

Which of the following suggest that lung damage induced by ozone may result from the oxidation of unsaturated fatty acids?

---

- 1) mitochondria in alveolar macrophages swell after exposure to ozone
- 2) the ozone syndrome can be duplicated by the injection of the ozonide of linoleic acid
- 3) there is a greater penetration of serum proteins from blood into the air space after exposure of the test animals to ozone
- 4) alpha-tocopherol (vitamin E) prolongs the life of animals that are continuously exposed to a constant concentration of ozone in the inspired air

## The toxic effects of hydrogen sulfide gas

---

- 1) may develop without warning because the gas is odorless and nonirritating
- 2) are treated, in part, by use of sodium nitrite
- 3) are attributable to sulfhemoglobin formation
- 4) may be noted in sewer workers



Hydrolysis of the glycoside amygdalin by enzymes in the gastrointestinal tract produces

---

- 1) benzaldehyde
- 2) glyceric acid
- 3) hydrocyanic acid
- 4) oxalic acid

Which of the following plants produce primarily cardiovascular effects when ingested?

---

- 1) lily-of-the-valley
- 2) jimson weed
- 3) rhododendron
- 4) dieffenbachia leaves

## The organophosphate and carbamate insecticides

---

- 1) have low dermal toxicities
- 2) are equivalently toxic for all insect pests
- 3) do not require metabolic activation to produce toxicity
- 4) are cholinesterase inhibitors

Chlorophenoxy herbicides have low mammalian toxicity BECAUSE they act on plant hormones and have no effect on animal hormones.

---

- A) if the statement and reason are true and related as to cause and effect
- B) if the statement and reason are true but not related as to cause and effect
- C) if the statement is true but the reason is false
- D) if the statement is false but the reason is true
- E) if the statement and reason are both false

Fluorocarbon refrigerants such as dichlorodifluoromethane are highly toxic chemicals BECAUSE they can deplete ozone in the stratosphere following photolysis and thereby allow increased penetration of solar ultraviolet light which is implicated in the initiation of skin cancer.

---

- A) if the statement and reason are true and related as to cause and effect
- B) if the statement and reason are true but not related as to cause and effect
- C) if the statement is true but the reason is false
- D) if the statement is false but the reason is true
- E) if the statement and reason are both false

(A) The ability of sodium silicate dusts to produce pulmonary fibrosis. (B) The ability of quartz dusts to produce pulmonary fibrosis.

---

A) if (A) is greater than (B)

B) if (B) is greater than (A)

C) if (A) and (B) are equal or nearly equal

Hemangiosarcoma is a

---

- A) benign neoplasm of endothelial cells
- B) malignant neoplasm of endothelial cells
- C) benign hematopoietic neoplasm
- D) malignant hematopoietic neoplasm
- E) malignant hepatic neoplasm

Which of the following can cause skin cancer?

---

- A) lead
- B) diethylstilbestrol
- C) soot
- D) vinyl chloride
- E) asbestos



Which of the following can cause vaginal adenocarcinoma?

---

- A) lead
- B) diethylstilbestrol
- C) soot
- D) vinyl chloride
- E) asbestos

Which of the following is least likely to detect liver disorders in rodents?

---

- A) SGPT activity
- B) serum alkaline phosphatase activity
- C) creatinine phosphokinase (CPK)
- D) serum bilirubin
- E) histopathology

Upon injection of an effective dose of nitrites, patients with hereditary methemoglobinemia

---

- A) will have more of their hemoglobin in its oxidized form than would a normal patient
- B) will have the same amount of their hemoglobin oxidized by the nitrite as will a normal patient
- C) may be helped by methylene blue
- D) B and C are correct
- E) all are correct

Which of the following *S. typhimurium* strains is sensitive to mutagens which induce pair substitutions?

---

- A) TA 1535
- B) TA 1537
- C) TA 1538
- D) all of the above
- E) none of the above

Ethanol produces its acute effects by

---

- A) specific action on cerebellum
- B) specific action on ganglia
- C) specific action on medulla
- D) local action on sensory nerves
- E) general action on central nervous system

Which of the following is useful in the treatment of methemoglobinemia?

---

- A) methylene blue
- B) nalorphine
- C) sodium nitrite
- D) amyl nitrite
- E) C and D

Blindness resulting from exposure to methanol is most likely the result of

---

- A) a direct effect of methanol on the optic nerve
- B) a direct effect of methanol on the retina
- C) an effect of a methanol metabolite (formate) on the optic nerve and lens
- D) an effect of a methanol metabolite (formate) on the retina and optic nerve
- E) none of the above

Vaginal cells from a rat exhibiting pseudo-pregnancy due to a toxic agent would be characteristic of the cells found in

---

- A) metestrus
- B) proestrus
- C) diestrus
- D) estrus
- E) ovulation



In the specific locus test for detecting chemical mutagens, the female is treated with the suspected mutagen prior to mating with an untreated male.

---

A) true

B) false

Which of the following is not an assay for dermal sensitization potential?

---

- A) guinea pig maximization test
- B) Klecak open epicutaneous test
- C) Buehler topical closed patch test
- D) Draize primary dermal irritation test
- E) split adjuvant technique

Interim clinical chemistry results from a chronic toxicity test on compound Y reveal elevated BUN and plasma creatinine levels. Based solely on these findings, which organ(s) is/are most likely affected by compound Y?

---

- A) liver
- B) kidney
- C) spleen
- D) lung
- E) B and D

The kidney appears to protect itself on a cellular level from heavy metal damage by

---

- A) decreasing the volume of rough endoplasmic reticulum
- B) decreasing the permeability of the cell membrane
- C) binding metals with RNA to form an inactive complex
- D) binding metals in lysosomes
- E) increasing the number of mitochondria

Which of the following statements is correct?

---

- A) percutaneous absorption is most efficient when the absorption agent is highly hydrophilic
- B) hexachlorophene can be more hazardous by dermal application than by oral administration
- C) most of the procarcinogens when applied dermally cause tumors at remote sites because the skin does not have an inducible cytochrome P-450 system
- D) percutaneous absorption is not influenced by the nature of the solvent
- E) photoallergic reactions are more common than phototoxic reactions

Which of the following produces liver necrosis without accumulation of lipids?

---

- A) thioacetamide
- B) carbon tetrachloride
- C) ethionine
- D) ethanol
- E) aflatoxin

Which one of the following is not correct?

---

- A) the early appearance of acid induced ocular injury is a good prognostic indication of the eventual outcome
- B) alkali induced ocular injuries which initially appear mild may progress to opacification, vascularization and perforation
- C) alkali injuries are more common in household accidents than acid injuries
- D) uveitis is frequently encountered with  $\text{NH}_4\text{OH}$  injuries because of rapid penetration of  $\text{NH}_4$  ion through corneal tissue
- E) concentrated  $\text{H}_2\text{SO}_4$  burns should not be treated by water flush because evolution of heat will further damage the eye

When is the ovum released from the ovary in mammals?

---

- A) at the primary oocyte stage
- B) when the second stage of meiotic division is triggered
- C) before the formation of the first polar body
- D) at the secondary oocyte stage
- E) at the tertiary oocyte stage



For a chemical to cause permanent sterility it must destroy all or nearly all of the following cell type

---

- A) spermatozoa
- B) spermatids
- C) secondary spermatocytes
- D) primary spermatocytes
- E) spermatogonia

In humans, CO binds to hemoglobin approximately

---

- A) 50 times more tightly than O<sub>2</sub>
- B) 450 times more tightly than O<sub>2</sub>
- C) 220 times more tightly than O<sub>2</sub>
- D) 110 times more tightly than O<sub>2</sub>
- E) 840 times more tightly than O<sub>2</sub>

The distribution of ventilation is best measured by which the following methods?

---

- A) closing volume
- B) airway resistance
- C) vital capacity
- D) nitrogen washout
- E) compliance

Which of the following conditions best describes aneuploidization?

---

- A) gain or loss of one or more intact genes
- B) an epigenetic factor in chemical carcinogenesis
- C) gain or loss of one or more intact chromosomes
- D) can be detected by the *Salmonella typhimurium* reverse mutation assay
- E) synonymous with clastogenesis

Which of the following is not a known skin sensitizer?

---

- A) ethylenediamine
- B) formaldehyde
- C) TCDD
- D) neomycin
- E) 2,4-dinitrochlorobenzene

Select the agent which is most commonly associated with the effect of severe CNS depression and ventricular fibrillation

---

- A) arsenic
- B) carbon monoxide
- C) carbon tetrachloride
- D) ozone
- E) ethylene glycol

Analysis of fructose levels in human seminal fluid can aid in the diagnosis of abnormal functioning of the

---

- A) epididymides
- B) Sertoli cells
- C) Leydig cells
- D) seminal vesicles
- E) germinal epithelium

An example of a drug(s) acting as a hapten to combine with a body constituent (e.g., serum protein) to produce a complete antigen, with resultant drug allergies is/are

---

- A) acetaminophen
- B) phenobarbital
- C) penicillin
- D) A and C
- E) none of the above



In fetal alcohol syndrome, similarities in facial features between humans and mice have been demonstrated and include

---

- A) narrow forehead
- B) short palpebral fissures
- C) small nose and small midface
- D) long upper lip with deficient philtrum
- E) all of the above

Which of the following is correct with regard to eye irritants?

---

- A) surfactants cause damage to protein principally through the lowering of surface tension
- B) prognosis for recovery immediately after exposure to mustard gas is always apparent
- C) the pH of acidic and basic solutions is the only factor for the type of lesion produced
- D) pain and dulling of the cornea after exposure to neutral solvents are both associated with lipid solubility
- E) lacrimators inhibit sulfhydryl enzymes but have only transient effects

Which of the following is/are functionally and/or physically defined T-cell mediated lymphokine(s)?

---

- A) interferon (type II)
- B) migration inhibition factor (MIF)
- C) macrophage aggregating factor
- D) blastogenic factor
- E) all of the above

Which of the following statements relative to the blood-brain barrier is not true?

---

- A) nonpolar, lipid-soluble compounds usually penetrate the barrier
- B) presence of pericapillary glia cells may be a reason for the barrier
- C) the barrier is more effective in the CNS of infants than in adults
- D) absence of intercellular pores and pinocytotic endothelial cells of the capillaries may be a reason for the barrier

4-isothiocyano-4'-nitrodiphenylamine was devoid of mutagenic activity in the *Salmonella typhimurium* Ames test with or without S9 activation. However, when large doses were given to rodents and primates, a mutagenic metabolite was detected in the urine. This apparent discrepancy is most satisfactorily explained by the fact that

---

- A) a series of catabolic reactions in the mucosa of the GI tract and the liver are required to produce the mutagenic metabolite
- B) the chemical is metabolized to the mutagen by microsomal enzymes in the bladder
- C) renal microsomal enzymes convert drug to an active mutagen
- D) GI microflora is responsible for the formation of the mutagenic metabolite
- E) there is no satisfactory explanation for the findings

Pneumoconiosis is a general term that describes

---

- A) the irritant-type responses to gases such as chlorine, ammonia and phosgene
- B) the syndrome of respiratory symptoms often observed in children of smoking parents
- C) an allergic reaction to a material after there has been a previous contact
- D) a characteristic pattern of lymphocytosis, hyperchromic anemia and thrombocytopenia associated with benzene exposure
- E) a class of occupational diseases in which the central factor is pulmonary fibrosis

An FDA segment I study is designed to detect effects

---

- A) caused by accumulated toxicity
- B) on parturition, lactation, neonatal development
- C) on embryonic development and organogenesis
- D) on mating, fertility, implantation, gestation, parturition, lactation, neonatal development
- E) on function and behavior in offspring

Which of the following are causal agents of chloracne?

---

- A) DDT and hexachlorobenzene (HCB)
- B) HCB and 2,4-dichlorophenoxyacetic acid (2,4-D)
- C) 2,4-D and 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD)
- D) TCDD and polychlorinated biphenyls (PCB's)
- E) PCB's and pentachloronitrobenzene (PCNB)



Which of the following items is not characteristic of the phenomenon known as "light hydrocarbon nephropathy (LHN)"?

---

- A) hyaline droplet formation in the renal proximal convoluted tubule cells
- B) granular cast formation due to the sloughing of cellular debris into the renal tubular lumen
- C) primarily observed following exposure to saturated aliphatic compounds that contain a ring in their structure
- D) appears to be species and sex dependent
- E) postulated to involve a chemical interaction with alpha-2u-globulin

Morphologic injury to the central nervous system follows excessive exposure to

---

- A) aminoglycoside antibiotics
- B) trialkyltin compounds
- C) bromobenzene
- D) anthracycline antibiotics
- E) none of the above

Which of the following structures is among the first to develop during gestation in the rat?

---

- A) brain
- B) aorta
- C) urogenital system
- D) pulmonary system
- E) digestive system

The major cause of lung cancer in the United States is

---

- A) exposure to automotive emissions
- B) exposure to coal-fired power plant emissions
- C) chest X-ray examination
- D) cigarette smoking
- E) exposure to asbestos fibers

A variety of substances may produce skin reactions in humans when the skin is exposed to sunlight, and in some cases to artificial light. These are referred to as photosensitivity reactions. Which of the following statements correctly characterizes these reactions?

---

- A) some of these reactions may occur on the first exposure to the chemical plus light (the so-called phototoxic mechanism)
- B) all photosensitivity reactions are immunologically mediated
- C) drugs behave as haptens, and as such are directly immunogenic
- D) metabolic products of drugs do not participate in photosensitivity reactions, as they are "detoxified"
- E) all of the above

Hemolysis has been recognized as resulting from exposure to each of the following except

---

- A) arsine
- B) boric acid
- C) naphthalene
- D) phenol (carbolic acid)

Which of the following toxicities would be expected in a worker exposed to toluene diisocyanate?

---

- A) chloracne
- B) asthma
- C) leukemia
- D) erethism
- E) hemolytic anemia

As a class of compounds, hepatic peroxisome proliferators are

---

- A) not carcinogenic in rodents
- B) negative in the Ames test
- C) positive in the CHO/hgprt locus test
- D) hypertriglyceridemic
- E) inhibitors of the  $\beta$ -oxidation of fatty acids



Retention of indocyanine green (ICG) is indicative of which type of liver injury?

---

- A) hepatitis
- B) hepatic steatosis
- C) centrilobular necrosis
- D) hepatic parenchymal cell damage
- E) cholestasis

Which of the following statements is not true for the Draize ocular irritancy test as conducted for the consumer product safety commission?

---

- A) this test has received considerable media attention partly because it is used to test cleaners, cosmetics and other home products
- B) a 0.1 ml volume of sample is placed into the conjunctival sac of the rabbit eye
- C) separate numerical scores are recorded for degree of corneal opacity, pannus, iritis, palpebral conjunctivitis and bulbar conjunctivitis
- D) scoring is required at 24, 48 and 72 hours after dosing, but not thereafter
- E) A and B

The susceptibility of animals to developmental toxicants varies with the developmental stage. Identify the characteristic vulnerabilities of different developmental periods.

---

- 1) preimplantation: embryo lethality
- 2) organogenesis: birth defects, embryo lethality
- 3) fetal: growth retardation, fetal death, transplacental carcinogenesis, functional deficits
- 4) neonatal: growth retardation, functional alterations

The theoretical time for a test compound gas to reach 99% equilibrium in an inhalation chamber for a selected target concentration is termed the T99. In addition to the equation constant what measurements are needed to calculate the T99?

---

- 1) chamber volume
- 2) chamber target concentration of the test compound
- 3) chamber airflow rate
- 4) molecular weight of the test compound

Which of the following are direct acting carcinogens?

---

- 1) N-2-fluorenylacetamide
- 2) triethylenemelamine
- 3) aflatoxin B1
- 4) ethylmethane sulfonate

2-naphthylamine has been demonstrated to be carcinogenic in

---

- 1) dog and rhesus monkey
- 2) man
- 3) mouse and hamster
- 4) rat

Which of the following is/are nephrotoxics?

---

- 1) halogenated hydrocarbons (chloroform, CCl<sub>4</sub> )
- 2) analgesics (phenacetin)
- 3) antibiotics (streptomycin, gentamicin)
- 4) heavy metals (mercury, chromium)

## Cirrhosis

---

- 1) can be induced in animals by chronic administration of carbon tetrachloride
- 2) is caused by diets containing high levels of choline, methionine, vitamin B12 or folic acid
- 3) can be caused in humans by chronic ingestion of ethanol
- 4) is clinically identical with siderosis of the liver



A patient with headache, nausea/vomiting, fatigue, jaundice hepatomegaly and oliguria may have an occupational disease caused by

---

- A) lead
- B) mercury
- C) benzol
- D) carbon tetrachloride
- E) hydrogen sulfide

In the United States in the last decade, the death rates from all cancers combined excluding lung cancer have

---

- A) decreased slightly or stayed level
- B) increased slightly or stayed level
- C) increased dramatically
- D) increased in males and decreased in females
- E) decreased in males and increased in females

Esterases and amidases are cytochrome P450 dependent enzymes.

---

- A) true
- B) false

It is a general principle of toxicology that once a chemical agent has been completely eliminated from the body, no manifestations will be evident in that person.

---

A) true

B) false

When poisons are swallowed, vomiting should not be induced if

---

- A) the poison is a corrosive chemical
- B) the patient is comatose
- C) the poison is a CNS depressant
- D) all of the above
- E) A and B

Which of the following is the tissue of choice to be collected for postmortem examination for the presence of cyanide?

---

- A) liver
- B) kidney
- C) brain
- D) whole blood
- E) fat

Hemodialysis is more useful in removal of materials with

---

- A) small volumes of distribution
- B) large volumes of distribution
- C) high lipid solubility
- D) high serum binding properties
- E) high urinary excretion rates

Aldrin and heptachlor are metabolized by MFO systems to corresponding intermediates which are more toxic than the parent compounds. One such class of intermediates is

---

- A) aldehydes
- B) epoxides
- C) ketones
- D) none of the above



Acute poisoning from methanol is more serious than that from ethanol because

---

- A) it is oxidized more rapidly than ethanol
- B) it is not capable of sparing protein in the body
- C) the presence of insulin retards the destruction of methanol
- D) it produces blindness
- E) methanol is an addicting drug

The deleterious effects of absorbed radium are due to

---

- A) neutron emission
- B) alpha particles
- C) beta particles
- D) gamma irradiation
- E) positron emission

A disease prevalence refers to the number of persons with the disease divided by the total number of persons at risk or in the population at a point in time.

---

A) true

B) false

What is the ratio of unionized to ionized agent for a weak acid (HA) with a pKa of 3.4 in the stomach at a pH of 1.4?

---

- A) 1000:1
- B) 100:1
- C) 10:1
- D) 1:1
- E) 1:100

A first-order decline in concentration of an agent in the blood means that the

---

- A) log concentration versus time plot gives a straight line
- B) log time versus drug concentration plot gives a straight line
- C) decline is not dependent on the concentration of the agent present
- D) concentration plotted against time is a straight line
- E) all of the above

The therapeutic efficacy of sodium nitrite in the treatment of patients with cyanide poisoning depends upon

---

- A) formation of methemoglobin
- B) an increase in oxidative phosphorylation
- C) an increase in coronary arterial blood flow
- D) a decrease in intracranial pressure
- E) relaxation of non-vascular smooth muscle

Hydrogen sulfide exposure is a principal health hazard in which industry?

---

- A) paper manufacture
- B) foundry worker
- C) glass manufacture
- D) coal mining
- E) automobile refinishing

In general, humans ingest considerably more lead from drinking water than from food.

---

A) true

B) false



Considerable evidence indicates that benzo(a)pyrene is metabolically activated to an electrophilic carcinogenic

---

- A) K region arene oxide
- B) free radical
- C) quinone
- D) diol epoxide
- E) superoxide

The animal species of choice for testing the carcinogenicity of an aromatic amine for cancer is the

---

- A) rat
- B) mouse
- C) dog
- D) guinea pig
- E) hamster

When the frequency of an abnormality is expressed as the number of animals (or persons) developing the abnormality during a specified time period, it is referred to as

---

- A) incidence rate
- B) prevalence rate
- C) specific rate
- D) standardized rate
- E) normalized rate

Which of the following mechanisms is/are related to the production of liver injury?

---

- A) accumulation of lipids
- B) interference with protein synthesis
- C) lipid peroxidation
- D) cholestasis
- E) all of the above

The lethality of ozone in mice is dependent on

---

- A) concentration
- B) duration of exposure
- C) age at exposure
- D) all of the above
- E) A and B only

The most important tissue for a forensic toxicologist to analyze in a suspected case of chronic heavy metal poisoning is

---

- A) brain
- B) lung
- C) hair and fingernails
- D) stomach contents
- E) blood

Angiosarcoma in humans is most often associated with exposure to which of the following?

---

- A) arsenic
- B) coke oven emissions
- C) vinyl chloride
- D) beryllium
- E) talc

Workers engaged in the manufacture of which of the following pesticides suffered neurological effects such as tremors and widened gaits, reduced sperm count and motility, and liver damage?

---

- A) DDT
- B) kepone (chlordecone)
- C) 2,4-dichlorophenoxyacetic acid (2,4-D)
- D) paraquat
- E) malathion



Methemoglobin in the blood is a biologic determinant for exposure to

---

- A) DDT
- B) phenol
- C) dioxane
- D) aniline
- E) none of the above

All of the following apply to the organophosphate insecticides except

---

- A) symptoms of acute systemic poisoning arise from the inhibition of acetylcholinesterase
- B) local effects can include miosis, bronchoconstriction, nasal discharge and sweating
- C) more frequently involved in human poisoning than any other class of insecticides
- D) includes lindane, endrin and toxaphene
- E) treatment of severe poisoning should include artificial respiration first, followed by administration of atropine and then 2-PAM

Biotransformation of foreign chemicals involves all of the following except

---

- A) oxidation, reduction and hydrolytic type reactions
- B) less polar and more lipophilic metabolites
- C) enhanced or reduced toxicity of toxicants entered in the body
- D) conjugation of metabolites of toxicants with endogenous substances such as glucuronic acid

Which of the following substances is regulated specifically under the toxic substances control act?

---

- A) polychlorinated biphenyls (PCBs)
- B) parathion
- C) butylated hydroxyanisole
- D) plutonium
- E) all of the above

The premanufacture notice (PMN) required under the toxic substances control act (TSCA)

---

A) allows no exemptions

B) applies only to materials produced in amounts exceeding 1 ton per year or 10 tons in total

C) forbids the manufacture or import of a new chemical for commercial purposes without notifying the EPA 90 days before beginning

D) requires the submission of a minimum data set as part of the PMN including acute studies and one study lasting a minimum of 28 days

E) exempts isolated site limited intermediates from the requirements of a PMN

When 300 mg of a substance is given intravenously, a plasma concentration of 60 mg/liter is obtained. The volume of distribution is

---

- A) 3 liters
- B) 5 liters
- C) 120 ml
- D) 200 ml
- E) 500 ml

Bioconcentration factor (BCF) refers to

---

- A) the catabolism of pollutants by soil bacteria
- B) the ratio of the concentration of a substance within an organism to that in the surrounding medium (usually water)
- C) the ability of a substance to concentrate up a food chain
- D) the measured concentration of a substance in an organism's tissues

Which of the following is/are not specially regulated by ambient air quality standards of the clean air act?

---

- A) total suspended particles
- B) photochemical oxidants
- C) sulfur dioxide
- D) PCB's
- E) all of the above



The ratio of the incidence rate in the population exposed to the risk factor, to the incidence rate in the population not exposed is

---

- A) attributable risk
- B) relative difference
- C) prevented fraction
- D) relative risk
- E) absolute risk

Which of the following statistical tests would be most appropriate to compare the mean values for 4 experimental groups when the groups are from normally distributed populations with the same variance?

---

- A) chi-square
- B) student t
- C) paired t
- D) Kruskal-Wallis
- E) one-way anova

Methods presently utilized by U.S. regulatory agencies for extrapolating dosages effective in inducing cancer in animals down to presumably "safe" doses for man have which of the following features in common?

---

- A) all assume that there is no absolutely "safe" dose, i.e. no threshold carcinogenic dose
- B) all assume a 10-fold decrement in risk per 10-fold decrement in dose
- C) all take into account species differences in biotransformation of carcinogens
- D) all take into account interactive effects of multiple carcinogens and anticarcinogens
- E) all take into account differences between route of administration to animals and most likely route of absorption by man

Federal government scientific policy regarding carcinogenesis asserts that

---

- A) no single mathematical procedure is recognized most appropriate for low-dose extrapolation in carcinogenesis
- B) pharmacokinetics and target organ dose are irrelevant in risk assessment
- C) quantification of the various sources of uncertainty involved in cancer risk assessment is not as important as the projection of the risk estimate itself
- D) exposure assessments should be independent of estimates of cancer risk for humans
- E) decisions on the carcinogenicity of chemicals in humans should be based only on data indicative of a positive response

Which of the following forces affecting molecular transfer are common to both intra-capillary and interstitial tissue compartments?

---

- 1) hydrostatic
- 2) osmotic
- 3) diffusion
- 4) orthostatic

Which of the following is/are correct?

---

- 1) oxidation of cytochrome P450 results in "active oxygen"
- 2) "active oxygen" is required for metabolism of all drugs
- 3) cytochrome P450 will complex with carbon monoxide
- 4) cytochrome P450 is required for metabolism of all drugs

## Chemical agents that are bound to plasma proteins

---

- 1) are not filtered through the glomeruli in the kidney
- 2) are usually bound irreversibly
- 3) constantly replenish the free plasma level of the agent
- 4) are thereby inactivated by biotransformation

NADPH (TPNH) is required as a cofactor for

---

- 1) the microsomal pathway of drug oxidation
- 2) conjugation by acetylation
- 3) the microsomal pathway of drug reduction
- 4) oxidation or reduction by liver alcohol dehydrogenase



In the NTP bioassay system the maximum tolerated dose is

---

- 1) the highest dose which can be administered without producing severe weight loss and/or mortality
- 2) used to avoid the effects of metabolic/physiologic overload in the test animals
- 3) specific to the test animal and route of exposure
- 4) defined as the acute oral LD50 dose in the test species

Immature animals are generally more sensitive than adult animals because of

---

- 1) less developed detoxification mechanism
- 2) decreased physical capabilities
- 3) greater body surface area per body mass
- 4) greater relative activity of mixed function oxidases

Which of the following statements are characteristic for both resistance and tolerance?

---

- 1) effects are readily reversible
- 2) result from natural selection processes in the group
- 3) are due to genetic factors (inherited)
- 4) always result in a decrease in the toxic response

## The food additive amendment of 1958 regulates

---

- 1) color additives
- 2) pesticide residues in or on raw agricultural commodities
- 3) substances which had been in use for many years without observed adverse effects much more stringently than before
- 4) pesticide residues in processed foods (e.g. tomato paste containing residues higher than those permitted in tomatoes)

## The color additive amendment of 1960 to the Food, Drug and Cosmetic Act

---

- 1) brought all colors except coal tar colors under the jurisdiction of the law
- 2) required re-evaluation of all colors
- 3) includes the Delaney clause
- 4) allowed the FDA to set limits on the amount of color used

## The biologic half-life

---

- 1) describes the physical half-life of ingested radioisotopes
- 2) is the retention half-time corrected for physical decay
- 3) is always shorter than the physical half-life
- 4) is a relevant parameter for the toxic action of incorporated radioisotopes

Characteristic(s) of a eutrophic lake include(s)

---

- 1) high concentration of oxygen throughout the entire depth of the water
- 2) turbidity
- 3) low amounts of nutrients present
- 4) overgrowth of algae

Which of the following toxicant-sign (symptom) pairs are correct?

---

- 1) Amanita phalloides - diarrhea and delayed jaundice
- 2) cadmium - anosmia, yellow teeth and kidney damage
- 3) kerosene - cough and rales
- 4) phosphate - "lumpy jaw"



The formation of oxygen radicals as a mechanism of action applies to which of the following?

---

- 1) paraquat
- 2) adriamycin
- 3) bleomycin
- 4) nitrofurantoin

Glomerular filtration is the most important mechanism of excretion into the urine for

---

- 1) toxicants with molecular weight less than 60,000
- 2) lipid soluble toxicants
- 3) toxicants not bound to plasma proteins
- 4) organic bases

Haber's rule (dose = conc x time) is most correct in an inhalation system involving

---

- A) extensively metabolized gases or vapors
- B) irritants
- C) smokes, dusts, aerosols
- D) nonmetabolized gases
- E) chlorinated solvents

## Organophosphate induced peripheral neuropathy

---

- A) can be reversed with 2-PAM
- B) is usually first manifest in nerves with the longest axons
- C) can be reversed with atropine
- D) B and C
- E) A and B

Which of the following is most likely to cause allergic contact dermatitis?

---

- A) isopropyl myristate
- B) coumarin
- C) formaldehyde
- D) quaternary ammonium salts
- E) benzyl alcohol

Which of the following is true about methyl salicylate?

---

- A) it is also known as oil of spearmint
- B) emesis or gavage is of no value in poisoning from oral intake
- C) it gives the urine, mouth and vomit a characteristic odor
- D) A and B
- E) all of the above

Characteristic signs of acute morphine poisoning include

---

- A) diarrhea
- B) pinpoint pupils
- C) depressed respiration
- D) A, B and C
- E) B and C

Atropine poisoning may produce all of the following signs or symptoms EXCEPT

---

- A) mydriasis
- B) hypothermia
- C) tachycardia
- D) hallucinations
- E) dry mouth and skin



In a hard rock miner with polyneuritis, exfoliative dermatitis and jaundice, the most likely diagnosis is chronic poisoning by

---

- A) silica
- B) beryllium
- C) fluorides
- D) arsenic
- E) asbestos

Which of the following is capable of causing severe hepato-renal damage?

---

- A) carbon tetrachloride
- B) methylene chloride
- C) toluene
- D) xylenes
- E) all of the above

Toxic effects of which of the following can be prevented by ethanol?

---

- A) methyl alcohol
- B) naphthalene
- C) disulfiram
- D) isopropyl alcohol
- E) sodium fluoroacetate

Which of the following is/are most effective for the treatment of cyanide poisoning?

---

- A) thiosulfate
- B) nitrites and thiosulfate
- C) methylene blue
- D) sodium bicarbonate
- E) oxygen and carbon dioxide

Severe headache is an early clinical sign of

---

- A) lead poisoning
- B) mercury poisoning
- C) chlorinated hydrocarbon insecticide poisoning
- D) organophosphate poisoning
- E) carbon monoxide poisoning

A gas which exerts its toxic effect mainly by irritation of the respiratory tract is

---

- A) carbon monoxide
- B) carbon dioxide
- C) nitrogen dioxide
- D) hydrogen cyanide
- E) helium

In populations, which of the following conditions is likely to be observed first in chronic fluoride poisoning?

---

- A) osteoporosis
- B) osteosclerosis
- C) mottled enamel
- D) osteomalacia
- E) skeletal fluorosis

Which of the following conditions has NOT been attributed to nickel?

---

- A) asthma
- B) lung cancer from nickel carbonyl
- C) nasal sinus cancer from nickel carbonyl
- D) dermatitis
- E) hepatocellular carcinoma



Paralytic shellfish poisoning is caused by ingestion of seafood contaminated with

---

- A) methyl mercury
- B) saxitoxin
- C) tetrodotoxin
- D) botulinum toxin
- E) domoic acid

Aflatoxins are toxic metabolites produced by strains of

---

- A) *Claviceps purpurea*
- B) *Aspergillus flavus*
- C) *Clostridium perfringens*
- D) *Aspergillus niger*
- E) *Fusarium moniliforme*

Which of the following vitamins and vitamin precursors has/ have been most commonly associated with adverse health effects following overdose?

---

- A) A and D
- B) riboflavin
- C) B2 and B6
- D) A, D and C
- E) C, biotin and pantothenic acid

Which of the following common garden plants contain(s) solanine?

---

- A) potato
- B) carrot
- C) turnip
- D) red beet
- E) A and C

Mycotoxin(s) that cause changes in bone marrow, widespread hemorrhage, anemia, granulocytopenia and increased susceptibility to infection in animals include

---

- A) zearalenone
- B) T-2 toxin
- C) citrinin
- D) aflatoxin
- E) fumonisin

The primary source of food contamination with phthalic acid esters is

---

- A) pesticide residues
- B) food processing procedures
- C) food packaging
- D) microorganisms and their metabolites
- E) environmental pollutants

Which of the following causes encephalopathy, peripheral neuropathy and anemia?

---

- A) cadmium
- B) carbon tetrachloride
- C) mercury
- D) vinyl chloride
- E) lead

The chelation agent of choice in the treatment of acute nickel carbonyl poisoning is

---

- A) calcium EDTA
- B) DTPA (diethylenetriamine pentaacetic acid)
- C) desferrioxamine
- D) dithiocarb (diethyldithiocarbamate)
- E) triethylenetetramine dihydrochloride



A metal chelator containing two thiol (SH) groups is

---

- A) dithiocarb
- B) BAL
- C) EDTA
- D) desferrioxamine
- E) penicillamine

Which of the following is/are NOT true of rattlesnake (*Crotalus* sp.) venom?

---

- A) is rich in enzymes
- B) composition may vary by season
- C) causes tissue damage
- D) contains potent neurotoxins
- E) B and C

Which of the following is the primary target organ for the toxicity of amphotericin B?

---

- A) liver
- B) lungs
- C) brain
- D) kidneys
- E) heart

All of the following are seen after the bite of the North American Moccasin snake (*Agkistrodon*) EXCEPT

---

- A) pain at the site of bite
- B) nausea and vomiting
- C) swelling and edema
- D) ecchymosis
- E) muscular weakness or paralysis

The major mode of energy transfer leading to biological damage from ionizing radiation is

---

- A) spallation (breaking apart) of  $^{12}\text{C}$  nuclei
- B) excitation of DNA atomic electrons
- C) electron-positron pair production
- D) electron capture
- E) ion pair production

Survivors of acute high dose whole-body radiation exposures are most likely to develop

---

- A) leukemia
- B) lung cancer
- C) mammary cancer
- D) pernicious anemia
- E) melanoma

Fractional deposition of dust particles in the respiratory tract is LEAST affected by the

---

- A) concentration of the dust particles in the inhaled air
- B) mass median aerodynamic diameter of the dust particles
- C) standard geometric deviation of the dust particles
- D) breathing pattern of the subject
- E) geometry of the respiratory tract

The initial interaction between ionizing radiation and a cell component that results in cell damage or death is between the radiation and

---

- A)  $\text{H}_2\text{O}_2$ , resulting in the generation of free radicals
- B) DNA, resulting in the generation of DNA adducts
- C) DNA, resulting in the induction of thymine dimers
- D) the cell membrane, resulting in changes in ion fluxes
- E) polysomes, resulting in disruption of protein synthesis



Silica is thought to exert its pulmonary toxicity by

---

- A) inhibiting the secretion of surfactant
- B) disrupting mucociliary clearance leading to increased retention of inhaled dusts
- C) rupturing the lysosomes of macrophages leading to cell lysis
- D) producing hypersensitivity pneumonitis
- E) producing byssinosis

After inhalation exposure to an irritating dust, bronchoalveolar lung lavage analysis revealed elevated alkaline phosphatase levels. This effect could be attributed to

---

- A) overt hepatotoxicity
- B) release of lysosomal enzymes
- C) lung edema
- D) lung fibrosis
- E) damage of type I pneumocytes

Inhalation exposure to which of the following has been associated with bronchial asthma?

---

- 1) chlorine
- 2) sulfur dioxides
- 3) toluene diisocyanate
- 4) western red cedar dust

Which of the following pairs correctly associate(s) a chemical with its principal toxic effect?

---

- 1) pentachlorophenol: impaired oxidative phosphorylation
- 2) strychnine: loss of spinal cord reflex inhibition
- 3) bipyridyl herbicides: superoxide formation
- 4) sodium fluoroacetate: blockade of Krebs cycle

## Acetylcholinesterase

---

- 1) occurs in erythrocytes and can be used as a marker of exposure to organophosphates
- 2) can become "aged" after inhibition by both organophosphates and carbamates
- 3) cannot be reactivated by 2-PAM after "aging" of the organophosphate-inhibited enzyme
- 4) is the target for organophosphate and pyrethroid insecticides

The accumulation of cadmium in the kidney and its resulting nephrotoxicity is associated with its induction of which of the following?

---

- 1) ceruloplasmin
- 2) glutathione
- 3) siderophilin
- 4) metallothionein

Which of the following is associated with chronic berylliosis?

---

- 1) extraction and smelting operations
- 2) pharmaceutical applications
- 3) manufacture of fluorescent lamps
- 4) manufacture of pesticides

Metals considered to be essential for humans include

---

- 1) iron, copper, and cobalt
- 2) selenium and molybdenum
- 3) manganese and zinc
- 4) antimony and lithium



## Asbestos

---

- 1) has been associated with the occurrence of pulmonary epidermoid carcinomas in
- 2) injection into the pleural cavities of rats produces mesothelioma
- 3) tumor production is not dependent on fiber length or diameter
- 4) is less active in tumor induction if it is pulverized

Which of the following is/are correct?

---

- 1) the toxicity of gasoline is related mainly to the concentration of benzene
- 2) aspiration pneumonitis is a major danger when gasoline is ingested
- 3) prolonged exposure to gasoline vapors can cause death due to respiratory failure because of severe pulmonary edema
- 4) emetics are contraindicated in the therapy for gasoline ingestion

Digitalis-like cardiotoxicity can be caused by

---

- 1) brown recluse spiders (*Loxosceles reclusa*)
- 2) coral snakes (*Micrurus* sp.)
- 3) puffer fish (*Tetradon Fugu* sp.)
- 4) Bufo toads (*Bufo* sp.)

Which of the following is/are correct about piperonyl butoxide?

---

- 1) it is synergistic with pyrethrins and rotenone
- 2) it has a very low acute oral toxicity in experimental animals
- 3) it is poorly absorbed from the gastrointestinal tract of experimental animals
- 4) it is teratogenic in rats

Which of the following is/are useful in the treatment of ingestion of inorganic mercury salts?

---

- 1) dimercaptosuccinic acid (DMSA)
- 2) 2,3-dimercapto-1-propanesulfonic acid (DMPS)
- 3) N-acetyl-D-penicillamine (NAP)
- 4) EDTA

## Benzene metabolism

---

- 1) does not lead to opening of the aromatic ring
- 2) results only in urinary phenol and catechol excretion
- 3) is affected by urinary pH
- 4) involves cytochrome P-450, epoxide hydratase and glutathione S-transferase

Acute, high level vapor exposure to many different organic solvents produces similar and rapid toxic effects, i.e. disorientation, confusion, paralysis, convulsion and respiratory or cardiac arrest. The similarity of the responses to solvents of diverse structure suggests

---

- 1) solvent vapor toxicity is mediated by hypoxia
- 2) physical interaction of organic solvents with CNS structures
- 3) that different organic solvents are metabolized to similar toxic compounds
- 4) equimolar concentrations of different solvents at the target tissues will produce effects of equal intensity

Insect growth regulators or juvenile hormones are a group of insecticides that

---

- 1) mimic compounds normally secreted by the corpus allatum in the insect brain
- 2) have target systems found in insects and not in mammals
- 3) control polymorphism in some insects
- 4) are neuropeptides



Which of the following are cardiovascular effects produced by barbiturates at toxic doses?

---

- 1) decreased myocardial contractility resulting in lowered cardiac output
- 2) decreased capillary permeability
- 3) peripheral pooling of the blood causing decreased venous return
- 4) Increased blood pressure

Which of the following statements is/are correct?

---

- 1) DDT has a longer biological half-life than methoxychlor in rats and man
- 2) blood DDE/DDT ratios greater than 1.0 indicate recent DDT exposure
- 3) DDT or its metabolites concentrate at the top of ecological food chains
- 4) DDT has the same effect on the nervous system as does nicotine

Exposure to commercial preparations of hydrocarbons, such as gasoline, can lead to

---

- 1) dermatitis
- 2) toxicity from constituents such as benzene
- 3) aspiration pneumonitis
- 4) coma

## The hexachlorocyclohexanes

---

- 1) are used primarily as agricultural defoliants
- 2) form toxic epoxides
- 3) are chemically unstable
- 4) show important steric differences in insecticidal activity

## Placental transfer of a chemical

---

- A) is independent of molecular weight considerations
- B) is facilitated by low lipid solubility
- C) is facilitated by high ionization at blood pH
- D) is facilitated by weak binding to transport plasma proteins
- E) is constant throughout pregnancy for a given species

The production of an irreversible cellular change which is a necessary but not sufficient condition for the development of cancer is called?

---

- A) promotion
- B) initiation
- C) latent period
- D) threshold dose
- E) none of the above

Ethylmethane sulfonate is a classical example of a/an

---

- A) point mutation agent
- B) clastogenic agent
- C) environmental carcinogen
- D) all of the above
- E) none of the above

In predicting fractional (percent) deposition of 100 micrograms of an aerosol in various parts of the airway, the most useful parameter to know would be

---

- A) count median diameter
- B) mass median diameter
- C) count mean diameter
- D) respiratory rate
- E) tidal volume



A toxic agent that affects spermatogenesis after the spermatocyte stage is NOT likely to act by

---

- A) delaying release of sperm (spermiation)
- B) inhibition of DNA synthesis
- C) toxicity to the Sertoli cells
- D) inhibition of nuclear condensation
- E) toxicity to the Leydig cells

Inhalation of which of the following may lead to pulmonary granulomas?

---

- A) lead
- B) mercury
- C) selenium
- D) beryllium
- E) thallium

Cough, fever and blood stained sputum is an expected consequence of industrial exposure to high concentrations of

---

- A) nitrogen dioxide
- B) zinc fumes
- C) cotton dust
- D) sulfur dioxide
- E) all of the above

Aspiration pneumonitis is a common clinical result after ingestion of

---

- A) methanol
- B) kerosene
- C) carbon tetrachloride
- D) carbon disulfide
- E) tetraethyl lead

Oxygen administration would be most helpful in

---

- A) severe anemia
- B) cerebrovascular accident
- C) cyanide poisoning
- D) shock due to severe hemorrhage
- E) carbon monoxide poisoning

Strychnine is an

---

- A) n-methyl-o-aspartate antagonist
- B) inhibitor of cytochrome oxidase
- C) antagonist of gamma-aminobutyric acid mediated inhibition
- D) antagonist of glycine mediated inhibition
- E) excitatory amino acid neurotoxin

Which of the following is an animal carcinogen NOT derived from fungus?

---

- A) aflatoxin B1
- B) sterigmatocystin
- C) griseofulvin
- D) ethionine
- E) ochratoxin

Hemorrhage is a characteristic toxic effect of which of the following?

---

- A) thallium
- B) benzene
- C) gasoline
- D) carbon tetrachloride
- E) warfarin



Which of the following causes proximal tubular necrosis in acute poisoning?

---

- A) carbon monoxide
- B) kerosene
- C) carbon tetrachloride
- D) carbon disulfide
- E) tetraethyl lead

"Ginger jake" paralysis resulting from ingestion of a contaminated ginger extract was caused by

---

- A) deltamethrin
- B) methanol
- C) triorthocresyl phosphate (TOCP)
- D) triethyltin
- E) acrylamide

Which of the following IS LEAST likely cause opacity of the cornea of the eye?

---

- A) caustic material splashed into the eye
- B) exposure to ionizing radiation
- C) exposure to infrared radiation
- D) exposure to ultraviolet radiation
- E) exposure to laser radiation

The term used to describe an increase in cell size is

---

- A) neoplasm
- B) dysplasia
- C) metaplasia
- D) hyperplasia
- E) hypertrophy

Lipofuscin accumulates in liver cells in chronic injury from carbon tetrachloride probably as a result of damage to

---

- A) external cell membranes
- B) mitochondria
- C) lysosomes
- D) endoplasmic reticulum
- E) nuclear membrane

Treatment of a pregnant rat with a testosterone antagonist would be expected to produce

---

- A) masculinized female offspring
- B) sterile female offspring
- C) feminized male offspring
- D) no effect on the offspring since embryos do not make testosterone
- E) no effect on the offspring since pregnant female rats do not make testosterone

The primary site of nephrotoxic action for analgesic mixtures (aspirin and phenacetin) appears to be

---

- A) distal convoluted tubule
- B) loop of Henle
- C) proximal convoluted tubule
- D) collecting duct
- E) glomerulus

What would be an appropriate response if you found a marginal increase in hydrocephaly only at maternally toxic doses in a conventional rat teratology study by giving the chemical on days 6-15 of gestation?

---

- A) repeat the study at the same dose
- B) repeat the study with the same or higher dose but administer on days 8-11 of gestation
- C) conclude the chemical directly induced hydrocephaly
- D) nothing
- E) stop all further studies with this drug



Which animal species is most frequently used to detect allergic contact sensitization (immunologically mediated cutaneous reactions to chemicals)

---

- A) rat
- B) guinea pig
- C) mouse
- D) miniature swine
- E) rabbit

Segment III reproductive toxicology study would assess a compound's potential to

---

- A) affect spermatogenesis
- B) affect late gestation and postnatal development
- C) cause embryotoxicity and teratogenicity
- D) affect early stages of gestation
- E) affect fertility

Exposure to which of the following will cause liver damage through lipid peroxidation?

---

- A) carbon monoxide
- B) natural gas
- C) kerosene
- D) carbon tetrachloride
- E) sulfur dioxide

Which of the following compounds activates one of the receptor types of the primary excitatory neurotransmitter in the CNS?

---

- A) allethrin
- B) pyriithiamine
- C) domoic acid
- D) glycine
- E) hexachlorophene

A patient exhibited respiratory distress and very low blood pressure shortly after being stung by a wasp. These were probably symptoms of

---

- A) IgE mediated anaphylaxis
- B) cardiotoxic properties of wasp venom
- C) IgM mediated anaphylaxis
- D) cell-mediated hypersensitivity
- E) delayed hypersensitivity

The criteria that are used for identifying a new teratogenic agent in humans include

---

- A) an abrupt increase in the frequency of a particular defect or association of defects (syndrome)
- B) coincidence of a defect with known exposure to an environmental change
- C) known exposure to an environmental change early in pregnancies yielding characteristically defective infants
- D) absence of other factors common to all pregnancies yielding infants with the characteristic defect(s)
- E) all of the above

Which of the following is characteristic of "altered foci" in rat liver?

---

- A) decrease in gamma-glutamyl transpeptidase activity
- B) increased glycogen storage
- C) iron exclusion after iron loading
- D) increase in glucose-6-phosphatase activity
- E) always progress to neoplasia

The postural hypotension caused by alpha adrenergic blockers, norepinephrine depleting agents, and ganglionic blockers is due to

---

- A) a direct vasodilation of blood vessels in the lower extremities
- B) insufficient cardiac output
- C) interruption of reflex sympathetic tone to the blood vessels of the lower extremities
- D) an increase in reflex parasympathetic tone of blood vessels in the lower extremities
- E) all of the above



Which of the following is/are human syndrome(s) characterized by monosomy?

---

- A) Down's syndrome
- B) Fanconi's syndrome
- C) xeroderma pigmentosum
- D) Turner's syndrome
- E) all of the above

Polycythemia has been observed in heavy drinkers of cobalt-contaminated beer because

---

- A) cobalt salts increase production of renal erythropoietic factor
- B) cobalt stimulates the adrenal medulla
- C) bivalent cobalt binds with an inhibiting erythrolysin
- D) B and C
- E) all of the above

At the end of a 90 day study in rats renal function tests showed no marked changes; however, histopathologic exam of kidney demonstrated evidence of structural changes. This is because

---

- A) the kidney is a dynamic organ and renal function tests are highly variable
- B) the histopathologic studies provide little information about renal integrity
- C) the kidney has the ability to compensate and regain normal function
- D) most nephrotoxics have their primary site of action only on the distal tubule
- E) collecting ducts are sensitive to nephrotoxics

Which of the following does NOT occur after the administration of acetazolamide?

---

- A) urine volume increases
- B) urine pH will become acidic
- C) urine concentrations of bicarbonate, sodium and potassium ions increase
- D) urinary chloride and ammonia concentrations will fall
- E) urinary protein concentration remains the same

Which of the following best characterizes chloracne?

---

- A) a sebaceous gland that becomes a keratinous cyst
- B) a sweat gland that becomes plugged with sebum
- C) an inflamed hair follicle
- D) a cutaneous granuloma
- E) allergic contact dermatitis

The physiological target within the eye for methanol toxicity is most likely

---

- A) rod cells
- B) cone cells
- C) all photoreceptor cells
- D) ganglion cells
- E) amacrine cells

What are the most likely detected adverse reproductive outcomes of human exposure to environmental toxicants?

---

- A) infertility in the male and abortion in the female
- B) Sertoli cell tumors in the male and premature labor in the female
- C) Leydig cell tumors in the male and intrauterine fetal growth retardation
- D) altered libido in both males and females
- E) abnormal sperm production and dysmenorrhea

Carcinogens that interact with and alter DNA are referred to as genotoxic or DNA reactive. Which of the following is considered non-genotoxic?

---

- A) benzidine
- B) phenobarbital
- C) aflatoxin
- D) 5-azacytidine
- E) nickel



Hepatic peroxisome proliferation in the rat has been produced by exposure to

---

- A) clofibrate
- B) diethylhexylphthalate
- C) trichloroethylene
- D) all of the above
- E) none of the above

The period of greatest teratogenic risk in humans (expressed in days after conception) is

---

- A) 44 to 80
- B) 20 to 56
- C) 38 to 74
- D) 29 to 63
- E) none of the above

The increase in teratogenicity of salicylate when coadministered with sodium benzoate in the rat is directly related to

---

- A) increased fetal blood volume
- B) decreased metabolism of salicylate
- C) decreased plasma protein binding of salicylate
- D) increased maternal blood volume
- E) increased excretion of water-soluble metabolites

## In hemolytic jaundice

---

- A) hepatic cells cannot excrete bilirubin as rapidly as it is formed and plasma concentration of free bilirubin rises
- B) obstruction of the bile duct occurs but the rate of bilirubin formation is normal
- C) skin appears yellow due to large quantities of bilirubin
- D) A and C
- E) B and C

Which of the following agents damage myelin-forming cells in the central and peripheral nervous system?

---

- 1) isoniazid
- 2) hexachlorophene
- 3) lead
- 4) taxol

Which of the following apply to the transport of substances across the skin?

---

- 1) all toxicants which move across the stratum corneum do so by passive diffusion and not by active transport
- 2) nonpolar solvents can diffuse through the lipid matrix between the protein filaments
- 3) polar substances can diffuse through the outer surface of protein filaments of the hydrated stratum corneum
- 4) the epidermis does not afford a significant barrier to the diffusion of toxicants

Which of the following describe(s) why the liver is a target organ for toxic substances?

---

- 1) the diverse metabolic capabilities of the liver enable it to metabolize many xenobiotics, but metabolism does not always result in detoxication
- 2) the liver plays an extensive role in intermediary metabolism and molecular synthesis, consequently interference with metabolic pathways may lead to toxic effects
- 3) the liver is exposed to relatively high concentrations of toxic substances absorbed from the GI tract
- 4) enterohepatic circulation can give rise to prolonged high concentrations of a toxic substance in the liver

Atopic diseases occur in persons with an inherited sensitivity to allergens. This can be diagnosed by

---

- 1) a careful review of symptoms in relation to the environment
- 2) direct skin (scratch) tests
- 3) indirect (passive transfer) skin test
- 4) provocative tests with the suspect antigen



Immunotoxic effects include

---

- 1) immunosuppression
- 2) alteration of host defense mechanisms
- 3) hypersensitivity or allergic reactions
- 4) decrease in killer cell count

Renal tubular damage has been documented to be the result of exposure to which of the following anesthetic agents?

---

- 1) procaine
- 2) ether
- 3) thiopental sodium
- 4) methoxyflurane

Referring to carcinogenicity, promotion is

---

- 1) rapid
- 2) slow
- 3) irreversible
- 4) reversible

Peripheral axonopathies may be caused as a result of overexposure to which of the following agents?

---

- 1) acrylamide
- 2) taxol
- 3) ethyl alcohol
- 4) 3,3'-iminodipropionitrile

The neutropenia seen most commonly following the administration of cancer chemotherapeutic agents arises from

---

- 1) hepatocellular damage
- 2) interruption of DNA synthesis
- 3) stimulation of erythropoietin release
- 4) inhibition of mitosis

Which of the following is no longer recommended for treating severe barbiturate poisoning?

---

- A) artificial respiration, oxygen
- B) parenteral fluids
- C) clear air way
- D) administer stimulants to counteract CNS depression
- E) maintain blood pressure, if necessary, with plasma expanders

Which of the following is NOT true for asbestos?

---

- A) industrial exposure is associated with asbestosis, cancer of the lung and mesotheliomas
- B) major clinical signs include diminished breath sounds, basilar crepitations, limited chest expansion and clubbing
- C) chrysotile fibers are biologically more active than either amosite or crocidolite
- D) asbestos exposure acts synergistically with cigarette smoking to cause lung cancer
- E) asbestos is not readily metabolized or excreted

An epileptic taking phenobarbital and diphenylhydantoin takes 16 gm of acetaminophen. You would expect the patient to

---

- A) recover relatively fast due to increased detoxication of acetaminophen
- B) show enhanced acetaminophen-induced hepatotoxicity
- C) show elevated plasma acetaminophen levels, but no liver failure
- D) excrete acetaminophen in the urine predominantly in the sulfate form
- E) show symptoms of barbiturate intoxication



To maintain a constant dose rate in a feeding study on a mg/kg/per day basis, the experiment must

---

- A) maintain a constant % composition of the test material in the feed
- B) maintain a constant ppm level of the test material in the feed
- C) periodically adjust the composition of the feed to maintain a constant intake of test material based on food consumption and rate of change in body weight
- D) limit the amount of treated feed available to keep food consumption at a constant level
- E) periodically adjust the composition of the feed to maintain a constant intake of test material based on food consumption

The majority of additives in the United States covered by the Food Additive Amendment are

---

- A) food colors
- B) synthetic and natural flavoring agents
- C) sweetening agents, such as, sucrose and corn syrup
- D) texturizing agents
- E) chelating agents

Which of the following species would be most likely to accumulate the highest levels of polychlorinated biphenyls?

---

- A) polychaete worms
- B) daphnids
- C) pelicans
- D) minnows
- E) catfish

The most common method of generating regression equations is the method of least squares. In this method, a set of differences is calculated, each difference is squared, and the solution is the equation which minimizes the sum of the differences. Which differences are used in the method of least squares when  $x$  is the independent variable?

---

- A) the differences between the observed  $y$  values when the  $x$  values have been put in ascending order
- B) the differences between the observed  $y$  values and the regression equation  $y$  values for given  $x$  values
- C) the differences between the observed  $y$  values and the regression equation  $x$  values for a given  $y$  value
- D) the differences between the observed  $x$  values and the regression equation  $x$  values for a given  $y$  value

E) the minimum distances between the observed  $y$  values and the regression line

The probit plot is most accurate for estimating the

---

- A) ED1
- B) ED50
- C) ED99
- D) no observed adverse effect level (NOAEL)
- E) threshold limit value (TLV) or maximum acceptable concentration (MAC)

The mean concentration of an airborne chemical that should not be exceeded during an eight-hour work day and 40-hour work week without risk of harm to nearly all workers is referred to as

---

- A) reference concentration (RfC )
- B) short term exposure limit (STE)
- C) acceptable daily intake (ADI)
- D) LC50
- E) threshold limit value/time weighted average (TLV / TWA)

Toxic effects from chronic exposure most frequently result when

---

- A) exposure to the agent is sudden and severe and absorption is rapid
- B) the agent is conjugated in the liver and excreted in the bile
- C) the agent is absorbed through the skin
- D) the rate of input of the agent exceeds the excretion rate
- E) the agent is metabolized to a more toxic agent



In the United States, which of the following types of new drugs would be required to receive the most extensive testing for toxic potential in animals before being submitted for clinical trial?

---

- A) a treatment for leukemia
- B) a radioactive preparation
- C) an antibiotic for gram negative organisms
- D) a non-prescription sedative
- E) a treatment for Parkinsonism

Disulfiram is used to block

---

- A) L-aromatic amino acid decarboxylase
- B) the oxidation of acetaldehyde
- C) the oxidation of formaldehyde
- D) alcohol dehydrogenase
- E) acetylcholinesterase

Inhibition of cytochrome oxidase is a major cause of toxicity of

---

- A) carbon tetrachloride
- B) methanol
- C) ethylene glycol
- D) cyanide
- E) warfarin

Sodium fluoroacetate and sodium fluoracetamide are powerful rodenticides which act by inhibiting

---

- A) the pentose shunt
- B) the citric acid cycle
- C) conversion of ADP to ATP
- D) conversion of  $\text{Fe}^{++}$  to  $\text{Fe}^{+++}$
- E) glycolysis

In a group of 200 people there were 10 new cases of gonorrhea per month. The incidence rate for gonorrhea in this group would be

---

- A) 0.05
- B) 2200/1000/yr
- C) 600/1000/yr
- D) 50/1000/yr
- E) not calculatable from the information given

In a carefully designed clinical trial, control groups which are necessary for valid conclusions

---

- A) can include only those patients who have not been treated for the disease under study
- B) can include placebo controls or the standard treatment for the disease
- C) will have their data normalized to remove potential bias
- D) require different criteria for evaluation than the treatment groups
- E) are unethical because patients must be treated with drugs when they consult a physician

Which of the following is NOT true for methemoglobinemia?

---

- A) may be endogenous or exogenous in origin
- B) under normal conditions, humans will have about 2% of their total hemoglobin in the methemoglobin form
- C) congenital erythrocyte abnormalities may result in steady-state levels of methemoglobin as high as 30-50%
- D) the condition results when conversion of ferrous to ferric hemoglobin occurs at a rate which exceeds the methemoglobin reducing capacity of the cell
- E) the enzyme system responsible for reducing >90% of the methemoglobin produced naturally is the NADPH methemoglobin reductase/P450 system

Sodium phosphate pollution of water results in

---

- A) adverse effects on fish schooling behavior
- B) increased occurrence of fish tumors
- C) loss of fecundity in fish populations
- D) teratogenicity in fish
- E) eutrophication



A target organ is an organ that

---

- A) first reacts to a chemical
- B) is first reached by a chemical
- C) sustains major toxicity
- D) accumulates the highest concentration of the chemical

The most likely source of interindividual variability in a species is

---

- A) genetic factors
- B) body weight and surface area
- C) geographical and environmental factors
- D) immunologic competence
- E) diet

According to the FDA GLP's published in 1978, the final report of a preclinical study must contain all of the following EXCEPT?

---

- A) name and address of the facility performing the work
- B) statistical methods used to analyze the data
- C) age and sex of each investigator involved in the study
- D) name, chemical abstract number or code number, purity, strength and composition of the test agent
- E) a description of all circumstances that may have affected the quality and integrity of the study

Threshold limit values (TLV) are

---

- A) normally expressed both as ppm and mg/m<sup>3</sup>
- B) confined to single chemical entities
- C) based on continuous exposure for a lifetime
- D) always based on toxicity
- E) all of the above

The Federal Hazardous Substances Act defines as corrosive

---

- A) any substance which evokes severe inflammatory response on application following initial exposure
- B) any substance that in contact with living tissue will cause destruction of tissue by chemical action
- C) any substance which on prolonged exposure causes inflammatory response
- D) any substance that will cause destruction of any non-keratinized epithelial tissues
- E) none of the above

The half-life of a chemical with first-order elimination characteristics is

---

- A) dependent upon dose
- B) dependent upon concentration
- C) independent of the dose
- D) proportional to the dose
- E) none of the above

Which of the following statements concerning cadmium toxicity is NOT true?

---

- A) occupational exposure may occur during welding
- B) the lungs readily absorb cadmium fumes
- C) tissue concentration of cadmium is a good indicator of chronic low-dose exposure
- D) there are several stages of cadmium poisoning, the most advanced being a very painful condition characterized by bone deformity (as in Itai-Itai disease of Japan)
- E) occupational cadmium exposure is not exacerbated by cigarette smoke

If the same contaminated diet were eaten by the following species, which species would receive a larger dose per unit body weight?

---

- A) mice
- B) rats
- C) rhesus monkeys
- D) dogs
- E) humans



Which of the following United States agencies require confirmation of the safety of a new product before marketing can be done?

---

- A) Environmental Protection Agency (EPA) and Occupational Safety and Health Administration (OSHA)
- B) OSHA and Consumer Product Safety Commission (CPSC)
- C) CPSC and Food and Drug Administration (FDA)
- D) FDA and EPA
- E) OSHA and FDA

Resistance to the pharmacologic effects of some drugs can be rapidly acquired following only a few doses of the agent over a period of hours. Subsequent doses of the agent fail to elicit the response regardless of the size of the dose. This phenomenon is known as

---

- A) tolerance
- B) idiosyncrasy
- C) physical dependence
- D) tachyphylaxis
- E) none of the above

Glucuronosyl transferase(s) is/are located predominantly in

---

- A) cytosol
- B) mitochondria
- C) endoplasmic reticulum
- D) peroxisomes
- E) lysosomes

The primary type of tumor observed in workers occupationally exposed to vinyl chloride was

---

- A) squamous cell carcinoma of the urinary bladder
- B) hepatocellular carcinoma
- C) neuroglioma
- D) angiosarcoma of the liver
- E) carcinoma of the lung

The occupational toxicologist should

---

- A) use a standard battery of tests for all chemicals
- B) use professional judgment and experience in evaluating a chemical
- C) consider conditions of use of the chemical
- D) B and C
- E) all of the above

A 3 year old child is found sitting in the middle of several house plants with leaves scattered everywhere. The child's lips and mouth are swollen. He has difficulty in talking and swallowing, and his respiration is labored. It is likely he has ingested

---

- A) oleander
- B) cyclamen
- C) rhododendron
- D) Dieffenbachia
- E) ficus

Some toxicants achieve their highest concentration at their site of toxic action. An example is

---

- A) paraquat in lung
- B) lead in bone
- C) chlorinated hydrocarbons in fat
- D) carbon monoxide in cerebellum
- E) TCDD in skin

Which of the following is NOT a correct pairing of toxic exposure with high risk groups?

---

- A) arsenic/individuals with high exposure to sunlight
- B) lead/children
- C) nitrate/infants
- D) asbestos/cigarette smokers
- E) A and C



Which of the following pesticide classes is most likely to leave persistent residues in soil?

---

- A) amide herbicides
- B) organophosphate insecticides
- C) chlorinated hydrocarbon insecticides
- D) nitrite herbicides
- E) carbamate insecticides

In an experimental study, subchronic inhalation exposure of rats to aniline did not result in statistically significant methemoglobin (MetHb) formation. Blood was sampled during sacrifice one day after the last exposure. Which of the following is the most likely reason that MetHb was NOT detected?

---

- A) aniline is not known to induce MetHb
- B) induction/adaptation of metabolism prevents MetHb formation
- C) the time of blood sampling is inappropriate
- D) the rat is an inappropriate species to test for MetHb formation
- E) the hematological laboratory made an error

Under conditions of first-order kinetics, repeated exposure to a toxicant will result in its accumulation until the rate of elimination equals the rate of absorption. On the average then, the whole body burden of a toxicant will be roughly equal to how many times the amount absorbed per half-life of elimination?

---

- A) 0.5
- B) 1.25
- C) 2.5
- D) 0.693
- E) 1.5

All of the following statements concerning the biologic assessment of exposure are true EXCEPT

---

- A) biologic monitoring of exposure includes an estimation of the internal dose
- B) monitoring the airborne concentration of an industrial pollutant provides a reliable estimate of the amount absorbed by a worker
- C) the concentration of polychlorinated biphenyls in blood is proportional to the amount accumulated in fatty tissue
- D) biologic monitoring of exposure is of practical value only when relationships between external exposure, internal dose and adverse effects are known

The addition of rabbits to the preclinical testing of drugs for teratogenicity was initiated due to

---

- A) retinopathies associated with some antimalarials early in the century
- B) birth defects associated with thalidomide
- C) early studies of the "internal" effects of radioactivity
- D) sulfanilamide research and passage of the Copeland bill in 1938
- E) none of the above

Which of the following chemical characteristics would NOT be associated with a tendency to be bioaccumulated?

---

- A) high lipophilicity
- B) high volatility
- C) low polarity
- D) high chemical stability
- E) halogen containing

## The "de minimus" concept

---

- A) means that the calculated risks are probably not real
- B) refers to a risk that represents  $1 \times 10^6$  excess cancers per lifetime
- C) refers to a risk that represents  $1 \times 10^6$  excess cancers per year
- D) refers to a risk that is small enough to be considered trivial and not worth regulating
- E) refers to any activity that results in less than one death per year

Which of the following cytochrome P450 isoforms is induced by polycyclic aromatic hydrocarbons?

---

- A) 1A1
- B) 2B1
- C) 2E1
- D) 2CII
- E) 4A1



Which of the following enzymes is NOT involved in mercapturic acid formation?

---

- A) acetylase
- B) epoxide hydrolase
- C) glutathione transferase
- D) glycine
- E) gamma-glutamyl transpeptidase

## The Monte Carlo procedure

---

- 1) is a statistical method used to distribute animals among dose groups and eliminate bias
- 2) results in a probability distribution description of risk
- 3) is most appropriate when group sizes are large, such as those found in carcinogenicity studies
- 4) is a statistical method which randomly selects values for each of many variables used as input for risk calculations

Contamination of aquatic environments by polycyclic aromatic hydrocarbons is of concern because

---

- 1) they are bioaccumulated by aquatic organisms from water, food and sediments
- 2) they are direct-acting carcinogens in several aquatic organisms
- 3) they have relatively long environmental half-lives
- 4) they are readily soluble in water

Emetics are contraindicated in cases in which the patient

---

- 1) has ingested a strong acid
- 2) has ingested a cleaning fluid
- 3) is in a coma or is semiconscious
- 4) has ingested a strong alkali

Which of the following are symptoms?

---

- 1) vertigo
- 2) paraesthesias
- 3) pruritis
- 4) fasciculations

Exposure of humans to airborne particles occurs via the

---

- 1) skin
- 2) mucous membranes
- 3) lungs
- 4) g.i. tract

## Probit analysis

---

- 1) adjusts a quantal response to an assumed normal distribution
- 2) can be used to calculate therapeutic indices
- 3) can be used to calculate LD50 value
- 4) requires that the dose be expressed logarithmically

To evaluate the environmental fate and transport of a chemical, which of the following is/are important considerations?

---

- 1) water solubility
- 2) biodegradation
- 3) sorption to soil
- 4) toxicity to aquatic invertebrates



Which of the following is NOT correct regarding cadmium toxicity?

---

- A) major sources of dietary cadmium include shellfish, such as mussels, scallops and oysters
- B) gastrointestinal absorption of cadmium is greater than respiratory absorption
- C) cadmium forms stable complexes with metallothionein, which can accumulate in the kidney to fairly high levels without toxic effect
- D) respiratory exposure to cadmium may result in obstructive lung disease
- E) renal tubular dysfunction resulting from chronic cadmium exposure appears to be irreversible

The grayanotoxins found in many species of *Rhododendron* produce effects similar to

---

- A) cyanogenic glycosides
- B) veratrum alkaloids
- C) lycopenes
- D) disulfiram
- E) taxol

Which of the following best summarizes lead poisoning and appropriate laboratory tests in addition to blood level determination?

---

- A) cough, substernal pain, dyspnea, pulmonary edema, tachycardia, tachypnea: check for lymphocytosis
- B) headache, dizziness, nausea, blurred vision, weakness, thready pulse, tachycardia, and cherry-red discoloration of mucous membranes: complete blood count (CBC)
- C) headache, dizziness, anorexia, muscle cramps, purpura and bleeding from gums: CBC and urine sulfate ratio
- D) headache, vertigo, weakness, unsteady gait, cyanosis, tremors and brown discoloration of urine: CBC and methemoglobin
- E) metallic taste, anorexia, constipation, irritability, pallor, tremors, abdominal tenderness: check for reticulocytes and look for increased porphyrins

Which one of the following is always contraindicated for treatment of poisoning by chlorinated pesticides?

---

- A) artificial respiration
- B) cleanse skin, remove from further exposure
- C) lavage
- D) epinephrine
- E) barbiturates for convulsions

Characteristic clinical symptoms of moderate organophosphate intoxication can include all of the following EXCEPT

---

- A) nausea, vomiting, diarrhea, urinary frequency
- B) salivation, diffuse sweating
- C) bronchoconstriction with respiratory impairment
- D) mydriasis
- E) skeletal muscular fasciculations and twitching

## Carbon monoxide

---

- A) is detoxified by biotransformation to formate
- B) denatures hemoglobin
- C) results from incomplete oxidation of carbon compounds
- D) combines irreversibly with iron bearing pigments
- E) has a maximum air concentration limited by its vapor pressure

## Organophosphate induced peripheral neuropathy

---

- A) can be reversed with 2-PAM
- B) is usually first manifest in nerves with the longest axons
- C) can be reversed with atropine
- D) B and C
- E) A and B

If sodium bicarbonate is given following phenobarbital ingestion, the excretion of phenobarbital is affected because the

---

- A) acid urine decreases amount of ionized drug
- B) acid urine increases amount of ionized drug
- C) alkaline urine increases amount of ionized drug
- D) alkaline urine decreases amount of ionized drug
- E) alkaline urine increases glomerular filtration rate



Symptoms of chronic intoxication with mercury vapor include

---

- A) colic, constipation, increased excretion of delta aminolevulinic acid, stippled erythrocytes
- B) emphysema, proteinuria, disturbance of calcium metabolism
- C) tremors, stomatitis, proteinuria, erethism
- D) gastrointestinal bleeding, anemia, convulsions
- E) emphysema, alopecia, wrist-drop (peripheral neuropathy)

Dimercaprol (BAL) is an effective antidote for all the following  
EXCEPT

---

- A) Paris Green (copper-aceto-arsenite)
- B) corrosive sublimate (mercuric chloride)
- C) arsenic trioxide
- D) sodium fluoride
- E) tartar emetic (antimony potassium tartrate)

Which of the following is/are capable of causing severe hepato-renal damage?

---

- A) carbon tetrachloride
- B) methylene chloride
- C) toluene
- D) xylenes
- E) A and B

All of the following compounds may produce sensitization of the myocardium to epinephrine EXCEPT

---

- A) carbon tetrachloride
- B) DDT (dichlorodiphenyltrichloroethane)
- C) gasoline
- D) chloroform
- E) carbon disulfide

The agent used to treat accidental poisoning with parathion is

---

- A) copper sulfate
- B) dimercaprol
- C) amobarbital
- D) atropine
- E) glycerol monacetin

## Pralidoxime (2-PAM)

---

- A) competes with atropine at muscarinic receptor sites
- B) enhances the hydrolysis of phosphorylated acetylcholinesterases
- C) is effective in reducing the severity of atropine poisoning
- D) is an effective antidote for all anticholinesterase insecticides
- E) B and D

The principal toxic effect observed in mammals exposed to excessive amounts of DDT is

---

- A) central nervous system stimulation leading to convulsion
- B) voluntary muscle paralysis
- C) bronchial constriction
- D) hepatocellular damage
- E) lowering of blood cholinesterase

Chlorine as a gas causes

---

- A) simple asphyxiation
- B) methemoglobinemia
- C) carboxyhemoglobinemia
- D) cholestasis
- E) pulmonary edema



Which of the following is NOT produced by botulinum toxin?

---

- A) paralysis of skeletal muscles
- B) involvement of the third cranial nerve with resultant diplopia and mydriasis
- C) dysphagia
- D) blockade of acetylcholine synthesis
- E) blockade of transmission of nerve impulses to the motor end plate

## Tetrodotoxin

---

- A) blocks sodium channels in electrically excitable tissues
- B) exerts a curare-like blockade at the neuromuscular junction
- C) destroys presynaptic nerve endings after an initial phase of transmitter release
- D) increases the  $\text{Na}^+$  permeability of electrically excitable membranes
- E) prolongs repolarization by extending the duration of  $\text{Na}^+$  permeability during the nerve action potential

Absorption of a drug into the circulation is usually most rapid following which type of administration?

---

- A) oral
- B) rectal
- C) subcutaneous
- D) intraperitoneal
- E) inhalation

Shaver's disease is a lung disease attributed to

---

- A) aluminum
- B) beryllium
- C) cadmium
- D) mercury
- E) lead

Which of the following antagonizes the action of vitamin K?

---

- A) sodium fluoroacetate
- B) alpha naphthylthiourea (ANTU)
- C) warfarin
- D) red squill
- E) digitalis

Anaphylaxis can be a major problem caused by the bite or sting of which of the following arthropods?

---

- A) fire ant (*Solenopsis richteri*)
- B) brown recluse spider (*Loxosceles reclusa*)
- C) black widow spider (*Latrodectus* sp.)
- D) honey bee (*Apis mellifera*)
- E) A and D

Which of the following conditions has NOT been attributed to nickel or nickel compounds?

---

- A) asthma
- B) lung cancer
- C) nasal sinus cancer
- D) dermatitis
- E) liver cancer

Itai-Itai disease in Japan was caused by

---

- A) contamination of water, soil, and diet with methyl mercury
- B) contamination of water, soil, and diet with cadmium
- C) dietary calcium deficiency
- D) lead-contaminated drinking water
- E) contamination of the diet with PCB's



Nitrosamines are LEAST likely to be found in

---

- A) smoked fish
- B) fried bacon
- C) hard boiled eggs
- D) hot dogs
- E) tobacco smoke

A mycotoxin that cause widespread hemorrhage, anemia, granulocytopenia and increased susceptibility to infection in animals is

---

- A) zearalenone
- B) T-2 toxin
- C) citrinin
- D) aflatoxin
- E) fumonisin

## Chronic beryllium disease

---

- A) frequently occurs years after exposure
- B) is prolonged in duration with no evidence of lasting cure
- C) is commonly progressive, especially if untreated
- D) may affect organs other than the lungs
- E) all of the above

## Polychlorinated biphenyls

---

- A) are generated by burning dioxins
- B) are found in the blood only after acute exposure
- C) exhibit marked differences in toxicity among isomers
- D) accumulate in fruit
- E) are not biodegraded

The clinical presentation associated with the ingestion of kerosene is usually

---

- A) depression of the central nervous system
- B) damage of the liver
- C) hydrocarbon pneumonitis
- D) ventricular arrhythmia
- E) renal damage

Which of the following does NOT result from chronic exposure to benzene?

---

- A) leukemia
- B) aplastic anemia
- C) uremia
- D) immunosuppression
- E) thrombocytopenia

The use of ethanol as a therapeutic agent in methanol intoxication is based upon the

---

- A) ability of ethanol to competitively block methanol penetration of the optic nerve
- B) ability of ethanol to increase the rate of excretion of oxylate
- C) competition with methanol as a substrate for alcohol dehydrogenase
- D) inhibition of aldehyde dehydrogenase conversion of glycoaldehyde to glycolic acid by ethanol
- E) ability of ethanol to increase the rate of methanol metabolism

The most commonly reported serious toxic side effect of non-steroidal anti-inflammatory drugs (aspirin, ibuprofen, naprosyn, etc.) is

---

- A) abnormal liver function
- B) gastric or duodenal ulcer
- C) acute interstitial nephritis
- D) photosensitivity
- E) tinnitus



The most important factor in the fractional absorption of solvents during inhalation exposure is the

---

- A) pulmonary ventilation rate
- B) diffusibility of the solvent through the alveolar - capillary membranes
- C) solubility of the solvent in blood and tissues
- D) solvent concentration gradient between inspired air and blood
- E) pulmonary perfusion

Fractional deposition of dust particles in the respiratory tract is LEAST affected by the

---

- A) concentration of the dust particles in the inhaled air
- B) mass median aerodynamic diameter of the dust particles
- C) standard geometric deviation of the dust particles
- D) breathing pattern of the subject
- E) geometry of the respiratory tract

Which of the following statements concerning tin is correct?

---

- A) most soluble and insoluble tin salts are well absorbed following oral administration
- B) tin is an essential element in humans
- C) trialkyl tins are the least toxic organotin compounds
- D) the major target site for triethyl tin is the CNS
- E) dietary tin increases gastrointestinal absorption of calcium and iron

Hydrogen chloride is

---

- 1) water-soluble so that, upon inhalation, harmful effects tend to be limited to the upper respiratory tract
- 2) often the cause of acute bronchitis, dyspnea and pulmonary edema upon exposure
- 3) painfully irritating so that exposed persons to escape exposure
- 4) a strong oxidizing agent

Which of the following statements is/are correct?

---

- 1) the concentration of ethanol in the breath is about 0.05% of the blood concentration
- 2) the oxidation products of methanol are more toxic to retina than methanol itself
- 3) pancreatitis and fatty liver are frequent complications in alcoholism
- 4) cross tolerance occurs between alcohol and barbiturates

Chelating agents for treating acute lead poisoning (in terms of both efficacy and toxicity) is / are?

---

- 1) dimercaprol (BAL)
- 2) dimercaptosuccinic acid (DMSA)
- 3) ethylenediaminetetraacetic acid (EDTA)
- 4) calcium disodium edetate

## The "therapeutic index"

---

- 1) gives an accurate indication of the safety of a drug
- 2) must be greater than 3 for a compound to be therapeutically useful
- 3) is usually presented with the clinical pharmacology of each new drug
- 4) requires data from two dose-response curves for its calculation

Lead contamination of food has occurred from

---

- 1) improperly glazed earthenware
- 2) colored decals on glasses
- 3) solder of tin cans
- 4) fertilizers



Food contains mutagens from

---

- 1) natural substances
- 2) synthetic compounds
- 3) processing of food
- 4) irradiation of food

Which of the following is/are difference(s) in the chronic toxicity of inhaled elemental mercury and the toxicity of ingested mercurous salts?

---

- 1) elemental mercury causes tremors, gingivitis, and increased excitability, whereas mercurous salts cause acrodynia, fever and increased activity of sweat glands
- 2) mercurous salts produce renal toxicity similar to mercuric salts whereas elemental mercury does not produce any renal toxicity
- 3) hypersensitivity reactions are more commonly associated with mercurous compounds than with mercury vapor
- 4) both compounds produce very similar chronic toxicity syndromes

## Asbestos

---

- 1) has been associated with the occurrence of bronchogenic carcinomas in humans
- 2) injection into the pleural cavities of rats produces mesothelioma
- 3) is less active in tumor induction if it is pulverized
- 4) tumor production is not dependent on fiber length or diameter

A farm worker has accidentally ingested the herbicide dinitroorthocresol. Following gastric lavage to remove any remaining herbicide from the stomach, the patient's therapy should include

---

- 1) fluid and electrolyte therapy to replace loss due to excess sweating
- 2) administration of ice baths to control fever
- 3) administration of oxygen to insure maximal oxygenation of the blood
- 4) administration of atropine sulfate to control excess exocrine gland secretion and gastrointestinal distress

Which of the following statements about radiation is/are true?

---

- 1) ionizing radiation gives rise, directly or indirectly, to chemical ionizations
- 2) ionizing radiation can produce chromosome aberrations in both somatic and male and female germ cells
- 3) the response of individual cells and types of cells to radiation is variable depending on the cell cycle and the oxygenation status of the cell
- 4) cells that frequently undergo mitosis are the least sensitive to radiation

## Benzene metabolism

---

- 1) does not lead to opening of the aromatic ring
- 2) results only in urinary phenol and catechol excretion
- 3) is affected by urinary pH
- 4) involves cytochrome P450, epoxide hydratase and glutathione S-epoxide transferase

## Respirable insoluble dusts

---

- 1) are toxicologically unimportant
- 2) can produce "lung overload" phenomenon
- 3) do not cause pulmonary inflammation
- 4) may cause pulmonary tumors

## Saxitoxin

---

- 1) produces paresthesias
- 2) is destroyed by cooking
- 3) is rapidly absorbed from the gastrointestinal tract and symptoms appear in minutes
- 4) enhances muscular strength



A rat with a respiratory rate of 100 breaths/min and a tidal volume of 1.3 ml has a minute volume equal to which of the following?

---

- A) 1.3 liters/min.
- B) 0.013 liter/min.
- C) 0.78 liters/min.
- D) 0.13 liters/min.
- E) none of the above

Which of the following species has the highest level of methemoglobin reductase activity?

---

- A) monkey
- B) dog
- C) cat
- D) rabbit
- E) mouse

Which of the following is associated with teratogenesis but not mutagenesis?

---

- A) radiation
- B) cell death
- C) inhibition of DNA synthesis
- D) alteration of the structure of DNA
- E) viral infection

Which of the following is the most sensitive test for hepatocellular necrosis?

---

- A) serum alkaline phosphatase activity
- B) complete blood count
- C) serum alanine aminotransferase activity
- D) serum bilirubin level
- E) blood urea nitrogen level

Which of the following statements is/are correct regarding tumor suppressor genes?

---

- A) these are mutated protooncogenes
- B) normal function of these genes is to inhibit cell proliferation
- C) these genes act in a dormant fashion
- D) these genes were discovered in experiments showing that cells exhibiting a normal phenotype arose from fusion of normal cells with tumor cells
- E) B and D

The state in which most chemicals cross the placenta is

---

- A) in the protein bound form
- B) as a free chemical
- C) as a metabolite
- D) in the ionized form
- E) in the conjugated form

A toxic agent that affects spermatogenesis after the spermatocyte stage is NOT likely to act by

---

- A) delaying release of sperm (spermiation)
- B) inhibition of DNA synthesis
- C) toxicity to the Sertoli cells
- D) inhibition of nuclear condensation
- E) toxicity to the Leydig cells

Teratogens appear to cause malformations most frequently when given

---

- A) just before birth
- B) during the last trimester
- C) no one stage of fetal development is uniquely sensitive
- D) during organogenesis
- E) before implantation of the blastula in the uterus



Blindness is a toxic effect produced by ingestion of

---

- A) methanol
- B) acetaldehyde
- C) ethylene glycol
- D) isopropyl alcohol
- E) all of the above

Which of the following substances is most likely to cause agranulocytosis?

---

- A) sulfur dioxide
- B) carbon monoxide
- C) benzene
- D) chloroform
- E) nitrobenzene

Acetaminophen induced liver injury is

---

- A) periportal
- B) midzonal
- C) centrilobular
- D) biliary
- E) diffuse

Which of the following causes proximal tubular necrosis in acute poisoning?

---

- A) carbon monoxide
- B) kerosene
- C) carbon tetrachloride
- D) carbon disulfide
- E) tetraethyl lead

"Gingerjake" paralysis resulting from ingestion of a contaminated ginger extract was caused by

---

- A) deltamethrin
- B) methanol
- C) triorthocresyl phosphate (TOCP)
- D) triethyltin
- E) acrylamide

Treatment of a pregnant rat with a testosterone antagonist would be expected to produce

---

- A) masculinized female offspring
- B) sterile female offspring
- C) feminized male offspring
- D) no effect on the offspring since embryos do not make testosterone
- E) no effect on the offspring since pregnant female rats do not make testosterone

Which of the following statements is NOT true?

---

- A) B-cells are precursors of antibody-producing plasma cells
- B) B-cells are involved in the production of allergic contact urticaria
- C) B-cells are not involved in immediate-type allergic reactions
- D) T-cells are important in delayed-type allergic reactions
- E) T-cells secrete lymphokines

The most reliable animal model, to date, for the study of immunologic reactions of the delayed hypersensitivity type in contact photosensitivity dermatitis is the

---

- A) rat
- B) rabbit
- C) guinea pig
- D) gerbil
- E) monkey



Inhaled particles are removed from the alveolus by

---

- A) Clara cells
- B) Type I cells
- C) Type II cells
- D) alveolar macrophages
- E) goblet cells

Which of the following compounds does NOT produce delayed neurotoxicity?

---

- A) triorthocresyl phosphate
- B) diisopropyl fluorophosphate
- C) leptofos
- D) mipafox
- E) tetraethyl pyrophosphate

Measurement of which of the following should NOT be used to assess glomerular function?

---

- A) para-aminohippurate excretion
- B) inulin clearance
- C) blood urea nitrogen
- D) urinary albumin
- E) plasma creatinine

Which chemical substance does NOT produce hemolytic anemia?

---

- A) toluene diisocyanate
- B) primaquine
- C) sulfanilamide
- D) acetanilid
- E) 2-butoxyethanol

Which of the following statements is NOT correct?

---

- A) many nephrotoxicants appear to have the primary site of action on (in) the proximal tubule
- B) the proximal convoluted tubule is the primary site of reabsorption of glucose and amino acids
- C) the pars recta (S3) has a greater capacity to absorb organic compounds than the distal tubule
- D) the loop of Henle is the site of damage produced by chronic administration of analgesic mixtures
- E) the collecting duct appears relatively insensitive to most nephrotoxicants

Which of the following would be appropriate to use as a positive control for S9 metabolic activation in a *Salmonella typhimurium* (Ames) assay?

---

- A) sodium azide
- B) mitomycin C
- C) nitrogen mustard
- D) 2-aminoanthracene
- E) all of the above

Which of the following is/are human syndrome(s) characterized by monosomy?

---

- A) Down's syndrome
- B) Fanconi's syndrome
- C) xeroderma pigmentosum
- D) Turner's syndrome
- E) all of the above

Which of the following compounds is carcinogenic without metabolic activation?

---

- A) benzo(a)pyrene
- B) dimethylnitrosamine
- C) N-acetoxy-2-acetylaminofluorene
- D) ethionine
- E) 2-acetylaminofluorene



The primary area of the brain where MPP<sup>+</sup> (the toxic pyridinium ion of MPTP, a contaminant of synthetic heroin) causes injury and death to dopaminergic neurons is

---

- A) cerebral cortex
- B) choroid plexus
- C) substantia nigra
- D) tectum
- E) none of the above

Acyclovir's antiviral action is primarily due to which of the following?

---

- A) it inhibits DNA synthesis by being phosphorylated by viral thymidine kinase
- B) it inhibits viral RNA-dependent DNA polymerase (reverse transcriptase)
- C) acyclovir-monophosphate is incorporated into the elongated viral DNA causing termination of biosynthesis of the viral DNA strand
- D) A and C
- E) A, B and C

Which of the following describe(s) why the liver is a target organ for toxic substances?

---

- 1) the diverse metabolic capabilities of the liver enable it to metabolize many xenobiotics, but metabolism does not always result in detoxication
- 2) the liver plays an extensive role in intermediary metabolism and molecular synthesis; consequently interference with metabolic pathways may lead to toxic effects
- 3) the liver is exposed to relatively high concentrations of toxic substances absorbed from the GI tract
- 4) enterohepatic circulation can give rise to prolonged exposure to a toxic substance in the liver

A test article is an inducer of liver microsomal enzymes and produces thyroid adenomas after long-term administration to laboratory rats. Which of the following physiological alterations would be associated with these changes?

---

- 1) thyroid follicular cell hyperplasia
- 2) thyroid hypertrophy
- 3) increased glucuronidation of thyroxine
- 4) increased serum thyroid stimulating hormone (TSH)

Which of the following are types of pneumoconiosis?

---

- 1) silicosis and anthracosis
- 2) talcosis and byssinosis
- 3) siderosis and aluminosis
- 4) stanosis and baggasosis

Atopic diseases in individuals with an inherited sensitivity to allergens can be diagnosed by

---

- 1) a careful review of symptoms in relation to the environment
- 2) direct skin (scratch) tests
- 3) indirect (passive transfer) skin test
- 4) provocative tests with the suspect antigen

Chemical teratogens may

---

- 1) be mutagenic
- 2) alter fetal development at doses which do not alter placental function
- 3) alter maternal health at dose levels below those altering fetal development
- 4) cause tumors

Which of the following is/are direct-acting carcinogen(s)?

---

- 1) ethylene imine
- 2) methyl methane sulfonate
- 3) mustard gas (bis(2-chloroethyl)sulfide)
- 4) myleran (1,4-butanediol dimethanesulfonate)



It is currently claimed that primary lung cancer is

---

- 1) caused by heavy cigarette smoking over a period of years
- 2) decreasing in incidence in both males and females
- 3) related etiologically to air pollution
- 4) highly predictable on the basis of genetics

Mushroom workers' lung is an example of

---

- 1) an allergic disease in which fungal spores are the offending antigens
- 2) an IgG-mediated disease
- 3) extrinsic allergic pneumonitis
- 4) an immune complex disease

Type IV allergic reactions may be

---

- 1) cytotoxic
- 2) cell mediated
- 3) anaphylactic
- 4) delayed hypersensitivity

Which of the following classes of chemicals is/are immunosuppressive?

---

- 1) alkylating agents
- 2) thiopurines
- 3) folic acid antagonists
- 4) certain antibiotics

Which of the following can cause a decrease in female fertility?

---

- 1) diethylstilbestrol
- 2) hexachlorophene
- 3) DDT
- 4) thalidomide

With respect to the mechanisms of occupational contact dermatitis, example(s) of potent sensitizers is/are

---

- 1) formaldehyde
- 2) poison ivy
- 3) nickel
- 4) poison oak

Prediction of the nephrotoxic potential of fluorinated anesthetic agents is commonly based on which of the following types of data?

---

- 1) the rate of release of fluoride from the drug by hepatic microsomal preparations
- 2) the concentration of metallothionein in the kidney
- 3) the oil gas partition coefficient for the drug
- 4) the rate of liberation of carbon monoxide from the drug by renal microsomal preparations

Which of the following cellular alterations are NOT reversible?

---

- 1) fatty degeneration
- 2) necrosis
- 3) metaplasia
- 4) neoplasia



Which of the following agents damage myelin-forming cells in the central and peripheral nervous system?

---

- 1) isoniazid
- 2) hexachlorophene
- 3) lead
- 4) taxol

Which of the following chemicals are known to produce phototoxic reactions?

---

- 1) furocoumarins
- 2) anthracene
- 3) tetracyclines
- 4) tocopherols

What is/are the most likely mechanism(s) by which glomerular nephrotoxics such as antibiotics induce proteinuria?

---

- 1) by increasing the diameter of pores in the glomerular basement membrane
- 2) by enhancing carrier-mediated transport of protein across the glomerular membrane
- 3) by inhibiting lysosomal protease activity
- 4) by reducing the number of fixed anionic charges on glomerular structural elements

Signs of advanced chloroquine-induced retinopathy may include

---

- 1) progressive scotoma
- 2) narrowing of the retinal arteries
- 3) night blindness
- 4) abnormal electroretinograms

The rate of dermal absorption of toxicants may be facilitated by

---

- 1) increasing the toxicant's lipophilicity (for example, 2,4-D acid change to 2,4-D ester)
- 2) increasing skin blood flow
- 3) choice of solvent/vehicle
- 4) increasing skin water content

Which of the following is/are true concerning transport of bilirubin from sinusoidal blood into the hepatocyte?

---

- 1) the process is competitively inhibited by bromosulfophthalein (BSP)
- 2) is a carrier mediated process
- 3) can not occur while bilirubin is bound to albumin
- 4) conjugation of bilirubin with glucuronic acid and transport occur simultaneously

Which of the following statements is/are correct?

---

- 1) a forward mutation refers to a mutational event that inactivates a wild-type gene
- 2) the Ames bacterial mutagenicity assay is an example of a mutational assay based on detection of forward mutations
- 3) a forward mutation in the hypoxanthine-guanine phosphoribosyl transferase gene (HPRT) results in resistance to 6-thioguanine
- 4) mutations in the thymidine kinase gene confer resistance to 8-azaguanine

In which order does rate of oxidative metabolism generally decrease?

---

- A) mouse, human, rat
- B) human, mouse, rat
- C) mouse, rat, human
- D) human, rat, mouse
- E) rat, mouse human



The most important first measure in treating an acute poisoning is to

---

- A) identify the poison
- B) assess the degree of loss of consciousness
- C) prevent further absorption of the poison
- D) maintain vital functions
- E) treat shock

Which of the following is/are correct for dose-response curves?

---

- A) the steeper the slope of a curve the lesser the hazard associated with the use of a compound
- B) 2 x LD50 dose is predicted to produce 100% lethality
- C) a shift in the dose-response curve to the right indicates decreased toxicity
- D) toxicity increases as LD50 increases
- E) all of the above

As a result of phase II metabolic reactions

---

- A) chemicals are usually rendered more water-soluble
- B) chemicals usually undergo reduction
- C) chemicals usually undergo oxidation
- D) chemicals are excreted at less rapid rates
- E) the molecular weight of the chemical is usually reduced

The most common method of generating regression equations is the method of least squares. In this method, a set of differences is calculated, each difference is squared, and the solution is the equation which minimizes the sum of the differences. Which differences are used in the method of least squares when  $x$  is the independent variable?

---

- A) the differences between the observed  $y$  values when the  $x$  values have been put in ascending order
- B) the differences between the observed  $y$  values and the regression equation  $y$  values for given  $x$  values
- C) the differences between the observed  $y$  values and the regression equation  $x$  values for a given  $y$  value
- D) the differences between the observed  $x$  values and the regression equation  $x$  values for a given  $y$  value

E) the minimum distances between the observed  $y$  values and the regression line

The mean concentration of an airborne chemical that should not be exceeded during an eight-hour work day and 40-hour work week without risk of harm to nearly all workers is referred-to as

---

- A) reference concentration (RfC)
- B) permissible exposure level (PEL)
- C) threshold limit value/time weighted average (TLV/TWA)
- D) A and C
- E) B and C

Chronic poisoning is best defined as which of the following?

---

- A) clinical illness resulting from absorption of an excessive dose of a chemical agent
- B) the sequelae of irreversible damage resulting from a severe acute intoxication
- C) progressively unfolding illness due to accumulative damage caused by a chemical agent
- D) delayed apparent illness due to progressive tissue degeneration following irreversible but initially subclinical injury
- E) illness resulting from the sudden liberation of toxic amounts of a chemical from inactive biologic storage sites

The most reliable method of identifying an agent responsible for poisoning is

---

- A) locating a toxic agent near the victim
- B) observation of the symptoms
- C) pathologic changes observed grossly or microscopically
- D) chemical analysis of the blood or tissues
- E) effectiveness of a specific antidote



Which one of the following phrases does NOT apply to the pharmacology of ethyl alcohol?

---

- A) the drug is metabolized primarily in the liver
- B) the drug is rapidly absorbed from the stomach
- C) an intermediate metabolite is formaldehyde
- D) central "stimulation" is actually the release from inhibitions
- E) the non-addict metabolizes about 10 ml/hour

Inhibition of cytochrome oxidase is a major cause of toxicity of

---

- A) carbon tetrachloride
- B) methanol
- C) ethylene glycol
- D) hydrogen sulfide
- E) warfarin

In a carefully designed clinical trial, control groups which are necessary for valid conclusions

---

- A) can include only those patients who have not been treated for the disease under study
- B) can include placebo controls or the standard treatment for the disease
- C) will have their data normalized to remove potential bias
- D) require different criteria for evaluation than the treatment groups
- E) are unethical because patients must be treated with drugs when they consult a physician

In which of the following is microsomal drug metabolism increased?

---

- A) infancy and old age
- B) Parkinson's disease
- C) diabetes
- D) organophosphate insecticide poisoning
- E) obstructive jaundice

Iatrogenic disease refers to disease resulting from which of the following?

---

- A) food additives
- B) air pollution
- C) one's occupation
- D) water pollution
- E) physician's activities

The following of an act, law, or proposal is referred to as

---

- A) monitoring
- B) liability
- C) good laboratory practices (GLP's)
- D) compliance
- E) quality control

Sodium phosphate pollution of water results in

---

- A) adverse effects on fish schooling behavior
- B) increased occurrence of fish tumors
- C) loss of fecundity in fish populations
- D) teratogenicity in fish
- E) eutrophication

The mechanism(s) responsible for the variation in species susceptibility to toxic agents include(s) differences in

---

- A) rate and pattern of metabolism
- B) biliary excretion
- C) plasma protein binding
- D) A and C
- E) A, B and C



Which of the following compounds has been suggested to produce liver injury via lipid peroxidation?

---

- A) dimethylnitrosamine
- B) ethylene dibromide
- C) phosphorus
- D) thioacetamide
- E) none of the above

A toxic effect NOT associated with acetaminophen poisoning is

---

- A) covalent binding to cellular macromolecules
- B) glutathione depletion
- C) phocomelia
- D) nausea
- E) A and D

If the same contaminated diet were eaten by the following species, which species would receive a larger dose per unit body weight?

---

- A) mice
- B) rats
- C) rhesus monkeys
- D) dogs
- E) humans

Sponsors must obtain new drug application (NDA) approval from the Food and Drug Administration (FDA ) prior to

---

- A) commercial marketing of the new drug in the United States
- B) filing an acceptable "notice of claimed investigational exemption for a new drug" (IND)
- C) manufacture of the clinical drug formulation for distribution
- D) A and C
- E) A and B

The metabolism of substances in the body may involve

---

- A) reduction allowing the substance to be excreted in the feces
- B) oxidation to a more water soluble form allowing the substance to be excreted in the urine
- C) hydrolysis which then permits excretion via the lungs
- D) conjugation with glucuronic acid and sulfate
- E) B and D

The "Delaney clause" regulates the sale or distribution of carcinogenic substances in

---

- A) drugs
- B) foods
- C) cosmetics
- D) A and B
- E) A, B and C

Some toxicants achieve their highest concentration at their site of toxic action. An example is

---

- A) paraquat in lung
- B) lead in bone
- C) chlorinated hydrocarbons in fat
- D) carbon monoxide in cerebellum
- E) TCDD in skin

What percentage of a normally distributed population is represented by the mean value  $+3$  or  $-3$  standard deviations?

---

- A) 33%
- B) 50%
- C) 68.3%
- D) 95.5 %
- E) 99.7%



The volume of distribution is

---

- A) large for agents that are highly bound to plasma proteins
- B) the ratio of half-life to total dose
- C) small for most toxic agents
- D) not applicable to compounds with nonlinear toxicokinetics
- E) large for agents that are actively sequestered in a specific organ

Which of the following is NOT a correct pairing of toxic exposure with high risk groups?

---

- A) arsenic/individuals with high exposure to sunlight
- B) lead/children
- C) nitrate/infants
- D) asbestos/cigarette smokers
- E) A and C

Under conditions of first-order kinetics, repeated exposure to a toxicant will result in its accumulation until the rate of elimination equals the rate of absorption. The steady state body burden of a toxicant will be approximately how many times the amount absorbed per half-life of elimination?

---

- A) 0.5
- B) 1.25
- C) 2.5
- D) 0.693
- E) 1.5

Apoptosis (programmed cell death) is characterized by all of the following EXCEPT

---

- A) high energy consumption
- B) condensation of nuclear chromatin
- C) cellular and nuclear swelling
- D) internucleosomal cleavage of DNA
- E) activation of highly conserved signal transduction pathways

Which of the following chemical characteristics would NOT be associated with a tendency to be bioaccumulated?

---

- A) high lipophilicity
- B) high volatility
- C) low polarity
- D) high chemical stability
- E) B and C

Short Term Exposure Limits (STELs) are

---

- A) worker exposure limits for a single 15 minute interval during normal work cycle
- B) promulgated by the United States Environmental Protection Agency
- C) based on carefully controlled studies in humans
- D) based on subchronic exposure studies in animals
- E) none of the above

## The "de minimus" concept

---

- A) means that the calculated risks are probably not real
- B) refers to an upper limit risk that represents  $1 \times 10^6$  excess cancers per lifetime
- C) refers to an upper limit risk that represents  $1 \times 10^6$  excess cancers per year
- D) refers to a risk that is small enough to be considered trivial and not worth regulating
- E) refers to any activity that results in less than one death per year

Ethyl methanesulfonate exerts carcinogenic effects by acting

---

- A) as a primary initiator
- B) as a promotor
- C) as a co-carcinogen
- D) as a hormone mimic
- E) by an unknown mechanism



Which of the following microsomal inducers increase(s) bile flow and excretion?

---

- A) benzo(a)pyrene
- B) 3-methylcholanthrene
- C) phenobarbital
- D) all of the above
- E) none of the above

If a chemical has a greater action on the central nervous system of a fetus than on the mother's central nervous system, one might expect that the drug

---

- A) has a high lipid solubility
- B) does not readily penetrate the blood brain barrier
- C) is not metabolized by oxidative enzymes of the liver
- D) has a low molecular weight
- E) is bound to plasma protein of the fetus

There are four general phases to the risk assessment process. Initially, data are collected and evaluated and toxicology and exposure assessments are completed. The final stage of the risk assessment process integrates all of the above into the

---

- A) integrated risk information system
- B) health effects assessment summary
- C) health effects advisory
- D) risk characterization
- E) health and environmental effects profile

General types of oxidation reactions catalyzed by cytochrome P450 is/are

---

- 1) aromatic and aliphatic hydroxylation
- 2) epoxidation
- 3) sulfoxidation
- 4) desulfuration

Exposure to 2,3,7,8-tetrachlorodibenzo-p-dioxin in humans may cause

---

- 1) hemolytic anemia
- 2) hydronephrosis
- 3) hepatic cirrhosis
- 4) chloracne

## A cohort study

---

- 1) is conducted to assess frequency with which diseases occur
- 2) is one in which exposed subjects are "paired" with unexposed subjects
- 3) may be either retrospective or prospective
- 4) solely relies on death certificates

## Probit analysis

---

- 1) adjusts a quantal response to an assumed normal distribution
- 2) can be used to calculate therapeutic indices
- 3) can be used to calculate LD50 value
- 4) usually expresses the dose logarithmically

Pesticide degradation mechanisms in the environment include

---

- 1) microbial metabolism
- 2) adsorption to soil clay particles
- 3) photodegradation
- 4) thermal decomposition



Known occupational respiratory carcinogens include

---

- 1) fibrous glass
- 2) asbestos
- 3) methanol
- 4) radon

The carcinogenicity of the following chemicals has been demonstrated in animals. Which of the following have/has also been linked epidemiologically to cancer in humans after occupational exposure?

---

- 1) trichloroethylene
- 2) styrene
- 3) chloroprene
- 4) vinyl chloride

Which of the following is/are true regarding the Maximum Tolerated Dose in carcinogenicity studies?

---

- 1) is estimated from subchronic studies
- 2) may overwhelm normal homeostasis and detoxication or repair mechanisms
- 3) provides maximum opportunity for detection of a neoplastic response
- 4) is a controversial issue concerning carcinogen bioassays

Contamination of aquatic environments by polycyclic aromatic hydrocarbons is of concern because

---

- 1) they are bioaccumulated by aquatic organisms from water, food and sediments
- 2) they are direct-acting carcinogens in several aquatic organisms
- 3) they have relatively long environmental half-lives
- 4) they are readily soluble in water

The risk assessment process presented by the U.S. National Academy of Sciences (1983) involves

---

- 1) hazard identification
- 2) dose-response assessment
- 3) exposure assessment
- 4) risk communication

Which of the following are Phase I reactions?

---

- 1) acetylation
- 2) hydroxylation
- 3) glucuronidation
- 4) demethylation

Sodium bicarbonate may be useful in the treatment of intoxication with

---

- 1) salicylate
- 2) phenobarbital
- 3) methyl alcohol
- 4) amphetamine

Fluoride can

---

- 1) produce a mottling of tooth enamel
- 2) produce fluorescent urine
- 3) increase bone density
- 4) increase the incidence of some forms of cancer



Which of the following statements is NOT correct?

---

- A) infants are deficient in rhodanese
- B) primaquine-sensitive patients cannot, when challenged, maintain sufficient reduced glutathione in their red blood cells
- C) primaquine-sensitive patients have an altered or deficient glucose-6-phosphate dehydrogenase
- D) dibucaine number of approximately 80 is considered normal
- E) vitamin K can displace bilirubin from plasma proteins

Which of the following statements is correct?

---

- A) phosgene forms during carbon tetrachloride combustion
- B) SO<sub>2</sub> is insoluble in water
- C) chlorine is a deep lung irritant
- D) HCl is a deep lung irritant
- E) dimethyl mercury has insufficient vapor pressure to be an inhalation hazard

Which of the following statements is/are correct?

---

- A) phenobarbital potentiates DDT toxicity
- B) red squill causes death in rodents due to electrolyte imbalance after severe vomiting episodes
- C) all cholinesterase-inhibiting insecticides react with the esteratic site of the enzyme
- D) B and C
- E) none of the above

Which of the following statements is/are correct?

---

- A) NO<sub>2</sub> causes pulmonary edema
- B) intermittent exposure to O<sub>3</sub> at low levels enhances CO diffusion
- C) a tracheal cannula will enhance pulmonary response to 0.5 PPM SO<sub>2</sub>
- D) B and C
- E) A and C

Which of the following statements is/are correct?

---

- A) SO<sub>2</sub> increases flow resistance and compliance
- B) SO<sub>2</sub> causes bronchoconstriction which can be blocked by parasympatholytics
- C) O<sub>3</sub> toxicity is worse at lowered temperatures
- D) A and B
- E) none of the above

Which of the following is NOT correct regarding exposure to agent orange containing TCDD?

---

- A) may produce ventricular fibrillation and death in dogs after massive doses
- B) is metabolized to 2,4-D and excreted in the urine of most mammals
- C) has no significant cumulative toxicity
- D) is highly, but reversibly bound to plasma proteins
- E) is a selective "hormone-type" weed killer

A 2-week-old infant was brought to the hospital with diarrhea, vomiting, opisthotonus and marked generalized erythema. Within three days, the general condition worsened. After frequent convulsive episodes, the patient became comatose. Desquamation was noted on many areas of the body. Aspirin and carbon monoxide poisoning were excluded. The most likely cause of the toxicity is

---

- A) zinc oxide
- B) nitrate-containing well-water
- C) boric acid
- D) ferrous sulfate
- E) phosphate detergent

Intramuscular injection of this compound often causes transient hypertension and tachycardia?

---

- A) dimercaprol (BAL)
- B) disodium EDTA
- C) calcium disodium EDTA
- D) penicillamine
- E) n-acetylpenicillamine



The mechanism of action of edathamil calcium disodium EDTA in lead poisoning is analogous to the action of

---

- A) chloride in bromide poisoning
- B) methemoglobin in cyanide poisoning
- C) nalorphine in morphine poisoning
- D) alkalosis in sulfanilamide-induced crystaluria
- E) atropine in organophosphate poisoning

The biotransformation product of methanol responsible for delayed acidosis is

---

- A) lactic acid
- B) formic acid
- C) acetone
- D) acetaldehyde
- E) acetic acid

Which of the following improves skeletal muscle function in anticholinesterase intoxication?

---

- A) atropine
- B) pralidoxime
- C) BAL
- D) A and B
- E) none of the above

Which of the following compounds causes a direct cardiac effect?

---

- A) red squill
- B) fluoroacetate
- C) warfarin
- D) methanol
- E) cyanide

A chronic exposure to fluoride in the drinking water that would probably lead to a moderate degree of fluorosis (mottled teeth but no functional impairment) is

---

- A) 10.0 ppm
- B) 3.0 ppm
- C) 1.5 ppm
- D) 0.3 ppm
- E) 0.01 ppm

Which of the following attributes is LEAST important in the selection of a drug for clinical use?

---

- A) margin of safety
- B) relative potency
- C) duration of action
- D) route of elimination
- E) efficacy

Following ingestion of a rodenticide, a dog exhibited severe tetanic convulsive seizures and died 1 hour later. Postmortem examination revealed cyanosis, dark blood, heart in diastole and subepicardial hemorrhages. The most likely cause of the symptoms and death of this animal is

---

- A) sodium fluoroacetate (1080)
- B) alpha naphthyl thiourea (ANTU)
- C) warfarin
- D) pindone
- E) thallium

Which of the following is the most likely source of high concentrations of nitrogen dioxide?

---

- A) coal burning smelting operations
- B) automobile exhaust
- C) decomposing fecal matter
- D) fertilizers
- E) silos



Coral snake venom is primarily a

---

- A) neurotoxin
- B) nephrotoxin
- C) hepatotoxin
- D) anticoagulant
- E) cardiotoxin

Compound X has a molecular weight of 150 and an IV LD50 in the rat is 300 mg/kg. If the respiratory minute volume is 200 ml/min, 50% of the inhaled material is retained and the test rats weigh 200g, estimate the LC50 for a 10 minute inhalation exposure to compound X

---

- A) 0.6 mg/liter
- B) 60 mg/liter
- C) 30 mg/liter
- D) 120 mg/liter
- E) 200 mg/liter

Which of the following food additives is NOT an emulsifying agent?

---

- A) cholic acid
- B) mono- and diglycerides
- C) ox bile extract
- D) sorbitol
- E) propylene glycol

Antimony differs most from arsenic with respect to

---

- A) treatment of choice in acute poisoning
- B) magnitude of workplace exposure limit
- C) capacity to cause pathologic changes in cardiac muscle
- D) inability to cause skin disease
- E) none of the above

A histamine-like syndrome is produced by

---

- A) tetrodotoxin poisoning
- B) scombroid poisoning
- C) ciguatera poisoning
- D) A and B
- E) B and C

Plants containing digitalis-like glycosides include

---

- A) foxglove (*Digitalis purpurea*)
- B) lily of the valley (*Convallaria majalis*)
- C) oleander (*Nerium oleander*)
- D) A and B
- E) A, B and C

Which of the following does NOT potentiate the hepatotoxicity of carbon tetrachloride?

---

- A) acetone
- B) acetaldehyde
- C) ethanol
- D) 2-butanol
- E) 2-butanone

Use of nonsteroidal anti-inflammatory agents (ibuprofen, piroxicam, naproxen) may lead to gastrointestinal side effects including bleeding and ulceration as a result of

---

- A) the inhibition of cyclooxygenase, alteration of platelet function and cholinergic effects
- B) the alteration of mucosal blood flow and HCl production via stimulation of histamine secretion
- C) their direct stimulatory effect on parietal cell HCl secretion and inhibition of prostaglandin synthesis
- D) the inhibition of cyclooxygenase and alteration of platelet function
- E) inhibition of arachidonic acid metabolism by inhibiting both cyclooxygenase and lipo-oxygenase



Which of the following rodenticides is most likely to cause severe acute liver necrosis?

---

- A) warfarin
- B) red squill
- C) alpha naphthyl thiourea (ANTU)
- D) phosphorus (elemental)
- E) thallium sulfate

The most important factors in alveolar deposition and clearance of dusts are

---

- A) particle size and solubility
- B) vapor pressure and shape of the particulate material
- C) particle toxicity and aerodynamic diameter
- D) particle size and the number of lung macrophages
- E) the induction of fibrosis and aerodynamic diameter of the particles

The toxicity of dinitrophenol results from

---

- A) irreversible demyelination of peripheral nerves
- B) uncoupling of electron transport from oxidative phosphorylation
- C) metabolic activation leading to bone marrow depression
- D) pseudoestrogenic activity
- E) induction of reactive astrogliosis and accompanying increased glial fibrillary acidic protein

The most dangerous inhaled radioactive materials are

---

- A) beta emitters
- B) alpha emitters
- C) gamma emitters
- D) those with highest energy regardless of type of emission
- E) elements of medium  $z$  number since their radiations have the most favorable charge to mass ratio

N-acetylcysteine is an effective antidote for acetaminophen overdose because it

---

- A) blocks the absorption of acetaminophen from the intestines
- B) accelerates urinary excretion of the unchanged drug
- C) blocks metabolism of acetaminophen to a toxic intermediate
- D) prevents the accumulation of the toxic intermediate
- E) binds to critical receptors on the mitochondria in the liver

Which of the following procedures is/are contraindicated in the treatment of intoxication with gasoline or kerosene?

---

- 1) emesis or gastric lavage
- 2) correction of fluid and electrolyte imbalances
- 3) administration of epinephrine
- 4) symptomatic and supportive care

## Alpha particles

---

- 1) are a type of ionizing radiation
- 2) lose energy when passing through biological tissue
- 3) are heavy charged particles
- 4) have linear energy transfer (LET) much greater than for gamma rays

Atropine blocks the

---

- 1) peripheral neuropathy of some organophosphates
- 2) peripheral neuropathy of acrylamide
- 3) nicotinic effects of organophosphates
- 4) muscarinic effects of organophosphates



Which of the following statements is/are true for chloromethane (methyl chloride)?

---

- 1) acute poisoning in humans may result in the following symptoms: dizziness, incoordination and convulsions
- 2) is a residue in food
- 3) is a reproductive toxicant in animals
- 4) is a liquid at ambient temperature and pressure

Which of the following must be metabolized to become active?

---

- 1) o,o-diisopropyl phosphorofluoridate
- 2) heptachlor
- 3) neostigmine
- 4) parathion

## Lead

---

- 1) is present in the diet
- 2) is absorbed mainly from the small intestine
- 3) is first distributed to the soft tissues, especially liver and kidneys, and then redistributed primarily to bone
- 4) compounds taken orally are well absorbed

Ethylene glycol is

---

- 1) a quick source of energy when ingested
- 2) metabolized to oxalate and precipitated in the kidney as the calcium salt
- 3) an excellent nontoxic solvent for many drugs
- 4) associated with metabolic acidosis when ingested

## Particle clearance mechanisms in the deep lung

---

- 1) can be overwhelmed by dust at extremely high concentrations leading to "lung overload"
- 2) can be overwhelmed by extremely high concentrations of so called nuisance dusts (e.g., carbon black) causing excess lung tumors in carcinogenesis bioassays
- 3) depend primarily on macrophage phagocytosis
- 4) are so effective that dusts, even at unphysiologically high concentrations, can be removed within weeks

Which of the following classes of drugs is/are absorbed from the gastrointestinal tract at a rate dependent only on the partition coefficient?

---

- 1) weakly acidic drugs
- 2) neutral-uncharged drugs
- 3) weakly basic drugs
- 4) quaternary ammonium drugs

## Exposure to the environmental air pollutant, ozone

---

- 1) results exclusively in bronchoconstriction due to smooth muscle contraction
- 2) is rapidly destructive to lung tissue
- 3) is itself an antigen
- 4) is immunosuppressive

Chronic exposure to arsenic trioxide is monitored in

---

- 1) hair
- 2) fingernails
- 3) urine
- 4) skin



Intentional food additives include

---

- 1) iron compounds
- 2) aluminum compounds
- 3) magnesium compounds
- 4) copper compounds

Which of the following is/are upper respiratory tract irritants?

---

- 1) ammonia
- 2) hydrogen fluoride
- 3) sulfur dioxide
- 4) chlorine

Which of the following is/are true about 2-PAM?

---

- 1) it is contraindicated in poisoning by the cholinesterase inhibitor, carbaryl
- 2) it reactivates the phosphorylated acetylcholinesterase
- 3) it is ineffective if treatment is delayed
- 4) it is very effective at penetrating the central nervous system

## Particles present in polluted air

---

- 1) are primarily cleared by mucociliary clearance if deposited in the tracheobronchial tree
- 2) from diesel exhaust, are relatively "inert" under ambient concentrations in experimental animals
- 3) frequently coexist with gases, resulting in enhanced pulmonary injury
- 4) cause greater injury if ultrafine than if larger in size for the same mass loading

Which of the following statements regarding berylliosis is/are true?

---

- 1) the proximate antigen is a protein conjugate
- 2) is a condition associated with the stimulation and proliferation of T-cells
- 3) produces an immune reaction as evidenced by a granulomatous inflammatory response
- 4) requires relatively large exposures

## Glycol ether

---

- 1) is both a reproductive and developmental toxicant in rodents
- 2) has a similar pattern of reproductive toxicity as ethyl ether
- 3) is a possible reproductive toxicity concern in humans
- 4) is not a reproductive toxicant

Which of the following fumigants interferes with respiratory activity by interaction with cytochrome oxidase and resulting inhibition of electron transport?

---

- 1) methyl bromide
- 2) phosphine
- 3) naphthalene
- 4) hydrogen cyanide

Hematopoietic toxicity is the primary toxic manifestation of chronic, low-dose exposure to which of the following?

---

- 1) short-chain alkanes
- 2) dichloromethane
- 3) carbon tetrachloride
- 4) benzene



Enzymes capable of metabolizing ethanol to acetaldehyde include

---

- 1) alcohol dehydrogenase
- 2) catalase
- 3) microsomal ethanol oxidizing system (cytochrome P450 2E1)
- 4) glutathione-S-transferase

Target tissues that are very sensitive to the acute effects of high doses of whole-body ionizing radiation include

---

- 1) bone marrow
- 2) liver
- 3) crypt cells in the intestinal tract
- 4) kidney

A teratogenic change in an embryo/fetus is usually induced during

---

- A) fertilization of the ovum
- B) the third trimester of pregnancy
- C) the period of organogenesis
- D) prior to implantation
- E) anytime during pregnancy

Which of the following metals produces chronic progressive pulmonary granulomatosis?

---

- A) manganese
- B) zinc
- C) beryllium
- D) strontium
- E) arsenic

Which of the following is most likely to produce liver damage?

---

- A) radioactive phosphorus
- B) ethylene
- C) carbon tetrachloride
- D) tetrachlorethylene
- E) aminopyrine

0.166 ml of acetone (specific gravity = 0.791 gm/ml; molecular weight = 58.08) is evaporated in a 29.8 liter container (under conditions where 1 mole of air is equal to 25 liters). What is the approximate acetone vapor concentration in the containers?

---

- A) 2 ppm
- B) 6 ppm
- C) 19 ppm
- D) 190 ppm
- E) 1900 ppm

The most important factor influencing the deposition pattern of particulates in the lung is

---

- A) aerodynamic size
- B) inter-radial shape
- C) hygroscopicity
- D) charge density
- E) spin state

The excretion of an organic cation (base) by the kidney theoretically could be enhanced if the subject is given

---

- A) sulfobromophthalein
- B) morphine
- C) sodium bicarbonate
- D) ammonium chloride
- E) probenecid



## Animal testing for transplacental carcinogenesis

---

- A) shows that potent carcinogens in adult animals are also potent transplacentally
- B) shows that genetics is unimportant in determining what tumors result from exposure to a carcinogen
- C) shows that the latency period is markedly reduced
- D) is a well-recognized predictor of human carcinogenicity
- E) does not reliably predict human risk

In a reproduction study, rats were treated with various dosages of compound X on gestation days 0-7 and allowed to deliver and raise their offspring to weaning. This study primarily tests the effect of compound X on

---

- A) implantation
- B) fetal growth and development
- C) parturition
- D) growth of pups following prenatal treatment
- E) none of the above

You have just reviewed the results of a subchronic animal toxicity study. Which of the following observations would LEAST likely lead you to conclude that the administered material may alter the immune function of the animal?

---

- A) thymic hypertrophy
- B) thymic atrophy
- C) lymphopenia
- D) splenic hypertrophy
- E) splenic atrophy

Which of the following agents exerts its primary nephrotoxic effects on the glomerulus?

---

- A) phenacetin
- B) amphotericin B
- C) cyclosporine
- D) mercuric chloride
- E) potassium bichromate

The symptoms of Wernicke's encephalopathy may be caused by the lack of

---

- A) folic acid
- B) glucose
- C) riboflavin
- D) thiamine
- E) niacin

If an acute exposure to a test chemical damages cells only in the premeiotic stages of spermatogenesis in man, abnormal sperm will start to appear in the ejaculate

---

- A) 5 to 8 weeks after exposure
- B) up to 11 weeks after exposure
- C) 11 to 15 weeks after exposure
- D) 1 to 4 weeks after exposure
- E) 26 weeks after exposure

How much of the plasma reaching the kidneys is filtered?

---

- A) 5%
- B) 20%
- C) 40%
- D) 70%
- E) 95%

Active secretion by the proximal tubule is responsible for elimination of which of the following

---

- A) glucose
- B) certain organic compounds
- C) nitrogenous waste
- D) chloride ions
- E) phosphate ions



In the two-stage model of tumorigenesis, the promotion stage differs from the initiation stage in that

---

- A) initiation is irreversible, while promotion is reversible
- B) initiation exhibits one-hit kinetics, while promotion exhibits multi-hit kinetics
- C) initiation usually results from alteration of the genetic information of a cell, while promotion results from alteration of the expression of the genetic information
- D) initiation probably exhibits no measurable threshold dose dependency
- E) all of the above

The lowest blood level of methemoglobin sufficient to produce visible cyanosis in humans is

---

- A) 0.1%
- B) 1%
- C) 2%
- D) 10%
- E) 30%

The principal difference in the kinetics of an antibody response following a secondary immunological stimulation as compared to the kinetics of a primary stimulation is

---

- A) shorter latency period until appearance of serum IgG
- B) production of IgG antibodies prior to IgM antibodies
- C) increase in serum antibody concentration is linear
- D) T lymphocytes as well as B lymphocytes differentiate
- E) there is no difference

What is the correct order of appearance for these structures, starting from the external side of the eye?

---

- A) cornea, lens, iris, ganglion cells, photoreceptors
- B) cornea, iris, lens, ganglion cells, photoreceptors
- C) lens, cornea, iris, photoreceptors, ganglion cells
- D) cornea, lens, iris, photoreceptors, ganglion cells
- E) cornea, iris, lens, photoreceptors, ganglion cells

Which of the following substances is the contact allergen found in plants of the *Toxicodendron* genus (poison ivy, oak, etc)?

---

- A) sesquiterpene lactone
- B) resorcinol monobenzoate
- C) hexamethylenetetramine
- D) diphenylguanidine
- E) pentadecylcatechol

Diabetes enhances the hepatotoxicity of carbon tetrachloride by

---

- A) alteration in hepatic gluconeogenic enzyme activity
- B) alteration in hepatic plasma membrane fluidity
- C) increased intracellular levels of free glucose
- D) induction of CYP2E1
- E) reduced portal blood flow leading to increased hepatic uptake of carbon tetrachloride

Which of the following is NOT an assay for mutagenic activity?

---

- A) Ames Salmonella assay
- B) mouse lymphoma assay
- C) recessive lethal test in *Drosophila melanogaster*
- D) mouse micronucleus assay
- E) mammalian specific locus assay

A classical teratology study (U.S. FDA Segment II) would detect an effect on

---

- A) the implantation rate
- B) the pregnancy rate
- C) dominant lethal mutations
- D) all of the above
- E) none of the above



A number of plant alkaloids, (e.g. colchicine, vincristine and taxol) produce axonopathy by

---

- A) disrupting the equilibrium of the association/dissociation cycle of microtubules within the axon
- B) inhibiting the formation of microfilaments
- C) inhibiting ATP synthesis, resulting in disruption of axonal metabolism
- D) blocking calcium movement within the cell resulting in the leakage of calcium by cells and subsequent disruption of microtubule formation
- E) binding to m-RNA and inhibiting protein synthesis

Which of the following is NOT true of the glomerular filtration rate (GFR)

---

- A) can be measured by inulin clearance from the plasma
- B) is a measure of the functional capacity of the glomerulus
- C) can be measured by PAH clearance from the plasma
- D) serum creatinine and BUN are insensitive indicators of GFR
- E) can be reduced by dehydration

During a thirteen-week repeat-dose toxicology study in rats with an *E. coli* derived recombinant protein, the blood level of the protein remained constant with continued exposure but the pharmacological effects of the molecule were no longer evident by the end of eight weeks of exposure. The most logical reason for this finding is

---

- A) elicitation of type III hypersensitivity
- B) IgE mediated anaphylaxis
- C) inactivation of the protein by non-specific peptidase
- D) formation of autoantibodies
- E) formation of neutralizing antibodies

Examples of direct acting carcinogens that do NOT require activation include

---

- 1) bis-(chloromethyl)ether
- 2) dimethyl sulfate
- 3) beta-propiolactone
- 4) 2-naphthylamine

Which of the following is/are correct regarding the acetylator phenotype?

---

- 1) inter-individual variation occurs in a bimodal distribution
- 2) slow acetylation of isoniazid leads to hepatic damage
- 3) rapid acetylation of 2-aminofluorene leads to greater DNA-adduct formation in mice
- 4) fast acetylation of sulfonamides leads to hemolysis

Which of the following is/are true of p53?

---

- 1) loss or inactivation of the p53 tumor suppressor gene is a common molecular lesion in human neoplasms
- 2) introduction of wild type p53 into p53-negative tumor cells inhibits proliferation by implementing a G1 arrest
- 3) cells from p53 knockout mice fail to arrest in G1 following gamma irradiation
- 4) p53 growth-arrest is mediated via p53-dependent transactivation of genes

Which of the following statements is/are true regarding myelinopathy?

---

- 1) compounds associated with injury of myelin include ethidium bromide, hexachlorophene, isoniazid and tellurium
- 2) after injury, remyelination occurs to a greater extent in the CNS compared to the PNS
- 3) intramyelinic edema is reversible during the early stages
- 4) functional consequences of demyelination are independent of the degree of the lesion

According to the International Agency on Research on Cancer (IARC), which of the following factors justify downgrading the weight of evidence for carcinogenicity from "sufficient" to "limited"?

---

- 1) evidence of carcinogenicity is restricted to a single experiment
- 2) there are unresolved questions regarding the adequacy of the design, conduct or interpretation of the study
- 3) the agent increases the incidence only of benign neoplasms or lesions of uncertain neoplastic potential, or of certain neoplasms which may occur spontaneously in high incidences in certain strains
- 4) the study did not use an occupationally-relevant route of administration



Which of the following is/are true regarding chemical injury to the cornea?

---

- 1) cationic detergents are more damaging than anionic detergents
- 2) organic solvents dissolve fats and usually cause pain on contacting the cornea
- 3) acid burns result from a combination of low pH and the capacity of the anion to combine with protein
- 4) lacrimators, like chloroacetophenone (tear gas), cause damage to the cornea

Which of the following is/are characteristic of peroxisome proliferators?

---

- 1) are human carcinogens
- 2) usually negative in short-term tests for genotoxicity
- 3) structurally similar compounds which produce diverse pleiotropic responses in experimental animals
- 4) are associated with hepatomegaly, increased expression of hepatic enzymes, and changes in lipid metabolism

Phototoxicity and photoallergy share which of the following characteristics?

---

- 1) structure and absorption spectrum of the xenobiotic are crucial determinants
- 2) dermatitis is restricted to light exposed areas
- 3) there is a dependency on wavelength and intensity of impinging light
- 4) incidence and severity of dermatitis are clearly dose-related

Developmental toxicology is concerned with

---

- 1) toxicity of the test chemical applied to male test animals for more than 3 months prior to breeding with untreated females
- 2) morphologic manifestations of defects caused by exposure to chemicals prior to birth
- 3) toxicity of the test chemical applied to female test animals for more than 3 months prior to breeding with untreated males
- 4) biochemical and functional lesions resulting from exposure to chemicals prior to birth

In chronic renal failure, extra-renal findings may include

---

- 1) enlarged parathyroids
- 2) pale streaking of heart and blood vessels due to mineralization
- 3) mottled, moist lungs and hydrothorax
- 4) soft bones

Which of the following agents has/have been shown to affect fertility in both experimental animals and humans?

---

- 1) phenol
- 2) phthalate esters
- 3) isoniazid
- 4) dibromochloropropane

Which of the following ion(s) is/are transported actively in the ascending limb of the loop of Henle?

---

- 1) sodium
- 2) bicarbonate
- 3) potassium
- 4) chloride

The frequency of baseline sister-chromatid exchange (SCE) can be altered by which of the following

---

- 1) fluorescent light
- 2) source and concentration of serum
- 3) dialysis of the serum
- 4) heat-treatment of the serum



Thymus-dependent lymphocytes are associated with

---

- 1) graft rejection
- 2) delayed type hypersensitivity
- 3) resistance to infectious agents
- 4) tumor cell cytotoxicity

Fertility tests or multigeneration studies are NOT satisfactory substitutes for a teratology test because

---

- 1) in fertility tests, females may cannibalize their young
- 2) dose levels may be lower because of the potential toxic effects on mating behavior
- 3) females are dosed pre- and post-fertilization
- 4) birth is by natural delivery

## Metaplasia

---

- 1) occurs when one type of differentiated cell replaces another type of differentiated cell
- 2) is usually associated with chronic irritation or endocrine imbalance
- 3) is usually considered an adaptive response to an adverse environment
- 4) may be a preneoplastic change

Cataracts may be induced by which of the following?

---

- 1) galactose
- 2) naphthalene
- 3) thallium
- 4) busulfan

Elevated levels of which serum enzymes would be expected in cholestatic liver injury but not in centrilobular necrosis in humans?

---

- 1) alkaline phosphatase
- 2) glutamic-pyruvic transaminase
- 3) gamma glutamyl transpeptidase
- 4) glutamic-oxalo-acetic transaminase

Which of the following metals can produce type IV allergic reactions?

---

- 1) nickel
- 2) chromium
- 3) gold
- 4) beryllium

Which of the following statements is/are true regarding renal clearance?

---

- 1) renal clearance of inulin is used to estimate glomerular filtration rate
- 2) clearance is defined as the quantity of plasma cleared of a substance per unit time
- 3) inulin is a polysaccharide that is freely filtered at the glomerulus and neither secreted nor reabsorbed
- 4) the initial step in the formation of urine is ultrafiltration of plasma in the glomerular capillaries

Which of the following chemicals produce both hepatocellular necrosis and fatty liver?

---

- 1) bromotrichloromethane
- 2) dimethylaminoazobenzene
- 3) galactosamine
- 4) urethane



Which of the following is/are true concerning Leydig cells?

---

- 1) synthesize and secrete testosterone
- 2) are located in the interstitial tissue between the seminiferous tubules
- 3) are sensitive to luteinizing hormone
- 4) convert testosterone into estrogen

In mammals, acute inhibition of acetylcholine esterase causes

---

- 1) lacrimation
- 2) salivation
- 3) miosis
- 4) CNS depression

## Dermal permeability in humans

---

- 1) remains relatively invariant as a function of age (with the exception of pre-term infants who lack a fully developed stratum corneum)
- 2) can be increased by covering the skin with an occlusive wrap
- 3) can vary greatly from place to place on the body
- 4) follows Fick's first law of diffusion if the skin is not damaged

Which of the following is/are true regarding use of the LD50?

---

- A) the most reliable way to establish dose levels for a chronic toxicity study is to determine the LD50 dose and then select a certain fraction of the LD50 as the highest exposure level for the chronic test
- B) there is no consistent relationship between chronic toxicity and the LD50
- C) the LD50 for laboratory animals is an accurate indicator of the relative hazard of the chemical for man
- D) A, B and C
- E) none of the above

he experimental dose range giving maximal information will include

---

- A) two no effect and two surely effective doses
- B) one no effect, one just effective and two surely effective doses
- C) two just effective and two surely effective doses
- D) one no effect and three surely effective doses
- E) four surely effective doses

The term used to describe the effect seen when the combined clinical effect of two agents is less than the algebraic sum of the anticipated individual responses is

---

- A) synergism
- B) additive
- C) potentiation
- D) antagonism
- E) cumulative effect

Occupational exposure to incompletely cured phenoplasts or depolymerized dust from phenoplasts may result in (Phenoplasts are polycondensation resins containing phenols and formaldehyde or other aldehydes)

---

- A) nonallergic irritant dermatitis due to the phenol
- B) allergic dermatitis due to free phenol
- C) allergic dermatitis due to formaldehyde
- D) pulmonary injury due to a cadmium stabilizer
- E) none of the above

Slow neutrons have a greater biologic effect than fast neutrons because they

---

- A) penetrate deeper into the tissues
- B) have a higher energy content
- C) have special affinity for the hematopoietic system
- D) are more easily captured by the nuclei of the atoms within the cells
- E) produce less secondary ionization following capture



The relationship between the primary effects of a chemical and the secondary effects is termed its

---

- A) margin of safety
- B) therapeutic index
- C) selectivity
- D) potency
- E) affinity

Degeneration of a cell can result in formation of vacuoles as a result of

---

- A) ionic imbalance
- B) matrix swelling of mitochondria
- C) protein accumulation in the vacuoles
- D) all of the above
- E) none of the above

Which of the following compounds is LEAST likely to potentiate carbon tetrachloride hepatotoxicity?

---

- A) methanol
- B) ethanol
- C) acetone
- D) benzene
- E) isopropanol

Which of the following is NOT associated with photochemical air pollution?

---

- A) O<sub>3</sub>
- B) NO<sub>2</sub>
- C) NO
- D) SO<sub>2</sub>
- E) O<sub>2</sub>

Which of the following is contraindicated for the treatment of amphetamine overdose?

---

- A) diazepam iv
- B) haloperidol iv
- C) ammonium chloride iv
- D) sodium bicarbonate iv
- E) ascorbic acid iv

The principal source of occupational arsenic poisoning is

---

- A) arsine
- B) arsenic trioxide
- C) phosphoste fertilizers
- D) sodium arsenate
- E) sodium arsenite

Section 5 of the Toxic Substances Control Act deals with which of the following

---

- A) pre-manufacture notice
- B) substantial risk notification
- C) issuance of test rules
- D) all of the above
- E) none of the above

Under which condition would a non-metabolized toxicant be expected to accumulate in the body following repeated exposure?

---

- A) when the interval between exposures is less than 5 half-lives
- B) when the rate of absorption is prolonged
- C) when the compound induces its own metabolism
- D) when the compound exhibits flip-flop kinetics
- E) when the bioavailability exceeds 50%



Ethyl alcohol is absorbed and passes into the central nervous system by the process of

---

- A) facilitated transport
- B) simple diffusion
- C) active transport
- D) osmosis
- E) facilitated diffusion

The maximum tolerated dose for a rodent oncogenicity study is

---

- A) 1/10 the LD50 for the compound under study
- B) the same for both sexes of a species
- C) the highest dose level in the subchronic (13-week) study
- D) estimated from a properly conducted subchronic (13-week) study
- E) determined by selecting a dose level which causes a significant body weight decrement

Some azo dyes may be cleaved releasing mutagenic and carcinogenic subunits following which of these metabolic reactions?

---

- A) reduction of the azo bond by intestinal bacteria
- B) acid hydrolysis in the stomach below pH 3.0
- C) cleavage in the urinary bladder by beta-glucuronidase
- D) N-demethylation by hepatic mixed function oxidases
- E) oxidative deamination by mitochondrial amine oxidases

The average total body burden of a toxicant at steady state is approximately

---

- A) 1.5-fold the amount administered per half-life of elimination
- B) 5.0-fold the amount administered per half-life of elimination
- C) 150-fold the amount administered per half-life of elimination
- D) 50-fold the amount administered per half-life of elimination
- E) none of the above

Exposure to ethylene glycol leads to

---

- A) eventual degradation path to glyoxylic acid
- B) several metabolites in urine and blood
- C) main path of glycolic acid is to glyoxylic acid
- D) initially unchanged ethylene glycol excreted in the urine
- E) all of the above

In a demographic study, a group of individuals having common characteristics such as sex, age or employment are defined as

---

- A) control population
- B) treatment group
- C) collective population
- D) cohort
- E) case population

The difference between a measured value and its corresponding true value parameter is called the

---

- A) sampling error
- B) level of significance
- C) standard error
- D) standard deviation
- E) normal distribution

The correlation coefficient ( $r$ ) expresses the relationship between two sets of variables ( $x$  and  $y$ ). Which of the following is true of  $r$ ?

---

- A)  $r$  is described in terms of the variances in both  $x$  and  $y$
- B)  $r$  is described in terms of the variances in  $y$  with respect to  $x$
- C)  $r$  is described in terms of the variances in  $x$  with respect to  $y$
- D)  $r$  is equal to the slope of the line of linear regression
- E)  $r$  is greater than or equal to zero



The ratio of the incidence in exposed workers compared to the incidence in unexposed workers is the

---

- A) relative attributable risk
- B) relative excess risk
- C) relative risk
- D) odds ratio
- E) none of the above

Which of the following statements regarding occupational exposure levels is NOT true?

---

- A) the ACGIH TWA is the average value for a normal 8-hour work day and 40-hour work week
- B) the ACGIH STEL is a value for a short period of time (usually 15 minutes)
- C) the ACGIH TLV-C is an average value not to be exceeded during a normal 8-hour day
- D) a TLV is defined as the airborne concentration of a substance without any adverse effect to nearly all workers exposed during a 40-hour work week
- E) OSHA has adopted some, but not all, of the ACGIH exposure standards

Nitrate toxicity includes all of the following EXCEPT

---

- A) shift to the left of the oxygen-dissociation curve
- B) reflex tachycardia
- C) oxidation of ferrous hemoglobin
- D) orthostatic hypotension
- E) cherry-red venous blood

Activated charcoal may reasonably be expected to

---

- 1) neutralize acids
- 2) sequester thallium salts
- 3) precipitate certain glycosides
- 4) adsorb alkaloids

A plasma half-life is the

---

- 1) time required for 50% of the dose of a drug to be absorbed into the blood stream
- 2) time required for 50% of a given dose of a drug to be metabolically altered
- 3) time required for 50% of the amount of a drug in the body to be excreted
- 4) time required for the plasma level of a drug to fall to 50% of the peak value following a single dose

The principal effect(s) of occupational exposure to freon is/are

---

- 1) pulmonary irritation
- 2) cardiac arrhythmia
- 3) primary hepato- or renal toxicity
- 4) narcosis or loss of consciousness

Which of the following factor(s) is/are generally important in the gastrointestinal absorption of toxicants?

---

- 1) specialized transport systems
- 2) the pH of the blood
- 3) stability in the acid of the stomach
- 4) intestinal lumen diameter

Which of the following are requirements of the Good Laboratory Practice (GLP) standards?

---

- 1) each testing facility shall maintain a current summary of training and experience of each person supervising or engaged in the conduct of a study
- 2) a testing facility shall have standard operating procedures in writing, setting forth study methods that management is satisfied are adequate to insure quality of the study
- 3) records, raw data and specimens shall be archived for at least 10 years except for materials which deteriorate and these shall be retained only as long as they are usable
- 4) each testing facility shall have a quality assurance unit



Mercapturic acids are

---

- 1) breakdown products of glutathione conjugates
- 2) readily excreted into the bile
- 3) formed in rat kidneys
- 4) formed by a microsomal transpeptidase

In a two-compartment open model the

---

- 1) semilogarithmic plot of tissue concentration versus time after exposure yields a non-linear curve
- 2) slope of the initial phase of the plasma concentration versus time curve can be resolved into a straight line using a technique referred to as methods of residuals
- 3) plasma concentration of the chemical in the distribution phase decreases more rapidly than in the post-distribution beta-phase
- 4) rate constants ( $K_{12}$  and  $K_{21}$ ) cannot be calculated from a single dose study

The steady-state distribution of a weak electrolyte across plasma membrane is determined by the

---

- 1) pKa of the electrolyte
- 2) total solute gradient across the membrane
- 3) pH of the media on each side of the membrane
- 4) thickness of the membrane

Drugs that are readily dialyzable are those that

---

- 1) bind to tissues
- 2) have a low volume of distribution
- 3) have a rapid rate of metabolism
- 4) have a slow rate of metabolism

Reproductive effects figure prominently in regulatory actions involving

---

- 1) ethylene oxide
- 2) chromium
- 3) dibromochloropropane (DBCP)
- 4) methyl bromide

Identify the matching synonyms for epidemiological study types

---

- 1) case control study - retrospective study
- 2) cohort study - retrospective study
- 3) cohort study - prospective study
- 4) case control study - matching study

## The linearized multistage procedure

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- 1) leads to a plausible upper limit to carcinogenic risk
- 2) gives an estimate of carcinogenic risk for which the true value cannot be zero
- 3) provides a prediction of carcinogenic risk which may not be realistic
- 4) is recognized as the most appropriate mathematical procedure for low-dose extrapolation in carcinogenesis

A reference dose, or RfD, is a toxicity value

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- 1) commonly used by regulatory agencies to assess noncarcinogenic effects
- 2) derived in part from MTD value(s) from chronic toxicity data
- 3) derived from a NOAEL or LOAEL for the most sensitive or critical toxic effect
- 4) specifically developed for evaluating chronic inhalation exposures



Probabilistic risk assessment is one methodology which explicitly deals with the uncertainties in health risk assessment. Which is/are advantages of probabilistic risk assessment?

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- 1) low dose extrapolations are not used
- 2) single point estimates for exposure are used, which simplifies the calculations and incorporates all uncertainties
- 3) incorporates uncertainty factors (UF) and modifying factors (MF) to derive a risk estimate
- 4) the risk estimate is expressed as a probability distribution which takes into account the range of uncertainty surrounding key variables

Mean residence time is

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- 1) a function of both distribution and elimination
- 2) a ratio of AUMC:AUC
- 3) is analogous to a chemical's biological half-life
- 4) directly related to the volume of distribution

Which of the following statements is/are true?

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- 1) under the U.S. OSHA Hazard Communication regulation, manufacturers are required to develop MSD's to appropriately communicate the hazard
- 2) the Food, Drug and Cosmetic Act obligates drug manufacturers to perform toxicological testing to evaluate safety of a new drug
- 3) the European Community obligates manufacturers importing new chemical substances to submit physical/chemical data and toxicological data if certain import quantities are exceeded
- 4) under the Toxic Substances Control Act, Congress empowered the EPA to require that manufacturers conduct testing of all new chemicals to permit an evaluation of their risks

Which of the following is true relative to the risk assessment process?

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- 1) mortality, reproductive effect and cancer are typical endpoints for hazard identification
- 2) exposure assessments take into account intensity, frequency, route and duration of exposures in populations at risk
- 3) risk characterization requires judgment as to levels of uncertainty in estimates of incidence of adverse effects in a given population
- 4) extrapolation from animal data to effects in humans is often essential in the risk assessment process

Which of the following statements is/are true for a river contaminated with Aroclor 1248 in the sediment?

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- 1) if the sediment has been in place for many years, capping or covering may be a desired remedial alternative, because it would prevent exposure to aquatic species which would occur with dredging activities
- 2) PCBs bioaccumulate in the food chain and a major pathway of exposure to human population will be through fish consumption
- 3) the major concern from a ecological perspective would be piscivorous wildlife such as mink
- 4) PCBs are known human carcinogens and therefore must be remediated to levels protective of public health

The antidotal use of which of the following is/are example(s) of dispositional antagonism?

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- 1) ipecac
- 2) atropine
- 3) osmotic diuretic
- 4) BAL

For which of the following is the U.S. Food and Drug Administration responsible for monitoring?

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- 1) a bacteriostatic in a moisturizing lotion
- 2) a fungicide in drinking water (field run-off)
- 3) a fungicide residue in apples (at harvest)
- 4) an algicide in swimming pool water

For which of the following agents is enzyme inhibition the principal mechanism of toxic action?

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- 1) paraoxon
- 2) fluoroacetate
- 3) dinitrophenol
- 4) aniline



Aquatic static acute toxicity tests provide practical means for

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- 1) deriving estimates of upper limit of concentrations producing toxic effects
- 2) evaluating relative sensitivity of different organisms to the same material
- 3) evaluating relative toxicity of large number of materials
- 4) developing understanding of the concentration-response relationship

## The "margin of safety" of a compound

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- 1) may be measured by the slope of its dose-response curves
- 2) is the magnitude of the range of doses involved in progressing from a threshold dose to a lethal dose
- 3) does not require parallelism between the slopes of the efficacy and toxicity curves to be meaningful
- 4) is a comparison of the estimated chemical exposure to its reference dose

Which of the following adverse effects is/are characteristic for exposure to styrene?

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- 1) narcosis
- 2) nausea
- 3) respiratory irritation
- 4) sterility in males

Which of the following plants does NOT contain toxic alkaloids?

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- A) monkshood - *Aconitum napel*
- B) foxglove - *Digitalis purpurea*
- C) poison hemlock - *Conium maculatum*
- D) green hellebore - *Veratrum viride*
- E) labernum - *Labernum anagyroides*

Which of the following has the highest oxidant value in "photochemical smog"?

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- A) carbon monoxide
- B) ozone
- C) sulfur dioxide
- D) peroxyacetyl nitrate
- E) 3, 4-benzpyrene

The average daily air concentration of a chemical which should not be exceeded if harm to an exposed worker is to be avoided, is referred to as

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- A) NOEL
- B) ED/50
- C) LD/50
- D) threshold limit value (TLV)
- E) LCt/50

Atropine poisoning may produce all of the following signs or symptoms EXCEPT

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- A) mydriasis
- B) hypothermia
- C) tachycardia
- D) hallucinations
- E) dry mouth and skin

A worker in a munitions plant developed symptoms of nausea and vomiting, hypertension, progressive liver damage and mandibular necrosis. The most likely cause of these symptoms is

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- A) potassium chlorate
- B) phosphorus (white)
- C) nitroglycerine
- D) sodium nitrate
- E) picric acid



Which of the following statements about dimercaprol is NOT true?

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- A) it contains two sulfhydryl groups
- B) it complexes with arsenic and mercury
- C) it is useful for promoting the excretion of lead
- D) it is used in the treatment of arsenic intoxication
- E) it has serious side-effects

Which of the statements or phrases is NOT true?

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- A) red squill - the toxic and pesticidal action against rodents is associated with its cardiac effects
- B) warfarin - a hydroxycoumarin compound which reduces the prothrombin levels in blood
- C) parathion - anticholinesterase insecticide
- D) dieldrin - a chlorinated hydrocarbon insecticide, more toxic than DDT
- E) 2-PAM (pralidoxime) - useful as an antidote for carbaryl (Sevin)

Biological activity of the methylenedioxyphenyl class of insecticide synergists (piperonyl butoxide)

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- A) initially inhibit microsomal mixed function oxidases, but can later induce them
- B) initially stimulate microsomal mixed function oxidases, then later inhibit them
- C) stimulate glutathione transferase activity
- D) A and C
- E) A, B and C

A basic drug with a  $pK_a$  of 3 is effective orally. Assume that the pH of the stomach is 1, the pH of the small intestine is 6 and the pH of the plasma is 7.4. Under these conditions, the drug would be expected to

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- A) be absorbed into the blood stream more rapidly from the stomach than from the small intestine
- B) be absorbed into the blood stream more rapidly from the small intestine than from the stomach
- C) be absorbed into the blood stream equally rapidly from the stomach and the small intestine
- D) not be absorbed from either the stomach or the small intestine
- E) pass principally from plasma into the stomach

The concentration gradient for passive diffusion between blood and tissues is determined by the concentration difference between

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- A) protein bound drug in the blood and free drug in the tissue
- B) protein bound drug plus free drug in the blood and free drug in the tissue
- C) free drug in the blood and free drug plus protein bound drug in the tissue
- D) free drug in the blood and free drug in the tissue
- E) protein bound drug plus free drug in the blood and protein bound drug plus free drug in the tissue

The renal clearance of a drug with a half-life of 78 minutes and an apparent volume of distribution of 14 liters is approximately

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- A) 1 ml/min
- B) 10 ml/min
- C) 100 ml/min
- D) 700 ml/min
- E) 7000 ml/min

Oximes, such as pralidoxime methiodide (2-PAM), have antidotal action in organophosphate poisoning by which of the following mechanisms?

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- A) competitive inhibition with acetylcholine at receptor sites
- B) physiological antagonism
- C) nucleophilic displacement of the organophosphorus inhibitor from the active site of acetylcholinesterase
- D) hydrolysis of phenylacetate
- E) none of the above

Lead poisoning inhibits heme synthesis leading to an increased urinary excretion of a number of heme precursors. Which of the following will appear first with increasingly severe poisoning?

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- A) uroporphyrinogen III
- B) delta aminolevulinic acid
- C) protoporphyrin III
- D) coproporphyrin III
- E) protoporphyrin IX



## FDA regulations for direct food additives

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- A) are set at a maximum limit for use based on specific review conducted by the agency
- B) allow manufacturers to use the minimum quantity required to produce the intended effect
- C) set limits by acceptance of guidelines developed by non-FDA review bodies
- D) A and C
- E) all of the above

Thiamine deficiency can be produced from the long term ingestion of

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- A) field horsetail
- B) foxglove
- C) deadly nightshade
- D) mushrooms
- E) rhubarb

Which of the following common garden plants contain(s) solanine?

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- A) potato
- B) carrot
- C) turnip
- D) red beet
- E) A and C

An inhalation toxicity study of chemical 'X' is being conducted with exposure concentrations of 1.0 and 5.0 mg/l. 'X' has a vapor pressure of 0.038 mm Hg (23 degrees C), a molecular weight of 240 gm/mole, and a density of 0.5. The chambers (4 cubic meters) are operated under dynamic airflow conditions at 800 l/min. A 23 degree C atmosphere saturated with vapors of 'X' would contain approximately which one of the following concentrations?

---

- A) 0.5 mg/l
- B) 2.0 mg/l
- C) 5.0 mg/l
- D) 20 mg/l
- E) 200 ppm

Sodium fluoroacetate (1080) poisoning is a result of blockage of which of the following?

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- A) cytochrome oxidase system
- B) synaptic transmission
- C) drug-metabolizing system
- D) Kreb's cycle
- E) glycolysis

Fat accumulation in liver cells following excessive halogenated hydrocarbon exposure is thought to be

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- A) due to a reduction in the amount of triglyceride lipoprotein complex resulting in reduced triglyceride removal from the cells
- B) due to mobilization of fatty acids from body stores causing increased triglyceride synthesis
- C) an early stage of cell necrosis due to altered mitochondrial function
- D) caused by an increased level of circulating catecholamines resulting in hepatocellular hypoxia
- E) directly related to increased intracellular availability of glycerol for triglyceride synthesis

Formation of nitrosamines in the gastrointestinal tract is blocked by

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- A) vitamin E
- B) vitamin C
- C) thiamine
- D) A and B
- E) A and C

Which of the following statements is true?

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- A) resistance to warfarin toxicity in certain strains of rats has been attributed to a decreased ability to metabolize warfarin to a more toxic metabolite
- B) warfarin is used clinically to treat peptic ulcer and polycythemia
- C) treatment of warfarin toxicity by either Vitamin K or Vitamin K oxide is equally effective
- D) warfarin's three direct actions: inhibition of prothrombin formation, capillary damage, and thrombocytopenia can all be attributed to a single moiety of the molecule, 4-hydroxycoumarin
- E) the pathologic lesion warfarin produces is in the blood, and the actual mechanism of action occurs in the liver



In carbon monoxide poisoning, which of the following statements is NOT true?

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- A) symptoms of carbon monoxide poisoning are: headache, weakness, nausea, dizziness and dimness of vision
- B) carboxyhemoglobin is formed
- C) patients with carbon monoxide poisoning have bluish-tinted skin
- D) oxygen is administered to revive a patient
- E) poor correlation exists between the blood content of carboxyhemoglobin and signs and symptoms in acutely poisoned humans

The deposition of asbestos fibers in the lung occurs mainly by the process of

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- A) interception
- B) impaction
- C) sedimentation
- D) diffusion
- E) no particular deposition process predominates

One of the reported effects attributed to the intake of molybdenum is

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- A) impaired vision and weakness in the limbs of cattle, a condition known as "blind stagger"
- B) a central nervous system condition similar to Parkinson's disease
- C) "Shaver's disease", characterized by emphysema and pulmonary fibrosis
- D) striped wool in black sheep when intake is alternated with copper
- E) none of the above

Which of following solvents have been implicated as causing peripheral neuropathy due to their conversion to a toxic metabolite?

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- A) n-hexane and methyl n-butyl ketone
- B) n-pentane and methyl propyl ketone
- C) 2-methyl pentane and methyl isobutyl ketone
- D) propane and methyl ethyl ketone
- E) Stoddard solvent and toluene

Fractional deposition of dust particles in the respiratory tract is LEAST affected by the

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- A) concentration of the dust particles in the inhaled air
- B) mass median aerodynamic diameter of the dust particles
- C) standard geometric deviation of the dust particles
- D) breathing pattern of the subject
- E) geometry of the respiratory tract

Which of the following correctly identifies a source and mechanism of toxic action of batrachotoxin?

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- A) Colombian arrow poison frog; preventing inactivation of sodium channels, resulting in sodium influx and persistent membrane depolarization
- B) puffer fish; blocking neural conduction by preventing movement of potassium ions across cell membranes, resulting in persistent polarization of the membranes
- C) dinoflagellates; blocking the potassium transport channels in the membranes of excitable cells
- D) brown recluse spider; blocking myoneural junctions, causing an ascending paralysis
- E) Western diamondback rattlesnake; preventing activation of potassium channels, resulting in persistent repolarization

In most cases, the acute toxicity of natural uranium (a mixture of U235 and U238) following inhalation would be due to

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- A) kidney toxicity
- B) beta radiation dose to the lung
- C) alpha radiation dose to the lung
- D) liver toxicity
- E) alpha radiation dose to the bone marrow

Which of the following classes of antibiotics is most commonly associated with ototoxicity and nephrotoxicity?

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- A) penicillins, e.g., ampicillin
- B) fluoroquinolones, e.g., ciprofloxacin
- C) aminoglycosides, e.g., gentamicin
- D) tetracyclines, e.g., chlortetracycline
- E) macrolides, e.g., erythromycin



Benzene metabolite(s) reportedly involved in its myelotoxicity include

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- A) muconaldehyde
- B) glycolic acid
- C) catechol
- D) bromobenzene oxide
- E) A and C

Which of the following is/are commonly used in calculating the acceptable daily intake (ADI) for a food additive?

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- 1) highest dose that gives rise to a no-observed-adverse-effect
- 2) likely human consumption (i.e. estimated daily intake)
- 3) one-hundred-fold safety factor
- 4) foods in which it will appear

## Poison oak dermatitis

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- 1) is a primary irritant dermatitis caused by pentadecylcatechols
- 2) is caused by a Type I hypersensitivity response to urushiols
- 3) shows cross reactivity with dermatitis caused by chromium salts
- 4) is caused by the same mechanism by which formaldehyde causes dermatitis

The following criteria are generally acceptable for the selection of the high dose for carcinogenicity studies of drugs

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- 1) 25-fold AUC ratio (rodent : human)
- 2) dose-limiting pharmacodynamic effects
- 3) saturation of absorption
- 4) maximum tolerated dose

The actions of ephedrine include

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- 1) alpha and beta adrenergic effects
- 2) bronchodilation
- 3) hypertension and arrhythmias at toxic doses
- 4) sympathomimetic effects

Which of the following block(s) uptake of iodide by the thyroid?

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- 1) coumarin
- 2) thiocyanate
- 3) salicylate
- 4) constituents of cabbage

The treatment for cyanide intoxication is based on the

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- 1) high affinity of cyanide for hemoglobin
- 2) conversion of hemoglobin to methemoglobin by nitrates
- 3) conversion of hemoglobin to methemoglobin by sulfite
- 4) detoxification of cyanide to thiocyanate

The treatment of a patient suffering from an overdose of aspirin may include which of the following?

---

- 1) gastric lavage or induction of emesis
- 2) administration of a respiratory stimulant
- 3) maintenance of plasma volume and electrolyte balance
- 4) administration of  $\text{NH}_4\text{Cl}$  to decrease urinary pH



Ethyl alcohol produces diuresis by

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- 1) increasing glomerular filtration rate
- 2) inhibiting carbonic anhydrase
- 3) blocking proximal absorption of sodium
- 4) inhibiting antidiuretic hormone secretion

Kernicterus may be produced in neonates by

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- 1) phototherapy
- 2) phenobarbital
- 3) protamine sulfate
- 4) sulfonamides

Which of the following is/are true for coal worker's pneumoconiosis (CWP)?

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- 1) The characteristic lesion of CWP is a focal macule associated with perifocal emphysema
- 2) the lungs of workers with CWP may be intensely black, but measurements of pulmonary function usually show minimal effect
- 3) exposure to coal dust may be complicated by simultaneous exposure to silicon dioxide leading to progressive massive fibrosis
- 4) CWP has not been associated with lung cancer because smoking is not allowed in mines

Which of the following is true of bagassosis?

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- 1) caused by a thermophilic actinomycosis
- 2) is uniformly fatal
- 3) is characterized in acute stages by a hypersensitivity pneumonitis
- 4) occurs more frequently in females

The voltage-dependent sodium channel is a primary target for

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- 1) brevetoxins
- 2) saxitoxins
- 3) ciguatoxins
- 4) gonyautoxins

## Recommended Dietary Allowances (RDA) are?

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- 1) applicable to intentional food additives
- 2) recommended in the USA by the Food and Nutrition Board of the National Academy of Science
- 3) set based on the NOAEL obtained from animal toxicity, data and an appropriate safety factor
- 4) set at levels sufficiently high to cover the need of almost all healthy individuals in specified age and sex category with the exception of the recommended intakes of energy

Important components of photochemical air pollution that lead to the production of smog include

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- 1) ozone
- 2) nitrogen oxides (NO<sub>x</sub>)
- 3) aldehydes
- 4) hydrocarbons

## In acetaminophen overdose

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- 1) greater percentage of the ingested dose as compared to the therapeutic dose is biotransformed by the cytochrome p-450 mixed function oxidase system to form an oxidizing electrophilic metabolite
- 2) mainly reactive nucleophilic metabolites are formed
- 3) cysteinyl residue of glutathione detoxifies the electrophilic metabolite by providing a nucleophilic thiol group
- 4) the nucleophilic metabolites are excellent substrates for glutathione-s-transferase resulting in glutathione depletion



BAL (British anti Lewisite) is used in the treatment of

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- 1) cadmium toxicity
- 2) selenium toxicity
- 3) tellurium toxicity
- 4) arsenic toxicity

Peripheral neuropathies have been reported in Italian and Japanese shoe factory workers, and in the printing and painting industries. This condition is apparently caused by a gamma-diketone metabolite from which of the following?

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- 1) N-hexane
- 2) methyl isobutyl ketone
- 3) 2-hexanone
- 4) 1,4-hexanedione

Which of the following statements is/are true?

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- 1) the roentgen (Coulomb per Kg of air) is a unit of radiation exposure
- 2) the Gray (rad) is a measurement of energy deposition in any medium by any type of ionizing radiation
- 3) the Sievert (rem) takes into consideration such modifying factors of dose as the quality of the radiation
- 4) the quality factor for various types of ionizing radiation is based on the linear energy transfer of the type of radiation

Acute, high level vapor exposure to many different organic solvents produces similar and rapid toxic effects, i.e. disorientation, confusion, paralysis, convulsion and respiratory or cardiac arrest. The similarity of the responses to solvents of diverse structure suggests

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- 1) solvent vapor toxicity is mediated by hypoxia
- 2) physical interaction of organic solvents with CNS structures
- 3) that different organic solvents are metabolized to similar toxic compounds
- 4) equimolar concentrations of different solvents at the target tissues will produce effects of equal intensity

Which of the following tests can best detect point mutations?

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- A) cytogenetics
- B) micronucleus
- C) specific locus
- D) dominant lethal
- E) heritable translocation

Which of the following is LEAST likely to detect liver disorders in rodents?

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- A) SGPT activity
- B) serum alkaline phosphatase activity
- C) creatinine phosphokinase (CPK)
- D) serum bilirubin
- E) histopathology

Which is NOT correct?

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- A) fetal hemoglobin is more easily oxidized than that of a one-year old child
- B) in utero diethylstilbestrol exposure has been associated with vaginal cancer
- C) isoniazid is likely to cause neurotoxicity in patients with atypical cholinesterase
- D) chloramphenicol is poorly metabolized in the newborn
- E) succinylcholine is a muscle relaxant

Allergic responses to chemicals result from damage to target cells that release which of the following?

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- A) histamine
- B) slow reacting substance (anaphylactoid srs-a)
- C) serotonin
- D) kinins
- E) all of the above



The state in which most chemicals cross the placenta is

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- A) in the protein bound form
- B) as a free chemical
- C) as a metabolite
- D) in the ionized form
- E) in the conjugated form

Osmolarity of urine is primarily controlled in the

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- A) glomerulus
- B) proximal tubule
- C) loop of Henle
- D) distal tubule
- E) collecting duct

The principal site of kidney damage by the analgesic mixture of aspirin and phenacetin is which of the following

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- A) proximal tubule
- B) distal tubule
- C) loop of Henle
- D) glomerulus
- E) arterioles

Which of the following statements is correct?

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- A) percutaneous absorption is most efficient when the absorption agent is highly hydrophilic
- B) hexachlorophene can be more hazardous by dermal application than by oral administration
- C) most of the procarcinogens when applied dermally cause tumors at remote sites because the skin does not have an inducible cytochrome p-450 system
- D) percutaneous absorption is not influenced by the nature of the solvent
- E) photoallergic reactions are more common than phototoxic reactions

Measurement of which of the following should NOT be used to assess glomerular function?

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- A) para-aminohippurate excretion
- B) inulin clearance
- C) blood urea nitrogen
- D) urinary albumin
- E) plasma creatinine

Which of the following substances is most likely to cause cirrhosis in the liver?

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- A) alpha-naphthylisothiocyanate (ANIT)
- B) folic acid
- C) halothane
- D) ethanol
- E) tannic acid

Which of the following agents exerts its primary nephrotoxic effects on the glomerulus?

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- A) phenacetin
- B) amphotericin B
- C) cyclosporine
- D) mercuric chloride
- E) potassium bichromate

The relative regional permeability of human skin has been described for hydrocortisone. Ranking from most permeable to least permeable is as follows

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- A) scrotum, forehead, palm, plantar foot arch
- B) scrotum, plantar foot arch, palm, forehead
- C) scrotum, palm, plantar foot arch, forehead
- D) all sites are relatively the same in permeability
- E) none of the above



Analysis of fructose levels in human seminal fluid can aid in the diagnosis of abnormal functioning of the

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- A) epididymus
- B) sertoli cells
- C) leydig cells
- D) seminal vesicles
- E) germinal epithelium

Which of the following is correct with regard to eye irritants?

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- A) surfactants cause damage to protein principally through the lowering of surface tension
- B) prognosis for recovery immediately after exposure to mustard gas is always apparent
- C) the pH of acidic and basic solutions is the only factor for the type of lesion produced
- D) pain and dulling of the cornea after exposure to neutral solvents are both associated with lipid solubility
- E) lacrimators inhibit sulfhydryl enzymes but have only transient effects

An FDA segment I study is designed to detect effects

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- A) caused by accumulated toxicity
- B) on parturition, lactation, neonatal development
- C) on embryonic development and organogenesis
- D) on mating, fertility, implantation, gestation, parturition, lactation, neonatal development
- E) on function and behavior in offspring

The phthalate acid ester di(2-ethylhexyl)phthalate is

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- A) a hepatic carcinogen in rats and mice
- B) a non-genotoxic carcinogen, (i.e. not known to have a direct effect on DNA)
- C) known to induce peroxisome proliferation which has been postulated to result in the formation of reactive oxygen species
- D) is a testicular toxicant in rodents
- E) all of the above

Glutathione tissue levels in the liver would be expected to markedly affect the hepatotoxicity of

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- A) acetaminophen
- B) bromobenzene
- C) carbon tetrachloride
- D) halothane
- E) A and B

Which of the following structures is among the first to develop during gestation in the rat?

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- A) brain
- B) aorta
- C) urogenital system
- D) pulmonary system
- E) digestive system

All of the following agents alter adrenergic neurohumoral transmission EXCEPT

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- A) methyldopa
- B) cocaine
- C) botulinum toxin
- D) phenylephrine
- E) phenoxybenzamine

Alopecia is an important clinical diagnostic clue in

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- A) thallium toxicity
- B) botulinum toxicity
- C) arsenic toxicity
- D) triorthocresyl phosphate toxicity
- E) methylmercury toxicity



Which statement concerning allergic contact dermatitis is true?

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- A) occurs as a result of a type III immune response
- B) the elicitation phase is the period when the sensitization develops
- C) the elicitation phase occurs when effector T lymphocytes become activated in response to a hapten-protein complex
- D) strong allergens are most often water soluble with molecular weights greater than 500
- E) the Index of Sensitivity is the relative capacity of an agent to induce sensitization

Restrictive pulmonary diseases are those diseases which result in

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- A) increased pulmonary compliance
- B) decreased pulmonary compliance
- C) increased resistance to air flow
- D) decreased resistance to air flow
- E) decreased blood flow due to pulmonary vasoconstriction

A nonsense mutation can result in which of the following?

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- A) a fully functional gene product
- B) the gain or loss of one or more base pairs in a gene
- C) a nonfunctional gene product because of premature termination of protein synthesis
- D) a fully functional gene with altered substrate specificity
- E) none of the above

Particles having a diameter of 5  $\mu\text{m}$  or larger when inhaled are largely deposited in the

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- A) tracheobronchial region
- B) nasopharyngeal region
- C) bronchioles
- D) alveoli
- E) none of the above

The interaction of alkylating agents with DNA can produce which of the following adducts?

---

- A) O6-alkylguanine
- B) C8-alkylguanine
- C) N7-alkylguanine
- D) all of the above
- E) A and C

The use of dantrolene in malignant hyperthermia is based upon its ability to

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- A) increase blood flow to muscles, thus increasing the rate at which heat is dissipated
- B) produce an increase in the respiratory rate which reduces the temperature of blood in the lungs and hence the body
- C) block release of calcium within muscle cells hence reducing muscle tone and heat production
- D) increase the rate of ATP synthesis by myocytes which results in muscle tone and heat production
- E) none of the above

Which of the following most accurately reflects the hierarchy of sensitivity to toxic agents from the most sensitive to the most resistant?

---

- A) Clara cells > Type II alveolar cells > Type I alveolar cells
- B) Type II alveolar cells > mucous cells > ciliated cells
- C) Type I alveolar cells > Clara cells > Type II alveolar cells
- D) mucous cells > ciliated cells > Type II alveolar cells
- E) ciliated cells > mucous cells > Type I alveolar cells

Thymic atrophy is a hallmark of toxicity which of the following classes of chemicals?

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- A) halogenated aromatic hydrocarbons
- B) polybrominated biphenyls
- C) dibenzodioxins
- D) polycyclic aromatic hydrocarbons
- E) all of the above



People with atopy have

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- A) diminished threshold for irritation and defective cell-mediated immunity
- B) diminished threshold for irritation and enhanced cell-mediated immunity
- C) increased threshold for irritation and enhanced cell-mediated immunity
- D) increased threshold for irritation and defective cell mediated immunity
- E) increased threshold for irritation and normal cell mediated immunity

Which of the following genotoxicity assays is based on a mutation at the thymidine kinase locus?

---

- A) Ames Salmonella assay
- B) mouse lymphoma assay
- C) Chinese hamster ovary assay
- D) mouse micronucleus test
- E) unscheduled DNA synthesis assay

The cell type that is primarily responsible for production of collagen in hepatic fibrosis is the

---

- A) Kupffer cell
- B) hepatocyte
- C) Ito cell
- D) monocyte, following its differentiation into a macrophage
- E) Sertoli cell

Which of the following is/are true integrative processes of the hypothalamo-pituitary-gonadal axis?

---

- 1) FSH and LH are glycoproteins synthesized and released from a subpopulation of the basophilic gonadotropic cells of the pituitary gland
- 2) luteinizing hormone-releasing hormone (LH-RH) acts on gonadotropic cells, thereby stimulating the release of FSH and LH
- 3) LH-RH and follicle-stimulating hormone-releasing hormone (FSH-RH) appear to be the same substance
- 4) gonadotropins are not important in the female

Which area(s) in the central nervous system are thought to have incomplete protection from blood-borne toxicants?

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- 1) choroid plexus
- 2) frontal cortex
- 3) area postrema
- 4) cerebellum

The theoretical time for a test compound gas to reach 99% equilibrium in an inhalation chamber for a selected target concentration is termed the  $t_{99}$ . In addition to the equation constant what measurements are needed to calculate the  $t_{99}$ ?

---

- 1) chamber volume
- 2) chamber target concentration of the test compound
- 3) chamber airflow rate
- 4) molecular weight of the test compound

Which of the following tests are routinely used to measure the effect of immunotoxicants on B-cell activity?

---

- 1) plaque-forming cell response to sheep red blood cells
- 2) complement depletion
- 3) immunoglobulin quantitation
- 4) delayed hypersensitivity reactions

Which of the following suggest(s) that DNA is a critical target in carcinogenicity?

---

- 1) many carcinogens are also mutagens
- 2) most, if not all, cancers display chromosomal abnormalities
- 3) many carcinogens can be metabolized to electrophiles that react covalently with DNA
- 4) nuclear DNA levels are markedly decreased in cancerous tissues



Which of the following compounds produce(s) cholestasis?

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- 1) amitriptylene
- 2) haloperidol
- 3) chlorpromazine
- 4) aflatoxin

The rate of dermal absorption of toxicants may be facilitated by

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- 1) increasing the toxicant's lipophilicity (for example, 2,4-D acid change to 2,4-D ester)
- 2) increasing skin blood flow
- 3) choice of solvent/vehicle
- 4) increasing skin water content

Which of the following statements concerning the kidney is/are true?

---

- 1) phenobarbital and salicylates will be cleared more rapidly from the plasma after IV administration of sodium bicarbonate
- 2) amphetamines will be cleared more rapidly from the plasma after IV administration of ammonium chloride
- 3) the mechanism for secretion of organic compounds is localized in the proximal tubule
- 4) because of high blood flow to the medulla, it is very sensitive to toxic insult

Which of the following statements is/are true?

---

- 1) the blood-brain barrier displays a developmental process of formation during which the selectivity of the barrier becomes greater
- 2) the blood-brain barrier is formed primarily by a specialized layer of glial cells surrounding the blood vasculature
- 3) tight junctions are present between adjacent vascular endothelial cells
- 4) small, charged molecules tend to penetrate the blood-brain barrier readily

Which of the following is/are true of p53?

---

- 1) loss or inactivation of the p53 tumor suppressor gene is a common molecular lesion in human neoplasms
- 2) introduction of wild type p53 into p53-negative tumor cells inhibits proliferation by implementing a G1 arrest
- 3) cells from p53 knockout mice fail to arrest in G1 following gamma irradiation
- 4) p53 growth-arrest is mediated via p53-dependent transactivation of genes

Which of the following is/are true regarding alpha-2u-globulin?

---

- 1) production of alpha-2u-globulin is a characteristic of some chemicals that produce renal tumors
- 2) alpha-2u-globulin is not seen in female rats, mice or most other laboratory animals
- 3) chemicals that induce alpha-2u-globulin formation do not generally react with DNA and are also negative in short term tests for genotoxicity
- 4) chemicals that induce the alpha-2u-globulin are generally of greater concern for humans than nephrotoxicants that do not induce this protein

Which of the following is/are a correct pairing(s) of a toxicant and its effect?

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- 1) hepatocellular carcinoma - aflatoxin B1
- 2) cirrhosis - ethanol
- 3) cholestasis - contraceptive steroids
- 4) veno-occlusive disease - pyrrolizidine alkaloids

## Congenital cleft palate

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- 1) is due to in utero exposure to the causative agent relatively late in the embryonic period of development in animal models
- 2) is primarily due to initial normal fusion of the palatal shelves and secondary necrosis at the fusion line
- 3) can be caused by antiepileptic drugs in humans and animal models
- 4) exhibits relatively constant spontaneous incidences across different human races and across different rat and mouse strains



When basing carcinogenic risk assessment on the assumption of no threshold, which of the following statements is/are correct?

---

- 1) the shape of the dose-response curve is likely to be linear at low doses
- 2) the response of the population follows a specific probability distribution function
- 3) a cancer can develop from a single cell that has undergone a number of transformations
- 4) the mechanisms causing the control group's cancer are different from those causing the treated group's cancer

Blood is carried to the liver by the

---

- 1) hepatic vein
- 2) portal vein
- 3) lymphatics
- 4) hepatic artery

Which of the following industrial chemicals are high risk hazards for type IV allergic reactions (cell-mediated, delayed hypersensitivity) ?

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- 1) trichlorethylene
- 2) isocyanates
- 3) vinyl acetate
- 4) mercaptobenzothiazole

Which of the following are direct acting carcinogens?

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- 1) N-2-fluorenylacetamide
- 2) triethylenemelamine
- 3) aflatoxin B1
- 4) ethylmethane sulfonate

In two stage skin carcinogenesis studies, tumor promoting activity has been shown by

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- 1) phenobarbital
- 2) simple phenols
- 3) retinoids
- 4) phorbol esters

Which of the following is NOT an example of a metabolic oxidation?

---

- A) the formation of an alcohol from an aliphatic hydrocarbon
- B) the conversion of an alcohol to an acid
- C) the formation of an epoxide
- D) the methylation of an amino group
- E) the formation of an aldehyde from an alcohol

Which of the following are true for saturable plasma clearance described by Michaelis-Menten kinetics?

---

- A) first order when plasma concentration,  $C$ , is less than the Michaelis constant
- B) first order when  $C$  is much larger than the Michaelis constant
- C) zero order when  $C$  is less than the Michaelis constant
- D) zero order when  $C$  is equal to the Michaelis constant
- E) the Michaelis constant is not useful for describing concentration dependent kinetics

The dose response curve is critical for which of the following?

---

- A) estimation of risk
- B) choice of species for study
- C) selection of an antidote
- D) determination of side effect frequency
- E) establishment of the frequency of dosing



The LD/50 of a mixture containing 25 parts of A (LD/50 = 250 mg/kg) and 75 parts of B (LD/50 = 750 mg/kg) was determined to be 625 mg/kg. The dose-response curves for a and b, each given alone, are parallel. Which statement is correct?

---

- A) the predicted LD/50 of this mixture is 1000 mg/kg. This represents antagonism
- B) the predicted LD/50 equals the observed LD/50, this represents summation
- C) the predicted LD/50 equals the observed LD/50, this is synergism
- D) the predicted LD/50 is 1000 mg/kg. This is synergism
- E) the predicted LD/50 = 500 mg/kg. This is antagonism

The probit plot is most accurate for estimating which of the following?

---

- A) ED1
- B) ED50
- C) ED99
- D) no observed adverse effect level (NOAEL)
- E) threshold limit value (TLV) or maximum acceptable concentration (MAC)

A substance of molecular weight 100,000 administered intravenously would generally be

---

- A) confined to plasma
- B) distributed in extracellular water
- C) distributed in intracellular water only
- D) distributed in total body water
- E) highly concentrated in cells

A first-order decline in concentration of an agent the blood means that the

---

- A) log concentration versus time plot gives a straight line
- B) log time versus drug concentration plot gives a straight line
- C) decline is not dependent on the concentration of the agent present
- D) concentration plotted against time is a straight line
- E) all of the above

A toxicant would be expected to accumulate in the body during a period of repeated exposure when the

---

- A) volume of distribution is large
- B) half-life is shorter than the exposure interval
- C) half-life is longer than the exposure interval
- D) toxicant is absorbed into soft tissues
- E) rate of absorption is greater than the rate of excretion

Under which condition would a nonmetabolized toxicant be expected to accumulate in the body following repeated exposure?

---

- A) when the interval between exposures is less than 5 half-lives
- B) when the rate of absorption is decreased
- C) when the compound induces its own metabolism
- D) when the compound exhibits "flip-flop" kinetics
- E) when the bioavailability exceeds 50%

Facilitated diffusion processes are characterized by

---

- A) metabolic energy requirement, 1st order kinetics at high doses, no accumulation against a net electrochemical gradient, structural specificity
- B) no energy requirement, 1st order kinetics at high doses, no accumulation against a net electrochemical gradient, structural specificity
- C) no energy requirement, zero order kinetics at high doses, no accumulation against a net electrochemical gradient, structural specificity
- D) energy requirement, zero order kinetics at high doses, accumulation against a net electrochemical gradient, structural specificity

E) no energy requirement, 1st order kinetics at high doses, no accumulation against a net electrochemical gradient, no structural specificity



Opisthotonus, which may occur following a toxic dose of strychnine, refers to

---

- A) paralysis of involuntary muscles
- B) labored breathing
- C) back arched in hyperextension with only the head and feet touching the ground
- D) muscles of the face and neck relaxed
- E) all of the above

Which of the following substances is regulated specifically under the Toxic Substances Control Act?

---

- A) polychlorinated biphenyls (PCB's)
- B) parathion
- C) butylated hydroxyanisole
- D) plutonium
- E) all of the above

Which of the following is NOT true? A material is considered to be hazardous under the OSHA Hazard Communication Standard if it has

---

- A) been listed as a carcinogen by the National Toxicology Program, the International Agency on Research on Cancer. Or by OSHA
- B) been listed in the Threshold Limit Values for Chemical Substances and Physical Agents in the Work Environment by the ACGIH
- C) has been found to be a health hazard based in at least one positive study conducted in accordance with established scientific principles
- D) a positive finding in the Ames microbial mutagenicity assay
- E) a dermal LD50 of 200 mg/kg or less in rabbits

When 300 mg of a substance is given intravenously, a plasma concentration of 60 mg/liter is obtained. The volume of distribution is

---

- A) 3 liters
- B) 5 liters
- C) 120 ml
- D) 200 ml
- E) 500 ml

Resistance to the pharmacologic effects of some drugs can be rapidly acquired following only a few doses of the agent over a period of hours. Subsequent doses of the agent fail to elicit the response regardless of the size of the dose. This phenomenon is known as

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- A) tolerance
- B) idiosyncrasy
- C) physical dependence
- D) tachyphylaxis
- E) none of the above

Barbiturates with a high lipid:water partition coefficient are largely

---

- A) bound to plasma protein, poorly filtered and readily reabsorbed from the renal tubular lumen
- B) bound to plasma protein, readily filtered and poorly reabsorbed from the renal tubular lumen
- C) unbound, readily filtered and poorly reabsorbed from the renal tubular lumen
- D) unbound, poorly filtered and readily reabsorbed from the renal tubular lumen
- E) none of the above

Which of the following statistical tests would be most appropriate to compare the mean values for 4 experimental groups when the groups are from normally distributed populations with the same variance?

---

- A) chi-square
- B) student t
- C) paired t
- D) Kruskal-Wallis
- E) one-way anova

Federal government scientific policy regarding carcinogenesis asserts that

---

- A) no single mathematical procedure is recognized as the most appropriate for low-dose extrapolation in carcinogenesis
- B) pharmacokinetics and target organ dose are irrelevant in risk assessment
- C) quantification of the various sources of uncertainty involved in cancer risk assessment is not as important as the projection of the risk estimate itself
- D) exposure assessments should be independent of estimates of cancer risk for humans
- E) decisions on the carcinogenicity of chemicals in humans should be based only on data indicative of a positive response



In determining whether the mean body weight of a group of 34 rats receiving drug X is significantly less than that of a group of 35 rats receiving drug Y, the investigator applied a chi-square test and selected alpha to equal 0.5. He reported a p value of 0.04. Which of the following is correct?

---

- A) the mean body weights showed no significant difference
- B) drug X produced a significant decrease in body weight compared to drug Y
- C) the experiment should be repeated using equal numbers of animals in each group
- D) a different statistical analysis should be used
- E) none of the above

The following is a matched list of agents and occupations associated with asthma. Which agent is NOT commonly associated with the occupation given?

---

- A) *Bacillus subtilis* enzymes -- detergent workers
- B) gum acacia -- printers
- C) psyllium -- pharmaceutical workers
- D) diisothiocyanate -- cabinet makers
- E) phthalic anhydride -- meat wrapper labellers

The concentration that should NOT be exceeded during any part of the working exposure is termed

---

- A) TLV: threshold limit value
- B) TWA: time-weighted average
- C) TLV-TWA: threshold limit value -time-weighted average
- D) TLV-STEL: threshold limit value -short-term exposure limit
- E) TLV-C: threshold limit value-ceiling

Which of the following is primarily used in the treatment of coumarin or warfarin overdose?

---

- A) atropine
- B) nonsteroidal anti-inflammatory agents
- C) N-acetylcysteine
- D) chlorpromazine
- E) vitamin K

As the administered dose of a xenobiotic increases, so does the probability of saturating the elimination processes of the compound, leading to conditions of zero-order elimination. The plasma concentration vs. time curve after single saturating doses is most likely to be

---

- A) concave as a semilog plot, linear as a cartesian plot
- B) linear as a semilog plot, convex as a cartesian plot
- C) convex as a cartesian plot, linear as a cartesian plot
- D) linear as a semilog plot, concave as a cartesian plot
- E) bi-exponential

The subfamily of cytochrome P-450 (CYP) enzymes that primarily metabolizes polycyclic aromatic hydrocarbons is

---

- A) CYP-450 1A
- B) CYP-450 2B
- C) CYP-450 2D
- D) CYP-450 2E
- E) CYP-450 3A

Short Term Exposure Limits (STELs) are

---

- A) worker exposure limits for a single 15 minute interval during normal work cycle
- B) promulgated by the United States Environmental Protection Agency
- C) based on carefully controlled studies in humans
- D) based on subchronic exposure studies in animals
- E) none of the above

Synthetic steroids (carcinogenic compounds) are allowed in food producing animals if

---

- A) no residues are detectable in edible tissue of treated animals
- B) an analytical method capable of detecting residues large enough to pose a lifetime risk greater than 1 in 1,000,000 has been developed
- C) "negligible risk" based on cancer potency data can be determined
- D) all of the above



A ten year old, 40 kg child ingested 101 ml of a 40% ethanol (specific gravity - 0.79) solution. If the volume of distribution ( $V_d$ ) for ethanol is 0.7 l/kg, what is the predicted peak blood ethanol level assuming instantaneous distribution?

---

- A) 1.14 g/l
- B) 2.85 g/l
- C) 0.114 g/l
- D) 1.14 g/ml
- E) 2850 mg/ml

Which of the following has NOT been implicated in causing "metal fume fever"?

---

- A) aluminum
- B) beryllium
- C) copper
- D) iron
- E) tin

Before you could perform a Western blot analysis of a tissue to determine if an enzyme of interest was present, you would need

---

- A) a cDNA or specific oligonucleotide for the gene
- B) an antibody specific for the enzyme in question
- C) an enzyme assay capable of selective detection of the activity of the enzyme
- D) a sample of intact DNA from the tissue
- E) both A and D

The mechanism of toxicity of phthalate esters is believed to involve

---

- A) alkylation of cellular DNA
- B) inhibition of acetylcholinesterase
- C) binding to a peroxisome proliferator-activated receptor
- D) hormesis
- E) necrosis of the renal glomeruli

Which of the following physical-chemical characteristics is generally most useful in estimating the environmental fate and effects of a non-volatile chemical?

---

- A) acidity constant
- B) Henry's Law constant
- C) octanol/water partition coefficient
- D) critical micelle concentration
- E) molecular weight

Characteristic(s) of a eutrophic lake include(s)

---

- 1) high concentration of oxygen throughout the entire depth of the water
- 2) turbidity
- 3) low amounts of nutrients present
- 4) overgrowth of algae

A high capacity to metabolize chemicals contributes to cell-specific acute toxicity for which of the following cell types?

---

- 1) hepatocytes
- 2) olfactory epithelial cells
- 3) lung Clara cells
- 4) skin squamous cells

The confidence interval for the mean of a normally distributed population

---

- 1) increases as the variability increases
- 2) will be greater for the 95% confidence limit than for the 99% confidence limit
- 3) decreases as the sample size increases
- 4) is calculated by means of the  $t$  statistic



The following test results are reportable to the EPA as required by section 8(e) of the Toxic Substances Control Act

---

- 1) birth defects observed in an oral teratology study
- 2) an acute oral LD/50 of 10 mg/kg
- 3) pronounced bioaccumulations in fish
- 4) a 20% decrease in organ weights in a subchronic feeding study

In a two-compartment open model the

---

- 1) semilogarithmic plot of tissue concentration versus time after exposure yields a nonlinear curve
- 2) slope of the initial phase of the plasma concentration versus time curve can be resolved into a straight line using a technique referred to as the method of residuals
- 3) plasma concentration of the chemical in the distribution phase decreases more rapidly than in the postdistribution beta-phase
- 4) rate constants ( $K_{12}$  and  $K_{21}$ ) cannot be calculated from a single-dose study

Reproductive effects figure prominently in regulatory actions involving

---

- 1) ethylene oxide
- 2) chromium III
- 3) dibromochloropropane (DBCP)
- 4) methyl bromide

Which of the following function as receptor-mediated intracellular "amplifiers"?

---

- 1) adenylate cyclase
- 2) phospholipase C
- 3) tyrosine kinase
- 4) acetylcholine esterase

Probabilistic risk assessment is one methodology which explicitly deals with the uncertainties in health risk assessment. Which is/are advantages of probabilistic risk assessment?

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- 1) low dose extrapolations are not used
- 2) single point estimates for exposure are used, which simplifies the calculations and incorporates all uncertainties
- 3) incorporates uncertainty factors (UF) and modifying factors (MF) to derive a risk estimate
- 4) the risk estimate is expressed as a probability distribution which takes into account the range of uncertainty surrounding key variables

Which of the following is/are correct regarding pKa, ionization and pH?

---

- 1) the extent of ionization in body fluid is dependent on the pH of the medium as well as the pKa of the material
- 2) pKa is the negative logarithm of the dissociation constant of a weak organic acid or base
- 3) when pH equals pKa, half the acid present is ionized and half is non-ionized
- 4) at a pH less than their pKa, acids are less completely ionized than  $\text{pH} > \text{pKa}$

Which of the following apply to the biotransformation of xenobiotics?

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- 1) Phase I reactions consist solely of oxidative reactions catalyzed by the cytochrome P450 system
- 2) different stereoisomers of a compound are metabolized identically
- 3) the major cause of hepatic toxicity with carbon tetrachloride is due to hepatic protein adduct formation by an oxidized metabolite of the solvent
- 4) Phase I reactions have been found to increase the toxicity of xenobiotics in some cases

Which of the following is/are example(s) of dispositional antagonism?

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- 1) ipecac
- 2) BAL
- 3) osmotic diuretics
- 4) atropine



Hallmarks of programmed cell death (apoptosis) include

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- 1) nuclear condensation and fragmentation
- 2) new RNA and protein synthesis
- 3) blebbing of the plasma membrane
- 4) phagocytosis by neighboring cells

Which of the following cause(s) toxicity at least in part due to formation of toxic oxygen radicals through the process of "redox cycling"?

---

- 1) cyanide
- 2) 1,2,3,6,-tetrahydro-1-methyl-phenyl-4-pyridine (MPTP)
- 3) methylene blue
- 4) paraquat

Which of the following regulatory programs is/are administered by the US Environmental Protection Agency (EPA)?

---

- 1) regulation of chemicals under the Toxic Substances Control Act (TSCA)
- 2) establishment of national emission standards under Section 112 of the Clean Air Act (CAA)
- 3) regulation of hazardous wastes under the Resources Conservation and Recovery Act (RCRA)
- 4) labeling of pesticides under the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)

Fumonisin produced by the fungus *Fusarium moniliforme* are a common contaminant of corn and corn products throughout the world. Fumonisin

---

- 1) have been implicated in the leukoencephalomalacia of horses
- 2) cause pulmonary edema in pigs
- 3) are hepatocarcinogenic in rats
- 4) have been linked epidemiologically with human esophageal cancer

Data from human epidemiologic surveys have shown an association between

---

- 1) hepatic angiosarcoma and vinyl chloride
- 2) bladder cancer and beta naphthylamine
- 3) mesothelioma and asbestos
- 4) dibromochloropropane and male infertility

Sodium bicarbonate may be useful in the treatment of intoxication with

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- 1) salicylate
- 2) phenobarbital
- 3) methyl alcohol
- 4) amphetamine

Emetics should NOT be used in the treatment of poisoning due to ingestion of

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- 1) acids
- 2) mercurial compounds
- 3) lye
- 4) fluoride

Which of the following statements are characteristic for both resistance and tolerance?

---

- 1) effects are readily reversible
- 2) result from natural selection processes in the group
- 3) are due to genetic factors (inherited)
- 4) always result in a decrease in the toxic response